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COLLABORATIVE RESEARCH

INVEST IN THE CARBON TRANSITION

Corporate carbon toolbox, 2026 update: transition credibility is the new alpha

Carbon has moved from ESG metric to financial variable. Investors are, by nature, exposed to companies whose business models are not aligned with a lower-carbon economy. Regulation, carbon pricing, capex needs and rising disclosure requirements are turning emissions into a direct cost-of-capital issue. The market is therefore shifting from looking only at historical emissions to assessing the credibility of transition plans: targets, execution, capital allocation, governance, and adaptability.

Despite regional differences, the direction of travel is clear: carbon is being priced. Frameworks such as SFDR 2.0 should heighten the focus on measurable transition progress and credible decarbonization strategies. This creates a growing divide between companies that are able to transition profitably and those likely to face rising costs, regulatory pressure, and valuation risk.

SG Corporate Carbon Toolbox provides a consistent framework to assess transition readiness across the STOXX 600. Built by SG analysts and overlaid with insights from Bernstein and Autonomous, our Toolbox combines GHG emissions data with sector expertise to evaluate both current carbon intensity and forward-looking decarbonization potential. It ranks companies on targets, execution, capex alignment, and governance, helping investors compare transition leaders and laggards across industries.

Our investment conclusion is straightforward: investors do not need to choose between returns and decarbonization. The most attractive opportunities are companies that are able to reduce carbon exposure while maintaining financial performance. Our Carbon Toolbox allows investors to build a basket offering good performance, with lower carbon intensity and more robust transition characteristics. At the sector level, the framework favors areas such as Consumer Staples and Healthcare, while pointing to greater caution on Materials and Utilities.

In short: carbon transition is becoming an investable alpha theme. As carbon costs rise and regulation tightens, companies with credible transition plans should be better positioned to protect margins, attract capital, and outperform in a carbon-constrained market.

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Contents

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Executive summary

Why this matters now: transition plans are becoming a core investment variable

Carbon has shifted from an ESG disclosure topic to a financial, regulatory and portfolio-construction variable. The uncertain timing of policy tightening makes unmanaged emissions a growing embedded risk. As a result, investors need to move beyond measuring past emissions and instead assess the credibility of companies' transition plans, including targets, capex, governance, technology pathways, policy exposure and their ability to maintain returns.

The science has crossed a threshold — and shortens the option value of delaying

2024 was the first year above 1.5°C (~1.55°C), with recent years confirming ongoing record warming and leaving only about three years of the remaining carbon budget at current emissions. For investors, the key implication is a rising likelihood of tighter policy action, making credible transition plans essential, as multiple mechanisms—not just carbon pricing—can reprice emissions.

Climate risk is already hitting balance sheets — and forcing action on the transition-risk issue

The physical risk is now real, with large natural catastrophe losses (USD224bn in 2025 after USD368bn in 2024) driving repricing across insurance, assets, credit and supply chains, and leading regulators to treat climate as a financial risk. For investors, ignoring carbon is a position, making it essential to identify companies able to adapt before adjustment costs are imposed.

Regulation is fragmenting — but the direction of carbon repricing is not

Regulatory non-alignment does not remove carbon risk but changes its form, with different regions moving at different speeds and creating a fragmented global framework. For globally exposed companies, this creates basis risk, as carbon exposure can arise across jurisdictions, leaving investors at risk if tightening occurs in any part of the system.

Capital is reallocating — and transition-plan credibility is separating beneficiaries from value traps

Global energy-transition investment reached a record USD2.3tn in 2025 (+8% yoy), led by electrified transport, renewables and grids, with clean-energy supply investment exceeding fossil fuels for a second year and with solid growth in the EU and US.

The key investment implication is to pinpoint companies whose capex, technology and operating models are aligned with tightening carbon constraints, especially as there are several major EU regulatory decisions arriving in 2026 (EU ETS review, CBAM, SFDR 2.0), making a bottom-up assessment of transition-plan credibility essential.

A game changer: SFDR 2.0 and the labelling of climate capital

Why SFDR 1.0 fell short

SFDR was designed as a disclosure regime, but Articles 6, 8 and 9 became de facto labels without consistent criteria, due to differing definitions of "sustainable investment."

This led to both greenwashing concerns and defensive reclassifications, while complex and inconsistent disclosures, data gaps, and limited comparability meant SFDR 1.0 neither prevented greenwashing nor did it direct capital effectively to the transition.

What SFDR 2.0 changes — and why it places added demands on corporate data

SFDR 2.0 replaces the Article 6/8/9 framework with three categories: Sustainable, Transition and ESG basics, shifting the focus from “already green” to “credibly greening,” and allowing capital to reach high-emitting sectors on a credible transition path.

The Transition category requires most investments to actively finance decarbonization, with each company demonstrating a credible plan, aligned capex, measurable progress, governance, and minimum safeguards. The label is based on tangible metrics, combining targets, execution, and capital allocation.

The reform introduces an ambition versus execution test at company level, making comparable transition data essential, while creating a system in which access to capital increasingly depends on having a credible, funded transition plan.

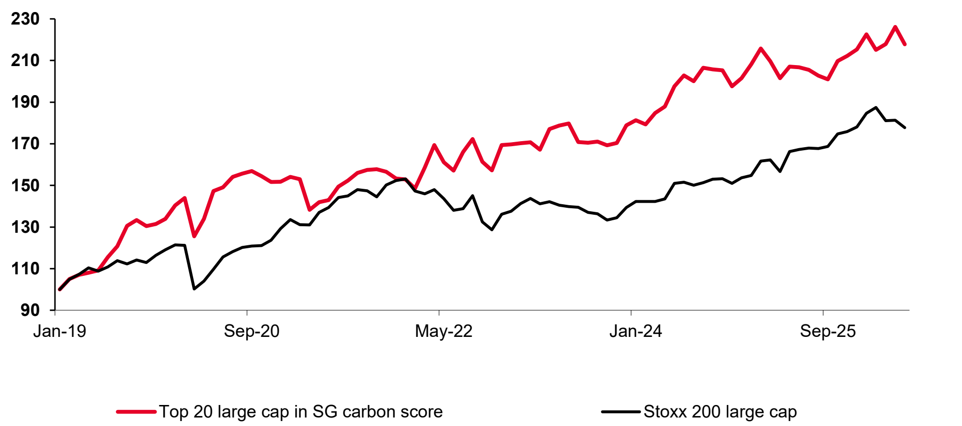
How to use our corporate carbon toolbox

Application 1: Using SG carbon score as a portfolio construction filter on the Stoxx 600 index

The SG carbon score can be used to build baskets of companies that are the best performers on carbon-transition efforts. The selection method consists of starting from a defined universe—such as STOXX 600 large caps—and selecting the two companies with the highest carbon scores within each sector to avoid sector bias, before assessing the performance of this basket over time.

For large caps, this basket outperformed its benchmark, the STOXX 200 Large Cap, over the 2019–June 2026 period, suggesting that large companies investing in the transition may offer better performance potential, as transition efforts help reduce costs.

Top 20 in SG carbon score vs Stoxx 200 Large Cap – June 2026

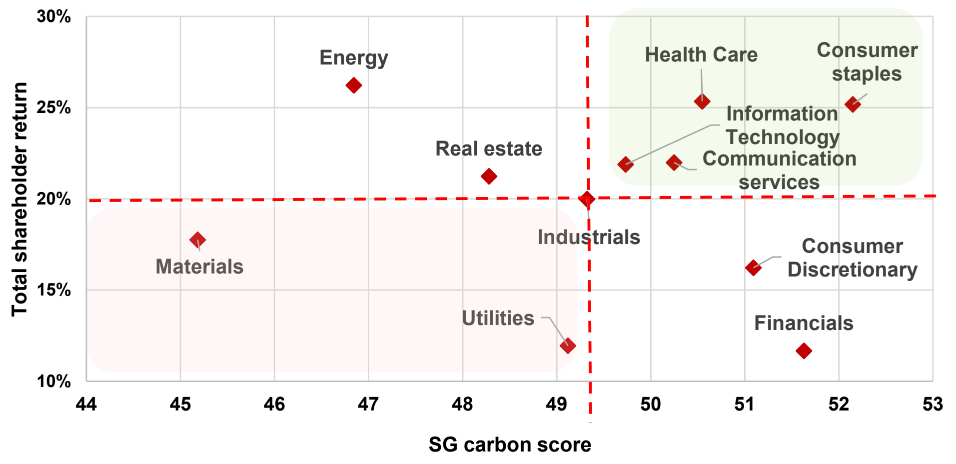


Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Historical data shows aggregated past stock performance, equally weighted. Past performance is not a guarantee of future performance.

Application 2: A tool for stock selection within the Stoxx 600

The SG carbon score enables asset managers to perform stock picking by identifying companies or sectors that combine both low carbon exposure and high market performance potential. By selecting companies with the highest SG carbon scores, alongside the best Total Shareholder Return (TSR) that blends expected share price appreciation (target price) with dividend yield over the next 12 months, based on Bernstein and Autonomous analyst estimates—investors can target stocks that combine robust decarbonization efforts with attractive financial returns.

SG carbon score vs Total Shareholder Return (TSR): sector view



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Our methodological framework

The SG Corporate Carbon Toolbox assigns a score to companies based on their carbon footprint and decarbonization efforts, combining these into a single corporate score that enables quantitative and qualitative comparison within a given universe, such as the STOXX 600. The SG score is built on four components—two backward-looking and two forward-looking—based on measurable metrics, including insights from Bernstein and Autonomous sector analysts.

The backward-looking criteria are *carbon intensity*, which measures a company's GHG emissions relative to its activity level, and *carbon intensity trend*, which captures the annualized reduction in Scope 1 + 2 emissions between the base year and the most recent data, reflecting past decarbonization performance.

The forward-looking criteria are *ambition*, which assesses the breadth of emission reduction targets through a reduction coefficient (annualized reduction weighted by Scope 1 + 2 emissions) and an analyst rating based on target quality, disclosure and alignment with frameworks such as SBTi; and *execution*, which evaluates both the progress achieved versus targets (performance rating) and the credibility of delivery through Bernstein and Autonomous analyst views, including green capex, green R&D and the alignment of board incentives with decarbonization objectives.

The final SG score is calculated as the average of the four sub-scores: carbon intensity, carbon intensity trend, ambition, and execution. It is applied across an equity universe—here, the STOXX 600—to provide a consistent and comparable ranking of companies based on both their historical carbon performance and forward-looking transition readiness.

Ambition and execution ratings

The table below summarizes, for each sector, the companies with the highest and lowest Ambition and Execution ratings, highlighting sector leaders and laggards. The results are presented in alphabetical order by sector to facilitate cross-sector comparison.

Sector leaders and laggards

Top Company	Top Ambition	Top Execution	Sector	Low Ambition	Low Execution	Bottom Company
Rolls-Royce Holdings Plc	4	4	Aerospace & Defense	2	3	Dassault Aviation SA
Mercedes-Benz Group AG	4	5	Automobiles & Auto Components	3	3	Stellantis NV
DNB Bank ASA	4	3	Banks	1	1	Bank Pekao SA
Heineken NV	5	4.5	Beverages	3	3	Anheuser-Busch Inbev SA/NV
Rockwool A/S-B SHS	4.5	4	Building materials	4	3.5	Kingspan Group Plc
Experian Plc	5	5	Business Services	2	1	Rentokil Initial Plc
ABB Ltd-Reg	4.6	5	Capital Goods	2.3	2.6	Kone Oyj-B
Arkema	4	4	Chemicals	2	2	Air Liquide SA
Eiffage	4.5	4	Construction	4.5	3.5	Vinci SA
FinecoBank SpA	5	3	Financial Services	3	3	M&G Plc
Nestle SA-Reg	5	5	Food	2	2	Chocoladefabriken Lindt-Pc
Axfood AB	5	5	Food & General Retail	1	2	Dino Polska SA
L'Oreal	3	5	HPC	2	5	Beiersdorf AG
Whitbread Plc	5	5	Hotels, Restaurants & Leisure	3	3	Sodexo SA
Axa SA	5	5	Insurance	3	2	Ageas
LVMH Moet Hennessy Louis Vui	4	5	Luxury Goods	3	2	Cie Financière Richemo-A Reg
Koninklijke Philips NV	5	5	MedTech	3	1	Demant A/S
Autotrader Group Plc	5	5	Media & Internet	3	2	CTS Eventim AA & Co KGaA
Glencore Plc	5	5	Metals & Mining	3	4	Rio Tinto Plc
Bp Plc	1	5	Oil & Gas Producers	5	1	Eni SpA
Gaztransport et Technigaz SA	4	5	Oil Services	2	1	Rubis
AstraZeneca Plc	5	5	Pharma & Biotech	2	2	Genmab A/S
Land Securities Group Plc	5	5	Real Estate	4	3	Covivio
ASML Holding NV	5	5	Semiconductors	4	5	BE Semiconductor Industries
Capgemini SE	4	5	Software & IT Services	1	1	Dassault Systèmes SE
Deutsche Telekom AG-Reg	3	5	Telecom	3	2	Telenor ASA
DHL Group	5	4	Transportation	5	1	Ryanair Holdings Plc
Getlink SE	4	4.5	Transportation Infrastructure	4	1.5	ADP
EDP SA	5	5	Utilities & Clean Energy	2	2	Enagas SA

Source: SG Cross Asset Research/Invest in the Carbon Transition

Why this matters now: transition plans are becoming a core investment variable

Carbon has moved from an ESG disclosure theme to a balance sheet, regulatory and portfolio-construction variable. The overriding reason for a bottom-up review of STOXX 600 transition plans is not that every jurisdiction has converged on the same rulebook — they have not. It is that the *direction of travel* is now clear enough, but the timing uncertain enough to make unmanaged emissions an embedded option for future regulations.

For investors, the risk is holding companies whose business model, capex and asset base are fundamentally misaligned with a world in which carbon costs, border adjustments, product standards or climate disclosure can tighten faster than expected. The analysis therefore needs to shift from measuring past emissions to judging the credibility of transition plans: targets, capex, governance, technology pathway, policy exposure and the ability to protect returns through the transition.

The science has crossed a threshold — and shortens the option value of delaying

2024 became the first calendar year with average temperatures above 1.5°C (~1.55°C versus 1850–1900), and 2025 ranked second or third hottest at ~1.43°C. The WMO confirmed 2015-2025 as the eleven hottest years on record. And, more decisively for the long-term goal, the 2023–25 three-year average has now exceeded 1.5°C for the first time, while the Global Carbon Budget 2025 puts the remaining 1.5°C budget at roughly three years at current emissions; atmospheric CO_2 first breached 430 ppm in May 2025 and global emissions hit a record ~53 $GtCO_2e$ in 2024.

For investors, the significance is less the individual climate datapoint than the *acceleration of policy probability*. Tighter physical constraints increase the likelihood that governments, regulators, customers and lenders will eventually force a sharper transition response. This is why transition plans matter: they are the corporate expression of preparedness in a world where the "do nothing" option is losing value. Carbon pricing is the foremost mechanism through which the "who pays?" question is answered, but it is not the only one; procurement rules, disclosure regimes, product standards, litigation and financing costs can all reprice emissions even before a uniform global carbon price exists.

Climate risk is already hitting balance sheets — and forcing action on the transition-risk question

Physical risk has stopped being abstract. Munich Re put 2025 natural-catastrophe losses at around USD224bn (~USD108bn insured), after a USD368bn year in 2024. The January 2025 Los Angeles wildfires and Hurricane Melissa are the kind of events now repricing insurance, real assets, sovereign credit and supply chains. Supervisors have drawn the obvious conclusion: through the NGFS and national regimes, central banks treat climate as a prudential risk, and physical and transition risk are now financially material rather than ethically optional.

For a portfolio, ignoring carbon is itself a position — and an increasingly expensive one. The key analytical step is to identify which companies can adjust their asset base, procurement, energy mix and capex before the cost of adjustment is imposed on them.

Regulation is fragmenting — but the direction of carbon repricing is not

The most important investment point is that regulatory non-alignment does not remove carbon risk; it changes its shape. Europe is at the frontier through the EU ETS, CBAM, ETS2, the 2040 target, and the SFDR rewrite. The UK is developing a parallel ETS and CBAM architecture. China has expanded its national ETS to steel, cement and aluminum and is set to move toward an absolute cap by 2027, while Japan, Korea, India, Vietnam and other Asian or emerging-market systems are moving at different speeds.

The US is the counter-example: no federal carbon prices, state-level carbon markets, and active federal anti-ESG pressure. For globally exposed European companies, this patchwork creates basis risk, not safety. A company can be unregulated at group level but exposed through a European plant, an imported input, an Asian supplier, a US state program, a customer procurement rule, or a bank covenant. The latent risk is owning companies that are fundamentally the 'wrong-way' on emissions when one part of that map tightens quickly.

Capital is reallocating — and transition-plan credibility separates beneficiaries from value traps

This is, on the numbers, the largest capital-reallocation story in markets. Bloomberg NEF estimates global energy-transition investment rose to a record USD2.3tn in 2025, up 8% year-on-year — USD893bn on electrified transport, USD690bn on renewables and USD483bn on grids. The EU grew 18% to USD455bn; even the US rose 3.5% to USD378bn despite federal headwinds; and clean-energy supply investment exceeded fossil-fuel supply for a second consecutive year.

The EU's Clean Industrial Deal reframes decarbonization as industrial strategy, mobilizing capital for clean made-in-EU production and linking climate policy to competitiveness. The practical implication is not simply to buy anything labelled "green." It is to identify companies whose capex, technology choices and operating models are aligned with tightening carbon constraints from those using transition language to defend a legacy asset base.

Why now, specifically? Beyond the fundamental, long-term case, the timing is tactical. The next two to three quarters will see a busy calendar of key decisions that are major determinants of both halves of the mechanism this note examines: the EU ETS comprehensive review, CBAM's first definitive year, and the SFDR 2.0 negotiations all land in 2026. The legislating done now informs positioning before the price signal and labelling architecture are reset. That is why the next section moves from macro carbon regulation to methodology: investors need a bottom-up, qualitative assessment of transition-plan credibility — ambition versus execution, capex alignment, governance, exposure to carbon prices and the ability to defend margins — before the market fully prices the regulatory path.

The investor side: SFDR 2.0 and the labelling of climate capital

Carbon regulation sets the real-economy price signal, while the disclosure regime determines how that signal is integrated into portfolios. The central development on the investor side is the overhaul of the EU Sustainable Finance Disclosure Regulation, SFDR 2.0, proposed by the Commission on 20 November 2025, which addresses the gap left by the first version: the absence of a credible way to label and channel climate capital. This invites two questions: why did the original regime fall short? And, what are the implications of the recast SFDR for issuers and investors?

Why did SFDR 1.0 fall short?

SFDR was conceived in 2019 (applying from March 2021) as a disclosure regime, not a labelling one. Articles 6, 8 and 9 were meant to describe how managers treat sustainability — integrating risks, promoting environmental or social characteristics, or pursuing a sustainable objective — not to certify a product as “green.” Lacking any official EU retail label, the market did the certifying itself: “Article 8” and “Article 9” hardened into de facto quality marks with no consistent criteria behind them. Because the pivotal term — “sustainable investment” — was left for each manager to define, two products carrying the same label could imply two very different things, leaving allocators without a comparable basis for selection.

That ambiguity produced precisely the outcomes the regime was meant to prevent. On one side, loose self-classification fed greenwashing concerns; on the other, regulatory tightening triggered defensive behavior. ESMA’s fund-naming guidelines and broader supervisory scrutiny drove successive waves of reclassification — more than 300 funds were downgraded from Article 9 to Article 8 at the end of 2022 alone, with further changes as the naming rules hit home. Managers increasingly judged that calling a transition-oriented strategy insufficiently “sustainable” carried reputational and legal risk, so many de-labelled rather than defend a borderline classification — the opposite of channeling capital into transition, leading to a general retreat from exactly those strategies the EU now wants to promote.

The disclosure machinery compounded the problem. Pre-contractual and periodic templates, the regulatory technical standards (RTS) and entity-level Principal Adverse Impact (PAI) statements were long, technical and heavily templated — too complex for retail investors to use, and too low-signal for institutions to rely on. The “do no significant harm” (DNSH) and good-governance tests proved genuinely hard to apply consistently, and persistent corporate data gaps made matters worse. The Omnibus I cut to the CSRD scope — removing roughly 90% of in-scope firms — threatens to drastically reduce the amount of corporate data on which fund-level disclosures depend. The Commission’s own Article 19 review concluded that disclosures were too complex, comparability was poor, and Articles 8 and 9 were being used as labels without criteria. The net result is clear: SFDR 1.0 neither stopped greenwashing nor reliably allocated capital to transition. Rectifying this fundamental shortcoming is the explicit purpose of the rewrite.

What SFDR 2.0 changes — and why it forces action on corporate data

The change: from "already green" to "credibly greening"

The SFDR rewrite replaces the Article 6/8/9 architecture with three product categories defined by their differing climate and sustainability impact:

- **Sustainable** — typically ~70% of the portfolio aligned with sustainability outcomes (clean industries: renewable energy, green infrastructure, low-carbon technologies).
- **Transition** — funding decarbonization pathways for companies not yet sustainable but on a credible path (high-emission sectors such as steel, cement and energy).
- **ESG basics** — integration of ESG factors without significant sustainability or transition claims.

The conceptual shift is most felt in the middle category and is more profound than it might appear. SFDR 1.0 implicitly asked "is the company green already?", concentrating labelled capital in technology, renewables and other already-low-carbon sectors. SFDR 2.0 asks "is the company moving toward green in a credible way?" and, by design, lets capital reach the high-emitting sectors where the transition has to happen — steel and cement, energy, airlines improving efficiency, utilities shifting their generation mix. The Commission defines the Transition category as investment in companies or projects that are not yet sustainable but are on a credible pathway to becoming so.

The Transition category in practice

The label is awarded at fund level but earned at company level. A Transition fund must allocate most of its capital — typically around 70% — to assets that actively finance change rather than passively hold "brown" exposure, ensuring the label is substantive rather than marketing. Each underlying holding must clear a credibility bar: a company qualifies as a transition asset only if it can demonstrate all of the following:

- **A credible transition plan**: science-based, sector-relevant targets on a defined timeline (e.g. 2030-2050), where "credible" means aligned with EU climate goals and the tenets of the Paris Agreement.
- **Capital alignment**: capex that backs the plan (renewables, electrification, process upgrades); strategy is not enough without money behind it.
- **Demonstrable progress**: interim milestones, not just long-dated goals, tracked through metrics such as emissions intensity and energy-mix change.
- **Governance and accountability**: board-level oversight of the transition and ESG KPIs linked to management incentives, to enforce execution discipline.
- **Minimum safeguards**: negative screening and no severe ESG violations or major controversies, so the label cannot be misused.

The category therefore works only if the essentials are present together — a credible plan, measurable progress and KPIs, and significant capital allocation toward the transition — and funds must disclose transition objectives, KPIs and progress over time. The practical test is intuitive: a coal company with no transition plan, or an oil firm expanding fossil capacity, is not eligible; a steel company investing in green hydrogen, committing to emission cuts and showing capex alignment is eligible — even though it is high-emitting today — precisely because its transition is credible.

The impact: a company-level, ambition-versus-execution test

What SFDR 2.0 really aims to solve is that ESG funds avoided high-emission sectors while the real economy did not decarbonize fast enough. The rewrite creates a safe, criteria-based label so managers can finance decarbonization without being exposed to greenwashing accusations.

The crucial overlap with our work is that the eligibility test resolves into two areas: **ambition** (the plan, the targets, the timeline) and **execution** (capex behind the plan, interim delivery, governance and accountability), which is exactly what our bottom-up framework scores at the company level.

The new regime will force asset managers to make this ambition-versus-execution judgement on every potential transition holding, and to disclose transition progress. Naturally, this implies that comparable and decision-useful company-level transition data becomes essential.

Our scoring is built to supply this exact input, aligning our franchise with the newly recast regulation. In effect, the rewrite raises the premium on standardized corporate disclosure (CSRD, ISSB). SFDR 2.0 and carbon pricing together create a dual pressure — higher operating costs and tougher access to capital — that makes a credible, funded, net-zero-aligned transition plan the price of entry for capital for heavy-emitting sectors.

The 2026 policy window

Beyond the fundamental, long-term case, the timing is tactical. The next two to three quarters will see a busy calendar of key decisions that are major determinants of both halves of the mechanism this note examines: the EU ETS comprehensive review, CBAM's first definitive year, and the SFDR 2.0 negotiations all land in 2026.

The global carbon map: regional approaches and the numbers

The market backdrop: an energy-complex re-rating, and a slower path to net zero

From an investor perspective, the current geopolitical environment — particularly Middle East conflict — has fundamentally raised the importance of the energy complex. The growing inclusion of defense stocks in ESG portfolios, especially in Europe, reflects a broadening definition of what qualifies as "sustainable," and energy could soon follow given its role in national security and economic stability. Recurring supply disruption — from Russia–Ukraine to Middle East tensions and the key shipping chokepoints — serves as a reminder of the strategic importance of hydrocarbons, while renewable replacement at meaningful scale remains years away, supporting demand for traditional energy firms.

That demand is amplified by long-term drivers: the rapid surge in AI-driven data-center infrastructure, rising electrification, and more frequent heatwaves fueling greater cooling demand — all pointing to a fundamentally more energy-hungry world. Of course, this creates a now well-recognized loop: higher energy consumption contributes to warming, which drives still greater energy demand. Fragmented carbon regulation and growing corporate pushback on aggressive targets together imply a more practical, slower path to net zero.

The net effect is a re-rating tailwind for energy — like defense — where valuations increasingly reflect long-term strategic relevance, not just cyclical earnings. Yet, with the world still on a 2.6–3.0°C trajectory, capital will increasingly favor companies actively enabling the transition. This strikes to the core of our "Invest in the Carbon Transition" report: identifying businesses that can bridge today's energy-security needs while executing credible decarbonization for the long term.

Carbon pricing is no longer local — it is being embedded in trade

Cap-and-trade systems are becoming the infrastructure of global climate policy. Emissions trading now covers roughly 22–26% of global emissions, having doubled coverage in recent years. The EU ETS has tightened its cap toward a 62% reduction by 2030 (versus 2005); China has expanded its national system from power into steel, cement and aluminum; and Japan is implementing a mandatory ETS from 2026. Coverage is broadening beyond power into harder-to-abate and consumer-facing sectors: shipping (EU ETS), buildings and road transport (EU ETS2), and heavy industry (China).

Carbon pricing is increasingly embedded in trade rather than confined to domestic policy. CBAM took effect from January 2026 across steel, cement, aluminum, fertilizers and electricity, requiring importers to pay a carbon cost equivalent to EU producers; the US, UK, Canada and Australia are considering similar measures. Price levels remain highly uneven: EU allowances trade around €80–100/t, against a global average nearer USD21/t — although that average has roughly doubled over the decade. Singapore, for example, has raised its carbon tax toward S$45 from S$25.

Europe continues to lead

Europe remains at the forefront of climate regulation: the European Green Deal targets net zero by 2050, with its Fit for 55 package aiming to reduce emissions by at least 55% by 2030. In addition, EU ETS is the world's largest carbon market covering heavy industries, power, and aviation, helping to set caps on emissions by industry and promote carbon prices through trading. Maritime is integrated and Buildings/road transport are being considered for implementation in the next couple of years. Europe has also introduced CBAM (carbon tax on imports), preventing carbon leakage from outside Europe. In this way, Europe applies the principle of penalizing the polluter.

The US is fragmented

In contrast, the US has withdrawn from the Paris agreement, has no effective net zero target, does not have a federal level carbon tax and the IRA continues to promote renewable energy via subsidies and tax credits. So, there is no system to penalize emissions (barring some state policies) and clean energy is commercialized via credits.

In May 2026, SEC rolled back the climate disclosure rule citing it as “overly burdensome”, Scope 1 + 2 emissions are no longer enforced, Scope 3 was never implemented, meaning companies are not even required to disclose emissions.

Asia is developing - and CBAM is pulling it forward

In Asia, which has the largest emission-reduction potential, the recent Middle East energy situation pushed many countries like India, China, South Korea to fall back on coal for energy security. The ETS market is still nascent and developing. China’s ETS covers power and heavy industry. Japan recently implemented a mandatory ETS covering ~60% emissions.

South Korea has a mature ETS market. India’s ETS is still in its early stages, currently based on intensity targets rather than a cap-and-trade system. That said, CBAM moved into the operational stage in 2026 and has asserted pressure on energy-intensive sectors like steel, cement, and aluminum, paving the way for domestic carbon pricing in these emerging markets.

Disclosure regimes are converging on transition - at different speeds

Carbon reporting is the connective tissue between the price signal and capital allocation. In the EU, CSRD makes carbon reporting mandatory and standardized for large companies, and the bloc’s green-finance taxonomy defines green and transition activities, aligning with SFDR 2.0’s transition focus. The US has moved the other way, with the SEC proposing to rescind its disclosure rule.

China is initiating a mandatory ESG-disclosure rollout from 2026 with its own green / transition taxonomy. The global baseline is increasingly IFRS S1 (sustainability risk) and S2 (climate), now being adopted across the UK, Japan, Australia, Singapore and several emerging markets. Outside the EU, the UK SDR mirrors SFDR 2.0’s intent — anti-greenwashing plus product labels (Sustainability Focus, Sustainability Improvers, Sustainability Impact) reported against ISSB standards. In short: the EU is the most advanced, Asia is expanding (particularly China, Japan, and South Korea), and the US is fragmented and state-led.

Most economies are set to underdeliver on their Nationally Determined Contributions (NDCs)

Our SG economists have estimated a 13.8 percentage point implementation gap on country-level GHG emissions-reduction goals (GHG emissions in 2025-35 – Projecting a higher implementation gap). Considering the major economies that we have highlighted in the table below cover almost 75% of global GHG emissions, we see that although emissions in advanced economies should fall by 2030, all those countries are expected to underdeliver on emissions expectations and not meet the 2030 NDC commitments. Equally, most emerging markets are also unlikely to meet the 2030 goals.

Summary of major economies' NDCs (recent update including various COP summits)

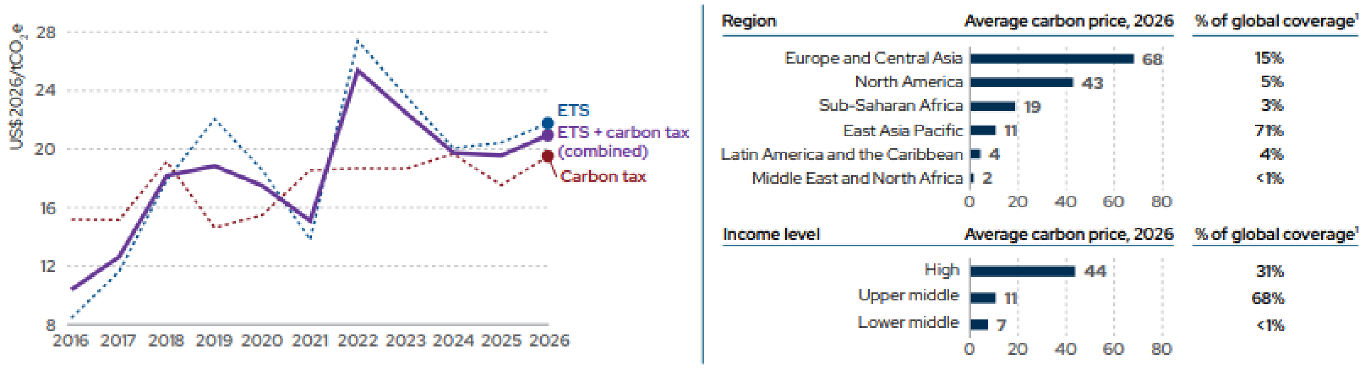
Economy	Carbon neutrality objective	Type of emissions (target)	Reference	Target year	Targeted reduction (%)	SG estimated gap to 2030 target (%p)	Comment
Australia	Yes, 2050	GHG	2005	2030	-43	25	Australia's 2025 NDC targets a 62–70% emissions cut by 2035 (from 2005 levels), reaffirms 2030 and 2050 goals, and expands renewables investment. While boosting clean energy certainty, the plan faces implementation hurdles and criticism for excluding fossil fuel exports.
EU27	Yes, 2050	GHG	1990	2030	-55	11	The European Union remains committed to climate action, reaffirming the phase-out of Russian gas by 2027. The EU targets a 55% GHG reduction by 2030 and net zero by 2050. However, slow transport electrification, relaxed car emissions standards, and political concessions—such as slower phase-out of free carbon allowances and potential delays to the ETS2 scheme—mean the EU is likely to miss its 2030 target by 12%. New 2035 and 2040 targets (up to 90% below 1990 levels) allow for international carbon credits, effectively softening ambition. Nearly half of future reductions are expected from ETS1, with ETS2 expanding to more sectors but raising energy costs and inflation. Changes to ETS2 and increased support for low-carbon technologies are likely, as the EU balances ambition with economic and political realities.
		GHG emissions in sectors covered by the ETS	2005	2030	-62		
Germany	Yes, 2050		1990	2030	-64.8	11	
France					-50.1	12	
Italy					-44.2	9	
Spain					-20.2	10	
Japan	Yes, 2050	GHG	2013	2030	-46	9	
UK	Yes, 2050	GHG	1990	2030	-68	8	The UK targets a 68% emissions cut by 2030 and an 81% reduction by 2035 (vs. 1990 levels). Despite strong policies, under-delivery on renewables, EVs, and heat pumps leaves the UK -8pp short of the 2030 NDC, increasing risks to the 2035 target.
US	Yes, 2050	GHG	2005	2030	-50/52	31	The Trump administration's reversal of US climate policy—withdrawal from the Paris Agreement, regulatory rollbacks, and fossil fuel promotion—undermines global efforts. Federal support for climate action and finance is diminished, though state and local initiatives may still drive some emissions reductions amid heightened regulatory uncertainty.
Canada	Yes, 2050	GHG	2005	2030	-40/45	36	Canada submitted its updated NDC targeting to reduce emissions by 45-50% below 2005 levels by 2035, building on the 2030 target of 40-45%. It is still significantly inadequate.
China	Yes, 2060	CO2 intensity	2005	2030	-65	0	China's 2035 climate target shifts to absolute GHG reductions, aiming for only a 7–10% cut from peak levels—widely seen as unambitious. Despite progress in carbon intensity and renewables, the pace is insufficient for deeper decarbonization, with development and energy security still prioritized over climate ambition.
Taiwan	Yes, 2050	GHG	2005	2030	-28±2	4	Taiwan aims to reduce GHG emissions by 28% in 2030 and by 36-40% in 2035 from 2005 levels. The progress is slow, with delayed renewable targets and ongoing industrial growth. Backloaded efforts and new carbon pricing are expected.
Indonesia	Yes, 2060	GHG	BAU	2030	-32	30	Indonesia's new NDC projects emissions to peak in 2030 and provides emission estimates for 2035. It emphasizes forestry reforms and renewables expansion. New regulations strengthen carbon markets and reopen international trading. Despite increased ambition, achieving targets requires major energy reforms, investment incentives, and overcoming coal reliance and regulatory hurdles.
India	Yes, by 2070	GHG intensity	2005	2030	-45	-3	India's April 2026 NDC targets a 47% emissions-intensity cut by 2035, only a marginal step-up from its 45% 2030 goal. While enhancing medium-term clarity and likely to be met comfortably, the ambition remains modest, misaligned with a 1.5℃ pathway and insufficient for achieving net-zero by 2070 despite strong clean-energy progress.
South Korea	Yes, 2050	GHG	2018	2030	-40	18	South Korea, under President Lee, proposes cutting emissions by at least 50% from 2018 levels by 2035, accelerating coal phase-out, and prioritizing renewables. Institutional reforms and economic growth are emphasized, but grid, permitting, and competitiveness challenges persist.
Russia	No	GHG	1990	2030	-25	9	
South Africa	Yes, 2050	GHG	Absolute, to GTCO2e 420	2030	-24	18	South Africa's new NDC sets 2035 emissions target of 320–380 $MtCO_2$-eq, relying on large renewable expansion, a strengthened just transition, and significant international finance. Experts argue the target is not ambitious enough, adaptation plans lack detail, and accountability is weak.
Turkey	Yes, 2053	GHG	BAU, base year 2012	2030	-41	32	Turkey submitted its new NDC in November 2025 aiming to reduce its GHG emissions (incl LULUCF) by 466 $MtCO_2$eq. compared to the BAU scenario, limiting emissions to 643 $MtCO_2$eq. (-42%) in 2035.
Saudi Arabia	Yes, 2060	GHG	2019	2030		119	Saudi Arabia has ambitious climate pledges under Vision 2030, targeting net-zero by 2060 and large emissions reductions via renewables, CCUS and hydrogen. However, continued fossil-fuel dependence, opaque targets and slow implementation mean it is likely to miss its 2030 goal by a wide margin.
Argentina	Yes, 2050	GHG	Absolute	2030		21	The strategic retreat from climate policy, prioritizing short-term economic and fossil-fuel interests, and abandoning credible planning have left the country structurally misaligned with climate goals and increasingly uncertain on its medium-term climate direction.
Brazil	Yes, 2050	GHG	2005	2030	-53.1	65	Despite halving Amazon deforestation and securing climate finance, progress toward 2030 NDC goals remains insufficient. The 2035 target pledges a 59–67% cut, but implementation and baseline clarity challenges persist, widening Brazil's projected 2030 shortfall.
Colombia	Yes, 2050	GHG	BAU	2030	-51	33	Colombia's new NDC targets a 26% emissions cut by 2035 (vs 2024 level) and limits deforestation, with advances in renewables and transport decarbonization. However, rising deforestation, fossil fuel dependence, fiscal constraints, and limited finance threaten achievement of its ambitious climate targets, highlighting implementation challenges.
Chile	Yes, 2050	GHG	BAU	2030	-45	4	Chile's 2025 NDC sets a 490 $MtCO_2e$ budget for 2031–35, with emissions peaking in 2025 and a 91 $MtCO_2e$ cap by 2035, only modestly stronger than 2030 goals. President Kast prioritizes growth-led investment, easing regulations and permitting, raising risks to medium-term climate ambition.
Mexico	Yes, 2050	GHG	BAU	2030	-35	26	Updated NDC sets absolute 2035 emissions caps at 364–404 $MtCO_2e$ (unconditional) and 332–363 $MtCO_2e$ (conditional), providing a clearer benchmark for long-term net-zero ambitions.
Total (75.4% of world GHG in 2030)						13.8	

Source: SG Cross Asset Research, UNFCCC, SG Cross Asset Research/Economics. Note, in this table, "GHG" means GHG excluding LULUCF

Carbon pricing differences across geographies

The highest level of direct carbon pricing is in Europe and Central Asia at US$ 68/tCO2e. These are followed by North America at US$ 43/tCO2e, and finally the East Asia Pacific region pricing, at US$ 11/tCO2e.

Average carbon price for ETFs, carbon taxes

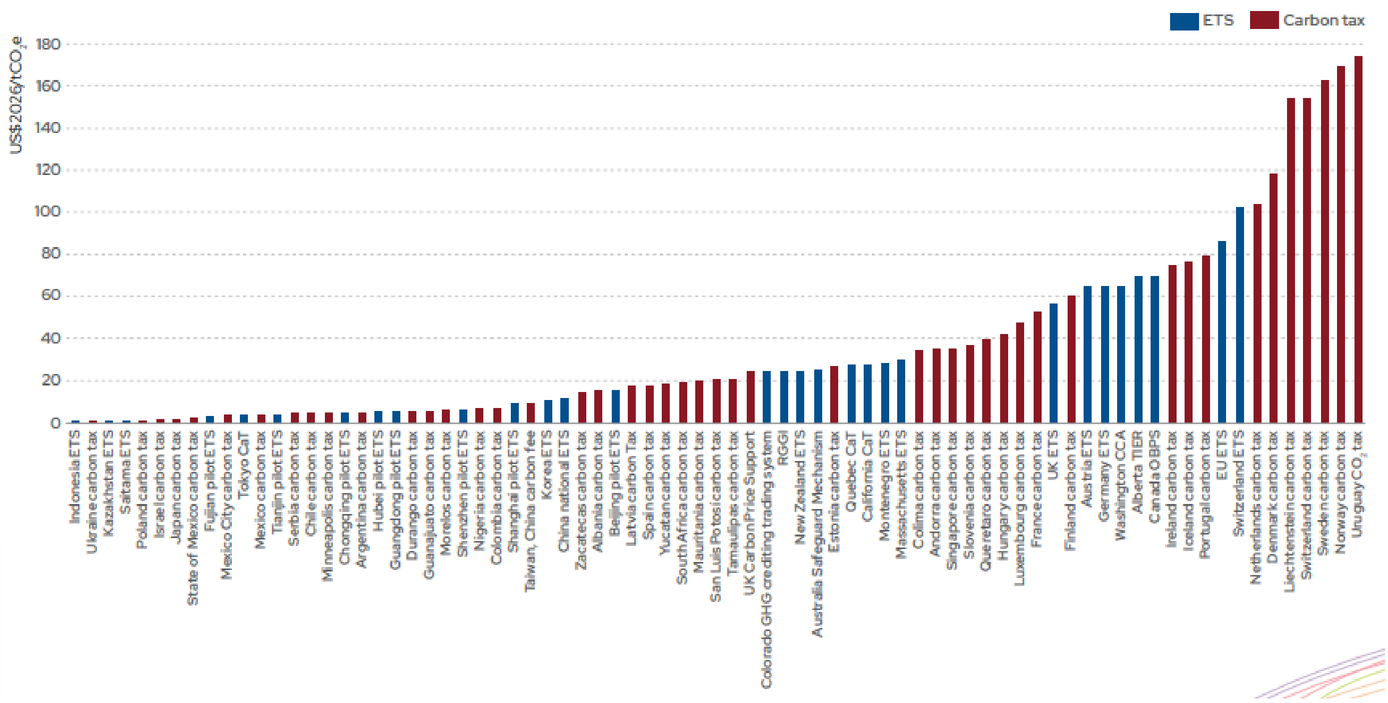


[1] Calculated as percentage of global GHG emissions covered by an ETS or carbon tax. Values exclude newly implemented ETSs in India, Japan and Viet Nam where carbon prices are not available at time of publication. Prices are as of April 1, 2026

Source: SG Cross Asset Research, World Bank (State and Trends of Carbon Pricing 2026)

And the carbon pricing differences can also be seen country by country, even with a region; see the chart below dated 1 April 2026.

Average carbon price for ETFs, carbon taxes



Source: SG Cross Asset Research, World Bank (State and Trends of Carbon Pricing 2026)

How to use the SG carbon score

Using the SG carbon score as a portfolio construction filter on the Stoxx 600 index

Construction of the peer group universe

The SG carbon score framework aims to monitor the performance of a basket of 20 companies selected based on their SG carbon score. These firms are distributed across ten sectors, with two companies representing each sector.

The selection process is based on the following criteria:

- The initial universe consists of the STOXX 600 index, excluding companies with insufficient data (resulting in 586 eligible firms).
- Companies are then classified by market capitalization into three segments:
  - Large caps: top third of the STOXX 600 by market capitalization
  - Mid-caps: middle third of the STOXX 600 by market capitalization
  - Small caps: bottom third of the STOXX 600 by market capitalization

To assess the relevance of the selection, the SG carbon score portfolios are compared against their corresponding STOXX 200 benchmarks. Three separate portfolios are built—one for each market capitalization segment—and each is evaluated against its respective benchmark index.

The 20 companies included in the analysis are selected based on their SG carbon score rankings. Historical price data for these companies spans from January 2019 to June 2026. Portfolio performance is calculated using market equally weighted returns across the selected stocks.

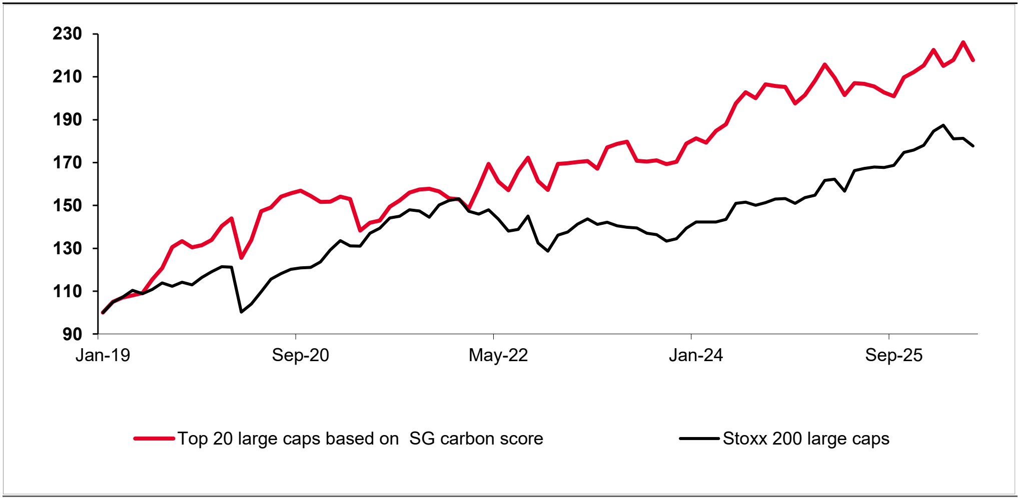
For each market cap segment, a portfolio is constructed using carbon scores and benchmarked against the appropriate STOXX 200 index. To reduce sector concentration risk and ensure diversification, the top two companies within each sector are selected. The performance of these portfolios is then analyzed over a five-year investment horizon.

Throughout this report, “baskets” refer to illustrative selections based on our carbon score tool and are not created as financial instruments.

Large caps: Stoxx 200 – Analysis of our performance

The results of the carbon filter applied to a universe of 200 large-cap stocks show that the resulting portfolio outperforms the Stoxx 200 large-cap index over the period from 1 January 2019 to 26 June 2026. To avoid sector biases in the portfolio construction, the filter was applied to a basket of 20 companies per sector, with the top two companies selected within each of the ten sectors. This leads to a final portfolio of 20 stocks.

Market performance: Top 20 large caps based on SG carbon score vs Stoxx 200 large caps – January 2019 to June 2026



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Historical data shows aggregated past stock performance, equally weighted. Past performance is not a guarantee of future performance.

Top 20 large caps based on SG carbon score (the higher the better): 1 January 2019 to 26 June 2026

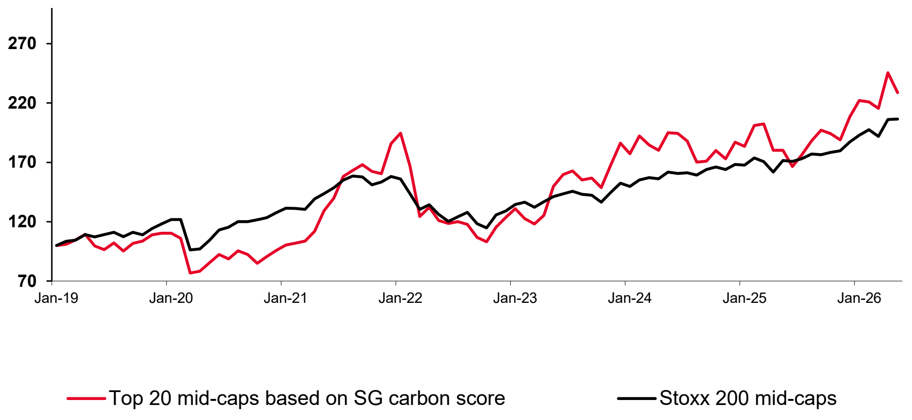
Company name	Sector	Market cap. (€bn)	Carbon score
DEUTSCHE TELEKOM AG-REG	Communication Services	128	60
TELEFONICA SA	Communication Services	21	53
INDUSTRIA DE DISENO TEXTIL	Consumer Discretionary	173	67
MERCEDES-BENZ GROUP AG	Consumer Discretionary	43	66
TESCO PLC	Consumer Staples	33	68
NESTLE SA-REG	Consumer Staples	230	65
ORLEN SA	Energy	33	54
REPSOL SA	Energy	23	53
AXA SA	Financials	90	69
LONDON STOCK EXCHANGE GROUP	Financials	47	69
ASTRAZENECA PLC	Health Care	252	73
KONINKLIJKE PHILIPS NV	Health Care	23	65
EXPERIAN PLC	Industrials	26	70
RELX PLC	Industrials	49	68
ASML HOLDING NV	Information Technology	594	70
SAP SE	Information Technology	166	61
GLENCORE PLC	Materials	69	65
FRESNILLO PLC	Materials	24	56
ENEL SPA	Utilities	100	67
ORSTED A/S	Utilities	26	67

Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

Mid-caps: Stoxx 200 – Analysis of our basket's performance

The results of the carbon filter applied to a universe of 200 mid-cap stocks indicate that the resulting portfolio outperforms the Stoxx 200 mid-cap index over the period from 1 January 2019 to 26 June 2026. To mitigate sector biases in the portfolio construction, the filter was applied to a basket of 20 companies per sector, selecting the top two companies within each of the ten sectors. This results in a final portfolio of 20 stocks, with the selected basket outperforming the benchmark.

Market performance: Top 20 mid-caps based on SG carbon score vs Stoxx 200 mid-cap index – 1 January 2019 to 26 June 2026



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Historical data shows aggregated past stock performance, equally weighted. Past performance is not a guarantee of future performance.

Top 20 mid-caps based on SG carbon score (the higher the better): 1 January 2019 to 26 June 2026

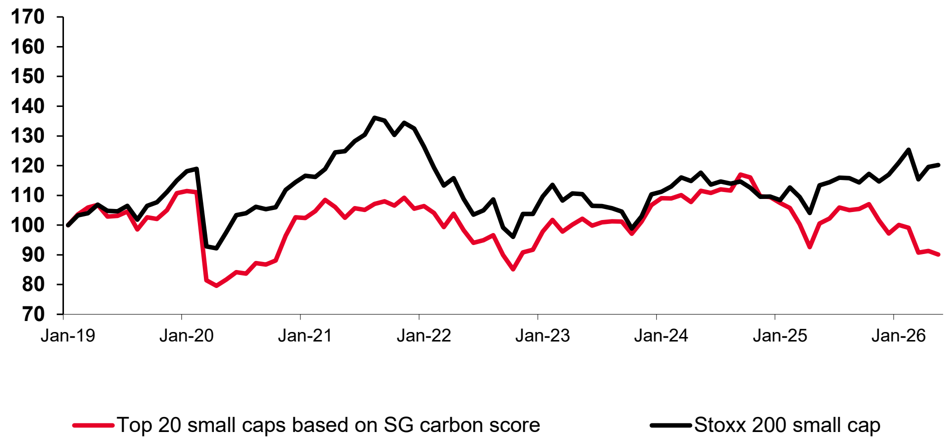
Company name	Sector	Market cap. (€bn)	Carbon score
KONINKLIJKE KPN NV	Communication Services	17	57
TELE2 AB-B SHS	Communication Services	11	57
RENAULT SA	Consumer Discretionary	8	65
LPP SA	Consumer Discretionary	8	61
PERNOD RICARD SA	Consumer Staples	17	61
HEINEKEN HOLDING NV	Consumer Staples	19	60
AKER BP ASA	Energy	17	52
GALP ENERGIA SGPS SA	Energy	14	51
AEGON LTD	Financials	12	67
NN GROUP NV	Financials	20	62
COLOPLAST-B	Health Care	12	58
STRAUMANN HOLDING AG-REG	Health Care	18	58
WOLTERS KLUWER	Industrials	13	74
FLUGHAFEN ZURICH AG-REG	Industrials	8	62
CAPGEMINI SE	Information Technology	15	68
SAGE GROUP PLC/THE	Information Technology	9	65
AURUBIS AG	Materials	8	51
STORA ENSO OYJ-R SHS	Materials	7	50
MERLIN PROPERTIES SOCIMI SA	Real Estate	10	58
UNIBAIL-RODAMCO-WESTFIELD	Real Estate	15	56

Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

Small caps: Stoxx 200 – Analysis of our basket's performance

The results of the carbon filter applied to a universe of 200 small-cap stocks show that the resulting portfolio underperforms over the period from 1 January 2019 to 26 June 2026. The small-cap universe includes the Utilities sector, which is not represented in the large- and mid-cap universes but is present among small-cap companies. As a result, the basket consists of 22 stocks, reflecting a selection across 11 sectors. This underperformance can be explained by the fact that small-cap companies are still in earlier stages of development and are primarily focused on expanding their business activities, while large- and mid-cap companies have reached a level of maturity that enables them to invest more significantly in decarbonization initiatives.

Market performance: Top 22 small caps based on SG carbon score vs Stoxx 200 small caps – 1 January 2019 to 26 June 2026



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Historical data shows aggregated past stock performance, equally weighted. Past performance is not a guarantee of future performance.

Top 22 small caps based on SG carbon score (the higher the better): 1 January 2019 to 26 June 2026

Company name	Sector	Market cap. (€bn)	Carbon score
AUTOTRADER GROUP PLC	Communication Services	4.5	71
ITV PLC	Communication Services	3.5	70
PANDORA A/S	Consumer Discretionary	6.9	65
WHITBREAD PLC	Consumer Discretionary	4.9	62
AXFOOD AB	Consumer Staples	5.1	73
BARRY CALLEBAUT AG-REG	Consumer Staples	6.9	55
SBM OFFSHORE NV	Energy	5.5	79
GAZTRANSPORT ET TECHNIGA SA	Energy	6.8	65
AVANZA BANK HOLDING AB	Financials	5.4	84
ALLFUNDS GROUP PLC	Financials	5.1	63
ZEALAND PHARMA A/S	Health Care	2.7	67
CAMURUS AB	Health Care	3.0	57
ARCADIS NV	Industrials	3.0	59
KION GROUP AG	Industrials	5.1	57
SOPRA STERIA GROUP	Information Technology	2.8	60
BECHTLE AG	Information Technology	3.8	50
THYSSENKRUPP AG	Materials	6.6	50
UMICORE	Materials	5.2	50
LAND SECURITIES GROUP PLC	Real Estate	5.6	62
GECINA SA	Real Estate	5.6	61
DRAX GROUP PLC	Utilities	2.9	55
TAURON POLSKA ENERGIA SA	Utilities	3.6	50

Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg. Past performance is not a guarantee of future performance.

Using SG carbon scores for stock selection

The calculation of carbon scores across all companies in the STOXX 600 enables asset managers to perform targeted stock-picking by constructing baskets that combine sustainability and financial performance.

Specifically, they can select companies with the strongest carbon scores, reflecting those most effectively committed to decarbonization, while simultaneously focusing on firms delivering attractive returns. This approach allows investors to identify stocks that not only lead in climate transition efforts but also generate solid total shareholder returns, which include both the expected share price appreciation (price target) and the dividend yield for the next 12 months.

The total shareholder return (TSR) formula is the following:

$$\text{Total shareholder return (TSR)} = \frac{\textit{Price Target}+\textit{Dividend}\ 12\ \textit{months ahead}}{\textit{Last Price}}$$

The price target is assigned by the Autonomous and Bernstein analysts who cover the stock; otherwise, it is based on the IBES (Institutional Brokers' Estimate System) consensus. The 12-month dividend is also provided by the IBES consensus.

Sector performance: SG carbon score vs total shareholder return (TSR)

Building on the 579 companies retained for the carbon score analysis, the TSRs are computed and averaged at the sector level alongside carbon scores. By combining these sector averages, we identify sector profiles that offer the most attractive trade-off between high carbon performance and strong TSR, thereby supporting sector selection.

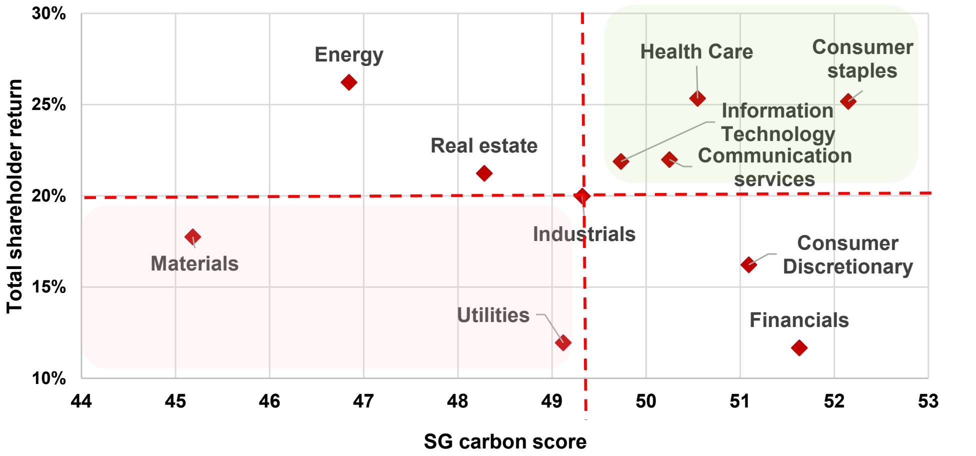
Average TSR/carbon score combination by Stoxx 600 sector: the higher the better

Sector	SG carbon score	Total shareholder return
Financials	51.63	12%
Consumer staples	52.15	25%
Energy	46.84	26%
Information Technology	49.73	22%
Materials	45.18	18%
Health Care	50.55	25%
Industrials	49.32	20%
Utilities	49.12	12%
Consumer Discretionary	51.09	16%
Communication Services	50.25	22%
Real Estate	48.28	21%
Average	**49.47**	**20%**

Source: Bloomberg, IBES, SG Cross Asset Research/Invest in the Carbon Transition, Bernstein analysis, Autonomous Research

By analyzing SG carbon scores alongside total shareholder returns across all Stoxx 600 sectors, Health Care, Consumer Staples, Information Technology, and Communication Services stand out as the most attractive sectors, as they combine above-average TSRs with strong SG carbon scores. Allocating to these sectors therefore helps maximize portfolio performance while reducing its carbon footprint. In contrast, Materials and Utilities appear less attractive, as they exhibit both below-average TSRs and below-average carbon scores. These sectors can therefore be excluded from the sector selection process.

SG carbon score vs total shareholder return (TSR): sector view



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Bernstein analysis, Autonomous Research

SG carbon score methodology

Methodology

The SG corporate carbon toolbox provides a score to assess a list of companies based on their carbon footprint and their efforts to decarbonize their activities. This scoring approach is an attempt to quantify these efforts with a rigorous scientific methodology, enabling a meaningful comparison between companies. It also allows for the ranking of firms while considering key factors such as their size and operational context, ensuring a more balanced and fair evaluation.

The framework focuses exclusively on carbon-related metrics, using high-quality GHG data at the company level, largely based on information reported by the companies themselves. It combines several calculated factors, including emissions levels, decarbonization targets, and the actions already implemented or planned to reduce carbon footprints. **These elements are then consolidated into a corporate score, providing both a quantitative and qualitative assessment of each company's carbon performance and enabling comparisons within a given universe, such as the Stoxx 600 Equity Index.**

The SG score is built around four key components: two backward-looking criteria and two forward-looking criteria. This structure allows the final score to capture both the decarbonization efforts already achieved by companies and the actions they are expected to undertake in the future. These components are based on factual data sourced from the financial reports of the target companies, as well as, importantly**, insights from Bernstein and Autonomous sector analysts, which are translated into tangible and measurable metrics.**

Backward/forward-looking criteria

▪ **Backward-looking:** this focuses on analyzing historical GHG (greenhouse gas) emissions data, including Scope 1, 2, and 3 emissions. Scope 1 covers direct emissions from a company's operations, Scope 2 includes indirect emissions from energy consumption, and Scope 3 encompasses all other indirect emissions across the value chain. However, our methodology only considers Scope 1 and 2 emissions, as ensuring the quality of Scope 3 emissions data remains a significant challenge. To avoid bias when comparing companies of different sizes and production capacities, we calculate carbon intensity, measured as emissions per unit of consolidated turnover. Additionally, the methodology tracks changes relative to a baseline year, typically 2019.

– **(1) Carbon Intensity:** calculated as the ratio of tons of CO2e GHG emissions in Scopes 1 and 2 to a company's consolidated revenue, both in 2025.

– **(2) Carbon Intensity Trend:** the percentage change in carbon intensity from 2019 to 2025.

▪ **Forward-looking:** most companies now set carbon-emission reduction targets, which we capture and normalize. We then use inputs from Bernstein (for non-financials) and Autonomous (for financials) to rate the ambition and execution of these targets.

– **(1) Ambition**: this reflects a company's commitment to its GHG reduction strategy. Around 8,000 corporations now disclose their carbon-emission targets, typically over a transition period of around ten years. To assess these targets effectively, we have collected specific information, such as the start date and the first interim target date, and the percentage reduction of GHG emissions in Scopes 1 & 2 over this specified period. We use two sub-criteria:

> a - **Reduction coefficient**: an assessment of a company's GHG reduction goal. From the reduction rate of GHG emissions in % for the announced perimeter disclosed by the company, we have annualized this reduction rate in % with the number of years between the baseline year and the first interim target year. Then, we have weighted the above % annualized reduction rate by GHG emissions in Scopes 1 & 2 only at the baseline year.

b - **Ambition rating**: this reflects the perspective of Bernstein/Autonomous sector analysts regarding the ambition of a company's GHG emissions reduction targets. **The sector analysts** evaluate the targets **on a scale of 1-5 with the best rating at 5**. The analysts assess the following factors in a best-in-class approach by sector:

- **Reduction coefficient** based on the annualized reduction rate weighted by GHG emissions in Scopes 1 & 2 at the baseline year. On a sector basis, the reduction coefficient ranks the ambition of a company's efforts to reduce its GHG emissions in Scopes 1 & 2 and so its carbon footprint.
- **Quality of disclosure.** Is the data measured or modelled? Are they limited to CO2, or do they include other greenhouse gases? Which scopes are included in the target (1+2, or 1+2+3)? How many interim targets are there in the published carbon reduction strategy? Is there a net zero strategy in 2050?
- **Science-based targets (SBTi) and independent control**. Does the company follow the science-based targets under the Science Based Targets Initiative (SBTi) or those of any other recognized consultants?

– **(2) Execution**: this score evaluates the management board's commitment to achieving the company's emission reduction targets and the board's strategic execution. The execution rating is based on a combination of quantitative data to build the **performance rating (for companies not covered by Bernstein/Autonomous)** and the **opinion of Bernstein/Autonomous sector analysts** (only for stocks covered):

a **– Performance rating:** this calculates the progress achieved by a company on its strategic plan for reducing GHG emissions in Scopes 1 & 2, from the base year to end-2024. We compare the annualized GHG emissions in Scope 1 + 2 reduction from base year to end-2024 with the annualized GHG reduction plan from base year to target year. The result is an outperformance or underperformance that we translate into a score.

b **– Execution rating:** the perspective of Bernstein/Autonomous sector analysts regarding the credibility of a company's GHG emission reduction strategy. **Opinion of sector analysts** – the analysts evaluate the targets **on a scale of 1-5, with the best rating at 5**. The following factors are considered in a best-in-class approach by sector:

- **Green capital expenditure (capex) as a percentage of total capex**: what proportion of the company's capital expenditure is allocated to green initiatives?
- **Green research and development (R&D) as a percentage of total R&D**: what percentage of the company's R&D budget is earmarked for green technologies and initiatives?
- **Board member incentives**: are the incentives for board members aligned with the decarbonization plan?

Data collection

▪ **Carbon data collection pipeline:** we utilize Bloomberg functions to improve the collection of carbon data, specifically for GHG emissions across Scopes 1 and 2. The carbon data prioritized in our analysis focuses on location-based Scope 1 and 2 GHG emissions. Location-based refers to emissions calculated using the average emissions intensity of the grids where energy consumption occurs, providing a clearer picture of the environmental impact tied to the specific energy mix of each location.

▪ **Missing data:** Not all companies report complete emissions data. Among the companies providing data on Bloomberg, some provide CO2 data without GHG data, while others offer market-based data instead of location-based data. To facilitate comparison, we employ a waterfall approach for data collection. This involves gathering five types of Scope 1 and 2 data from Bloomberg: location-based GHG, market-based GHG, location-based CO2, Bloomberg GHG estimates, and industry-average implied emissions. In the absence of location-based data, we sequentially move to market-based data, followed by CO2 data, and then to Bloomberg estimates and industry averages, ensuring that all collected data are recognizable and traceable.

– **Waterfall approach:** the functions used for our Scope 1 and 2 data extraction, following the waterfall approach outlined above, are detailed below.

Carbon data collection methodology by GHG emissions scope using Bloomberg formulas

Collection of Scope 1 data	Collection of Scope 2 data
FORMULA 1	**FORMULA 1**
GHG_SCOPE_1	GHG_SCOPE_2_LOCATION_BASED
FORMULA 2	**FORMULA 2**
CO2_SCOPE_1	GHG_SCOPE_2_MARKET_BASED
FORMULA 3	**FORMULA 3**
GHG_SCOPE_1_ESTIMATE	CO2_SCOPE_2_LOCATION_BASED
FORMULA 4	**FORMULA 4**
GHG_SCOPE_1_IND_IMPLIED_ESTIMATE	GHG_SCOPE_2_ESTIMATE
	FORMULA 5
	GHG_SCOPE_2_IND_IMPLIED_ESTIMATE

Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

SG carbon score calculation formula

The SG carbon score is calculated based on four company-level criteria: carbon intensity, carbon intensity trend, ambition, and execution. Each criterion is then converted into a score using a **min-max scaling** approach applied to a defined universe of companies, such as the Stoxx 600 index. **The final score is therefore the average of the four individual scores:**

– **(1) Carbon intensity score:** the carbon intensity score for a company (n) is calculated using the formula:

$$\boldsymbol{Carbon\ intensity\ score_n = 100 * \frac{\log Carbon\ intensity_n - Max}{Min - Max}}$$

where Max and Min represent the maximum and minimum logarithmic carbon intensity values across all companies, ensuring the score is normalized on a scale of 0 to 100.

– **(2) Carbon intensity trend score:** the carbon intensity trend score for a company (n) is calculated using the formula:

$$\boldsymbol{Carbon\ intensity\ trend\ score_n = 100 * \frac{Max - CITn}{max - min}}$$

where Max and Min represent the maximum and minimum carbon intensity trend CIT values across all companies, ensuring the score is normalized on a scale of 0 to 100.

– **(3) Ambition score:** the ambition score is the average of two components that are scored: the reduction coefficient and the ambition rating, both of which are calculated using min-max normalization formulas based on the reduction coefficient and the ambition rating.

The reduction coefficient and the ambition rating for company (n) are scored using the formulas:

$$\begin{cases} \textit{Reduction coefficient score}_n = 100 * \dfrac{\textit{Reduction coefficient} - \textit{Min}}{\textit{max} - \textit{min}} \\ \textit{Ambition rating score}_n = 100 * \dfrac{\textit{Ambition rating} - \textit{Min}}{\textit{max} - \textit{min}} \end{cases}$$

The ambition score for company (n) is then calculated using one of the following formulas depending on whether the company is covered by Bernstein/Autonomous analysts or not:

$$\textit{Ambition score}_n = \begin{cases} \dfrac{\textit{Reduction coefficient score}_n + \textit{Ambition rating score}_n}{2} \\ \textit{Reduction coefficient score}_n \quad \text{if issuer is not covered} \end{cases}$$

– **(4) Execution score:** the execution score for company (n) is then calculated using one of the following formulas depending on whether the company is covered by Bernstein/Autonomous analysts or not:

$$\textit{Execution rating score}_n = 100 * \frac{\textit{Execution rating} - \textit{Min}}{\textit{max} - \textit{min}} \quad \text{if issuer is covered}$$

When execution ratings are unavailable from Bernstein analysts, the execution score is replaced by a performance score, which is also calculated using min-max normalization. The performance score for a company (n) is derived according to this formula:

$$\textit{Performance score}_n = 100 * \frac{\textit{Performance coefficient} - \textit{Min}}{\textit{max} - \textit{min}} \quad \text{if issuer is not covered}$$

Carbon score calculation

	Formula	Comments	Acronyms	
Strategy + Coverage	Carbon score = (CI + CIT + A + E) / 4	A = (R+D)/2	CI = Carbon intensity trend	D = Disclosure
Strategy + No coverage	Carbon score = (CI + CIT + A + P) / 4	A = R	A = Ambition	E = Execution
No strategy + No coverage	Carbon score = (CI + CIT + A + P) / 4	A = 0; P=0	R = Reduction	P = Performance

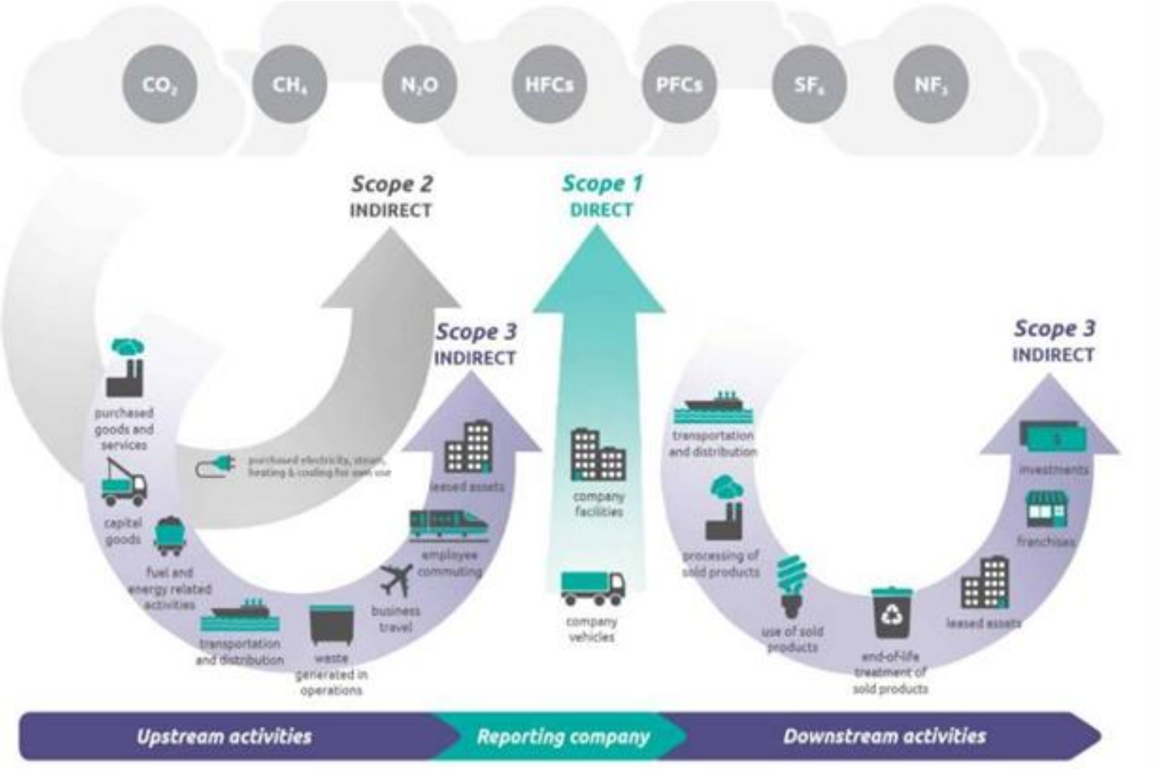
Source: SG Cross Asset Research/Invest in the Carbon Transition

Weakness in Scope 3

GHG emissions are currently classified into three scopes:

- **Scope 1:** direct emissions from sources owned or controlled by a company.
- **Scope 2:** indirect emissions from the generation of purchased or acquired electricity, heat, steam, or cooling consumed by a company.
- **Scope 3:** all other indirect emissions that occur in the value chain of the reporting company, including both upstream and downstream emissions.

GHG emissions in the three scopes



Source: SG Cross Asset Research/Invest in the Carbon Transition

Scope 3 has a paradoxical profile: it represents the bulk of the three scopes' value but is currently the least consistent due to the following issues.

- **Scope 3 emissions are not fully under control.** Among the Scope 3 categories, GHG emissions are generated mainly by the supply chain and the use of sold products, neither of which are fully under a company's control. As a result, it is difficult for a company to build a reliable transition plan that includes Scope 3.
- **Double accounting of GHG emissions remains an issue.** For example, adding Scope 1, 2, and 3 upstream GHG emissions (i.e. from the supply chain) using the same methodology for car manufacturers and auto parts makers leads to double accounting of emissions.
- **No homogeneous historical data.** The chart below details Valeo Group's Scope 3 emissions between 2013 and 2022, which we analyzed in our September 2024 report as follows:

There are 15 categories of Scope 3 emissions. The CSRD encourages breaking down Scope 3 emissions into these 15 categories of the carbon score and specifically mentions the need to disclose data on all the 15 categories. (For additional information about CSRD, please refer to Invest Without Carbon - September 2024.)

List of Scope 3 categories

Table [5.3] List of scope 3 categories

Upstream or downstream	Scope 3 category
Upstream scope 3 emissions	1. Purchased goods and services 2. Capital goods 3. Fuel- and energy-related activities (not included in scope 1 or scope 2) 4. Upstream transportation and distribution 5. Waste generated in operations 6. Business travel 7. Employee commuting 8. Upstream leased assets
Downstream scope 3 emissions	9. Downstream transportation and distribution 10. Processing of sold products 11. Use of sold products 12. End-of-life treatment of sold products 13. Downstream leased assets 14. Franchises 15. Investments

Source: SG Cross Asset Research/Invest in the Carbon Transition

Unfortunately, there is currently little consistency among companies regarding the publication of these categories. Indeed, few companies publish data across all categories. Moreover, the reported categories are not the same for each company, and the publication rate per company changes each year. It is therefore impossible to compare Scope 3 emissions, either at the company level or sector level.

Ambition & execution ratings

We collect contributions from Bernstein and Autonomous sector analysts who have updated their ambition and execution ratings for this new edition of the 2026 Carbon Survey. In partnership with Bernstein's sector analysts, SG Macroeconomic Strategy Research has developed two forward-looking ambition and execution ratings for inclusion in our corporate carbon toolbox, which is part of our carbon product range. These two ratings are fully integrated into our corporate carbon toolbox and play a key role in our overall ranking methodology.

Ambition rating

Ambition reflects a company's commitment to decarbonization and its willingness to reduce its carbon footprint over time. Our ambition rating combines the quantitative metrics used to derive the reduction coefficient with the qualitative assessment of Bernstein and Autonomous sector analysts for the stocks they cover.

– **Reduction coefficient:** around 8,000 corporations now disclose their carbon emission targets, typically over a transition period of around ten years. To assess these targets effectively, we have collected specific information, such as the start date and the first interim target date, and the percentage reduction of GHG emissions in Scopes 1 & 2 over this specified period. We then calculate a reduction coefficient based on the annualized reduction rate weighted by GHG emissions in Scopes 1 & 2 at the baseline year.

– **Opinion of sector analysts:** sector analysts express their views by rating targets on a 1–5 scale, where 5 is the highest score, using a best-in-class approach within each sector. Their evaluation takes into account several elements: the reduction coefficient, which gauges the ambition to cut Scope 1 and 2 GHG emissions based on the annualized reduction rate weighted by baseline emissions; the quality of disclosure, including whether data are measured or modelled, the range of gases covered, the scopes included, the presence of interim targets and a 2050 net-zero objective; and the use of science-based targets and external validation, assessing alignment with frameworks such as the Science Based Targets Initiative (SBTi) or other recognized standards.

Execution rating

This rating evaluates the management board's commitment to achieving the company's emission reduction targets and the board's strategic execution

– **Performance rating:** this calculates the progress achieved by a company on its strategic plan for reducing GHG emissions in Scopes 1 & 2, from the base year to 2025. We compare the annualized GHG emissions in Scope 1 & 2 reduction from base year to 2025 with the annualized GHG reduction plan from base year to target year. The result is an outperformance or underperformance that we translate into a score.

– **Opinion of sector analysts:** analysts rate targets on a 1–5 scale, with 5 as the highest score, using a sector-based best-in-class approach, and assess several factors including the share of low-carbon capex within total capital expenditure, the proportion of green R&D within overall R&D spending, and the extent to which board incentives are aligned with the company's decarbonization strategy.

EXHIBIT 1: **Ambition ratings 1 to 5 (the best)**

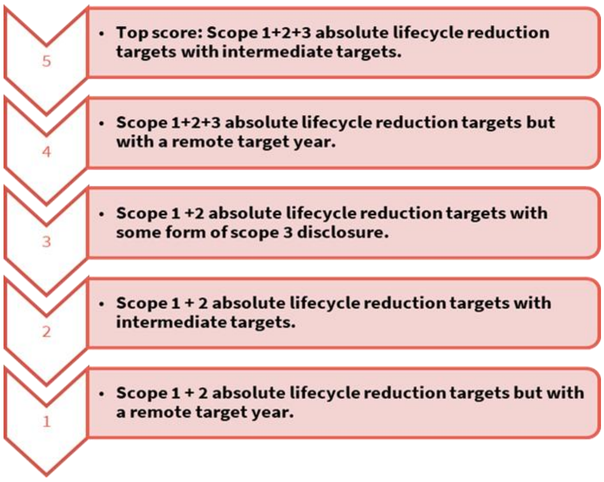
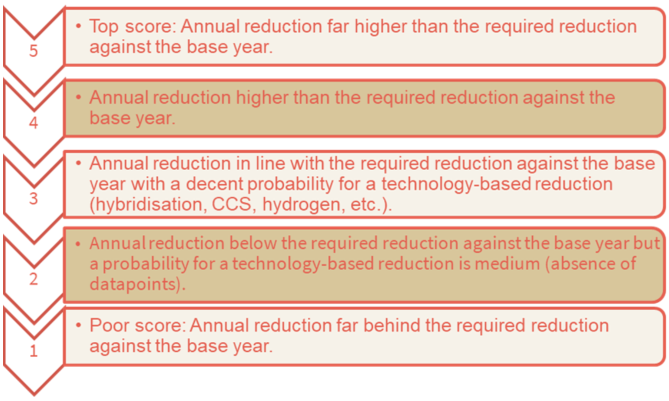


EXHIBIT 2: **Execution ratings 1 to 5 (the best)**



Source: SG Cross Asset Research/Invest in the Carbon Transition

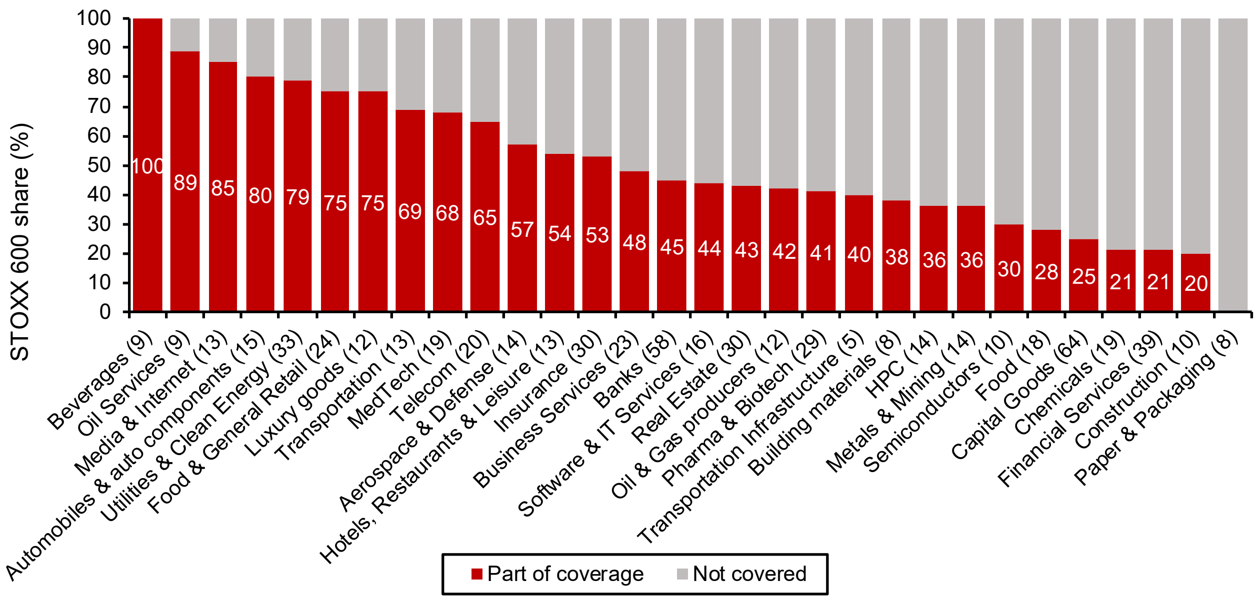
Company-level carbon signals: ambition & execution ratings

In this section, we analyze the net-zero transition strategies of companies across our Bernstein and Autonomous European coverage, as reflected in the carbon ambition and execution ratings assigned by our equity sector analysts.

The scope

The analysis rates 286 companies that are covered by our equity research analysts and overlap with the STOXX 600 Index (encompassing 48% of the index). In 13 sectors out of 30, we cover more than half of the companies in the respective sectors.

Overlap of companies in our carbon analysis vs Stoxx 600



Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research

Overview of ambition and execution ratings

Ambition: companies achieved an average ambition rating of **3.7 out of 5**. While **22%** earned the highest possible score of **5**, a further **35%** received a strong rating of **4**. Overall, **57%** of companies scored **4 or above**, indicating a high level of ambition in their decarbonization commitments. However, the remaining companies have an opportunity to strengthen the scope and rigor of their target-setting frameworks.

Execution: the average execution rating stands at **3.5 out of 5**. Approximately **19%** of companies achieved a perfect execution score of **5**, while **33%** scored **4 or higher**, suggesting a strong likelihood of delivering on their decarbonization objectives. That said, a high execution score does not necessarily correlate with high ambition. Some companies demonstrate strong capabilities to achieve their targets, but those targets may be relatively less ambitious. Therefore, it is important to consider ambition and execution ratings together to assess overall decarbonization leadership.

Only **7%** of companies achieved a perfect '5' score on both ambition and execution, demonstrating excellence in both target-setting and delivery. A larger group, **37%** of companies, scored **4 or above** across both dimensions, indicating that more than a third of all companies exhibit strong performances in establishing meaningful decarbonization goals and implementing credible measures to achieve them.

Despite these positive results, notable gaps remain. Around **20%** of companies received high ambition scores (**4 and above**) but lower execution scores (**3 and below**), suggesting that while they have set robust emissions-reduction targets, they face challenges in translating these commitments into tangible progress. Conversely, **15%** of companies scored highly on execution but relatively low on ambition, indicating a strong ability to deliver on their goals but less ambitious decarbonization pathways or a need for greater transparency around their climate strategies.

Share of companies on overlay of ambition and execution ratings (1: the worst, 5: the best)

		Execution ratings					
		1	**2**	**3**	**4**	**5**	*Total*
Ambition ratings	**1**	2%	0%	0%	0%	0%	*2%*
	2	1%	2%	3%	0%	1%	*8%*
	3	0%	6%	13%	10%	3%	*32%*
	4	0%	5%	9%	**14%**	**7%**	*35%*
	5	1%	1%	4%	**9%**	**7%**	*22%*
	Total	*4%*	*15%*	*29%*	*33%*	*19%*	

Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research

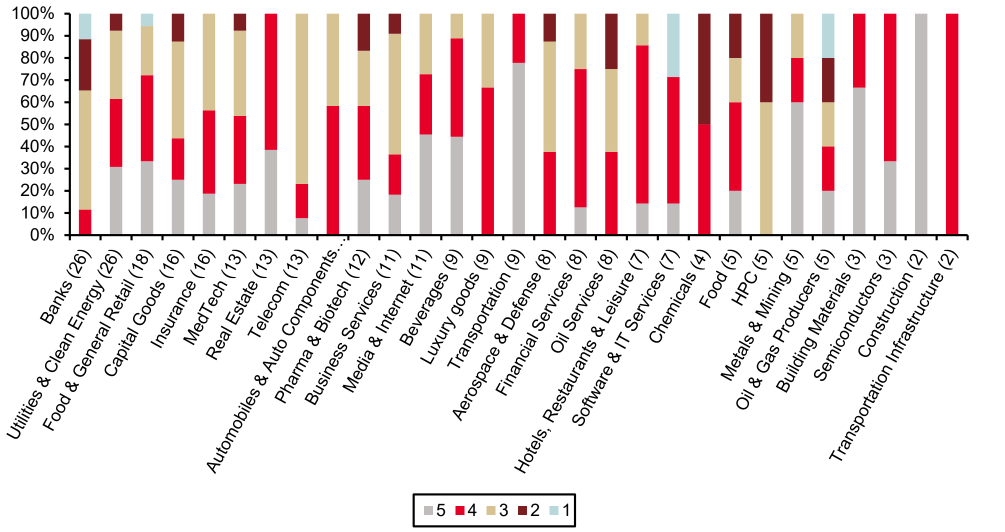
Ambition rating by sector

Most sectors exhibit a relatively balanced distribution of decarbonization ambition scores. To better understand the sectors that have the greatest influence on the overall results, we examined the **ten largest sectors by company coverage**, which together account for approximately **60%** of the total sample. Within this group, **Banks, Capital Goods, Food & General Retail, MedTech, Pharma & Biotech, and Utilities & Clean Energy** display a broad dispersion of companies across at least four ambition-rating categories, indicating a well-distributed range of scores within each sector.

Among the ten largest sectors, **Real Estate** stands out, with **100% of companies achieving an ambition rating of 4 or above,** followed by **Food & General Retail** (72%) and **Utilities & Clean Energy** (62%). In contrast, **Banks** and **Telecom** are more heavily concentrated in the mid-range decarbonization targets, with 54% and 77% of companies, respectively, receiving an ambition rating of 3.

Despite the generally positive distribution of scores, several major sectors had no companies that achieved the highest ambition rating. These include **Aerospace & Defense, Automobiles & Auto Components, Banks, Luxury Goods, and Oil Services**, suggesting further scope for strengthening decarbonization targets and enhancing alignment with leading climate-transition frameworks.

Distribution of ambition ratings by sector (1: the worst, 5: the best)



Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research. Footnote: Number in bracket () represents count of companies covered in each sector

Execution rating by sector

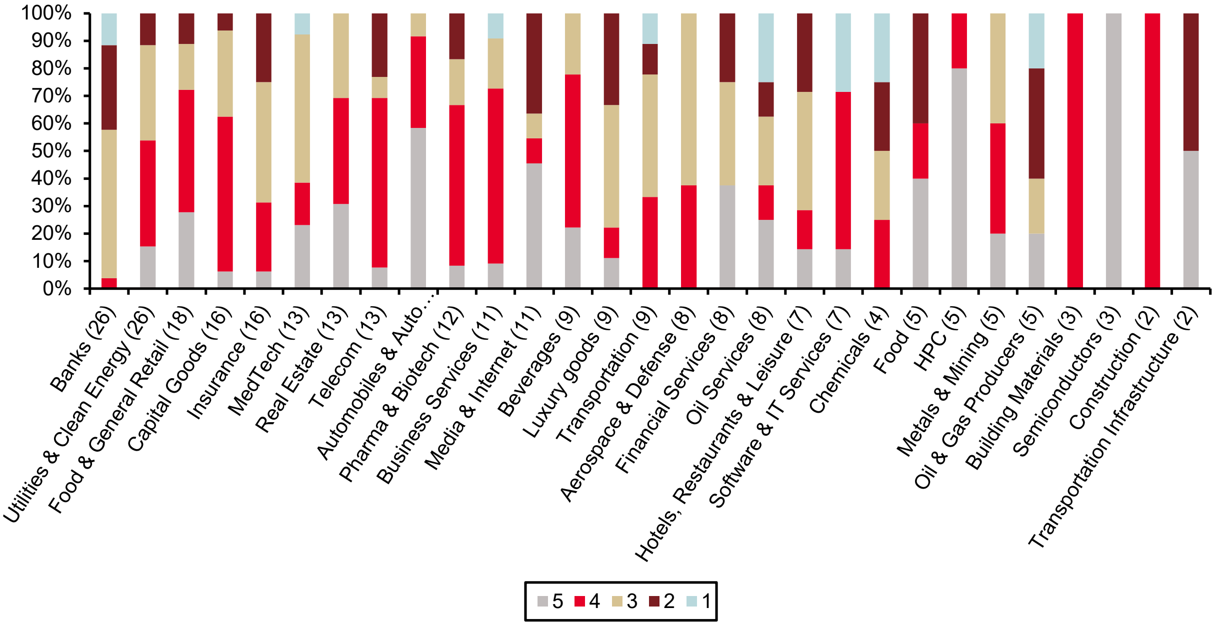
Compared with ambition ratings, a smaller proportion of companies achieved high execution scores, with **52%** receiving ratings of **4 or above**, versus **57%** for ambition. Across sectors, execution ratings are largely concentrated in the **3** and **4** categories, with **62%** of companies falling within these two rating bands.

Focusing on the ten largest sectors by coverage, which together account for approximately 60% of our universe, we find that Food & General Retail, Insurance, MedTech, Pharma & Biotech, Telecom, and Utilities & Clean Energy exhibit a broad distribution of companies across at least four execution-rating categories, reflecting a broad range of progress expected in implementing decarbonization strategies within these sectors. Within this group, Automobiles & Auto Components leads in terms of the proportion of companies scoring 4 or above (92%), followed by Food & General Retail (74%), Real Estate (69%), Telecom (69%), and Pharma & Biotech (67%).

Semiconductors stands out with **100%** of companies achieving the highest execution score of **5**, while **HPC** also demonstrates strong performance, with **80%** of companies receiving a top rating.

At the other end of the spectrum, several major sectors had no companies that attained a perfect execution score. These include **Aerospace & Defense, Banks, Chemicals, and Transportation**, highlighting potential challenges in translating climate commitments into measurable and demonstrable progress toward decarbonization goals.

Distribution of execution ratings by sector (1: the worst, 5: the best)



Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research. Footnote: Number in bracket () represents count of companies covered in each sector

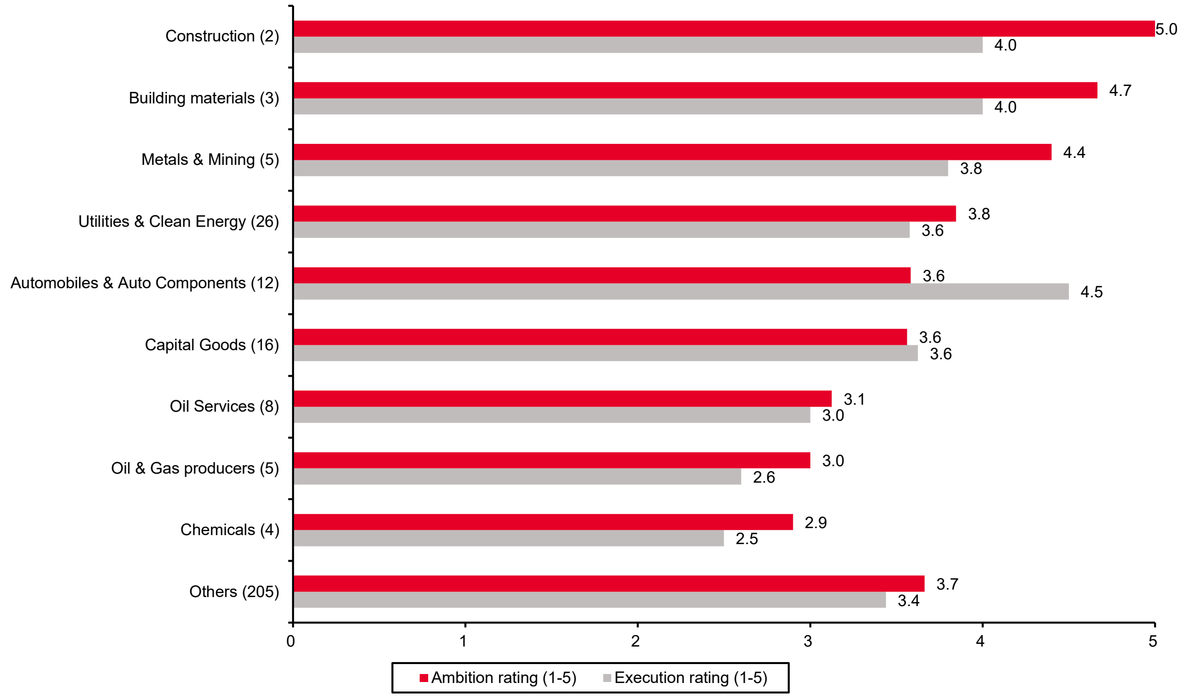
Ambition and execution ratings (carbon-intensive sectors)

The chart below illustrates how the most carbon-intensive sectors—typically those with higher energy consumption and a critical role in the low-carbon transition—perform on our ambition and execution ratings. Across nearly all these sectors, average ambition scores exceed execution scores, with the exception of **Automobiles & Auto Components** and **Capital Goods**.

Average ambition ratings among carbon-intensive sectors range from **2.9 to 5.0**. **Construction** leads with a perfect ambition score of **5.0**, followed by **Building Materials** (**4.7**) and **Metals & Mining** (**4.4**). These strong ratings reflect the increasing commitment of companies in these sectors to set more rigorous decarbonization targets in the near-to- medium term and to align them with recognized frameworks such as the Science Based Targets initiative (SBTi).

Average execution ratings in carbon-intensive sectors range from **2.5 to 4.5**. **Automobiles & Auto Components** emerges as the top performer with an execution score of **4.5**, followed by **Building Materials** (**4.0**) and **Construction** (**4.0**). These sectors demonstrate relatively strong implementation of decarbonization strategies, supported by a combination of initiatives such as low-carbon investments, management-linked climate incentives, renewable energy adoption, and measurable progress toward emissions-reduction targets.

Ambition and execution ratings (1: the worst, 5: the best): Carbon-intensive sectors



Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research. Footnote: 'Others' denote remaining sectors altogether. Number in bracket () represents count of companies in each sector

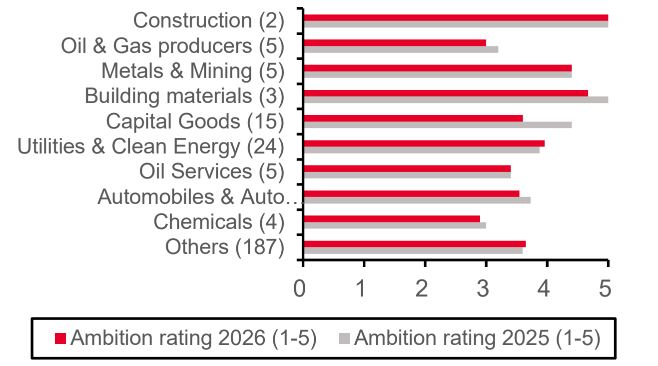
Ambition and execution ratings (vs previous year): carbon-intensive sectors

Compared with our 2025 Sustainability Research report, *Europe's Carbon Moment: A System Under Pressure*, we observe only modest changes in the ambition and execution ratings. Of the **286 companies** included in this year's analysis, **261** overlap with the prior-year universe. For these companies, average ambition ratings remained broadly unchanged, while **execution ratings improved** marginally, by **2%** overall.

However, the picture is more mixed across carbon-intensive sectors, where average **ambition ratings declined** by **5%** in this update. Several sectors experienced more pronounced shifts. **Capital Goods** recorded one of the largest declines in ambition (**-18%**), although this was partly offset by a **6%** improvement in execution. **Building Materials** saw declines in both ambition (**-7%**) and execution (**-8%**), while **Oil & Gas Producers** also experienced reductions in ambition (**-6%**) alongside weaker execution (**-7%**). Notable changes in execution include **Chemicals** (**-33%), Metals & Mining** (**-10%**).

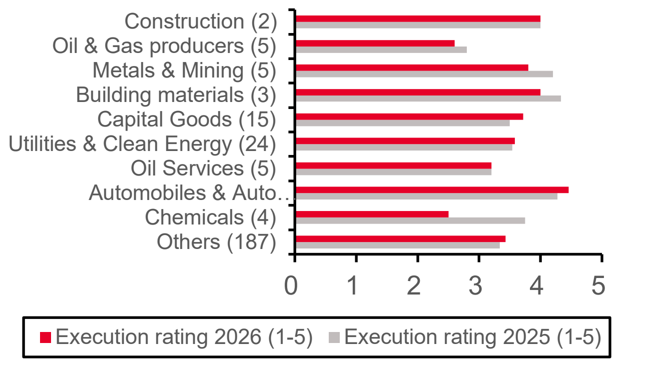
The chart below focuses exclusively on carbon-intensive sectors and includes only those companies that were covered in both this year's analysis and last year's publication, allowing for a like-for-like comparison of changes in ambition and execution over time.

Ambition rating (2026 vs 2025) (1: the worst, 5: the best)



Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research. Footnote: The exhibit reflects companies included in both 2026 and 2025 analyses. 'Others' denote remaining sectors altogether. Number in brackets () represents count of companies covered in each sector

Execution rating (2026 vs 2025) (1: the worst, 5: the best)



Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research. Footnote: The exhibit reflects companies included in both 2026 and 2025 analyses. 'Others' denote remaining sectors altogether. Number in brackets () represents count of companies covered in each sector

Best positioned companies by sector on ambition and execution rating

Based on the quantitative assessment by our equity analysts, every sector has at least one top performer on a combination of carbon ambition and execution rating, with some sectors featuring multiple companies sharing the leading position. Out of the 46 companies highlighted in the exhibit below, **21 companies achieved a perfect score of '5' for both ambition and execution ratings**. Notably, several sectors had no companies that attained the highest ambition score, reflecting gaps in the strength of the decarbonization targets. Similarly, in some sectors, no company received a perfect execution score, indicating challenges in demonstrating a high likelihood of achieving their stated targets.

Best-positioned companies by sector in ambition and execution ratings (1: the worst, 5: the best)

Company name	Sector	Ambition rating (1-5)	Execution rating (1-5)
Rolls-Royce	Aerospace & Defense	4	4
Thales	Aerospace & Defense	4	4
Mercedes-Benz	Automobiles & Auto Components	4	5
Michelin	Automobiles & Auto Components	4	5
Pirelli & C	Automobiles & Auto Components	4	5
Renault	Automobiles & Auto Components	4	5
Volvo	Automobiles & Auto Components	4	5
DNB Bank	Banks	4	3
KBC Group	Banks	4	3
NatWest	Banks	4	3
Heineken	Beverages	5	5
Pernod Ricard	Beverages	5	5
Rockwool	Building Materials	5	4
Experian	Business Services	5	5
ABB	Capital Goods	5	5
Arkema	Chemicals	4	4
Eiffage	Construction	5	4
Avanza Bank	Financial Services	4	5
Nestle	Food	5	5
Axfood	Food & General Retail	5	5
Inditex	Food & General Retail	5	5
Tesco	Food & General Retail	5	5
Whitbread	Hotels, Restaurants & Leisure	5	5
Haleon	HPC	3	5
L'Oreal	HPC	3	5
AXA	Insurance	5	5
LVMH	Luxury Goods	4	5
Autotrader	Media & Internet	5	5
ITV	Media & Internet	5	5
Wolters Kluwer	Media & Internet	5	5
Koninklijke Philips	MedTech	5	5
Glencore	Metals & Mining	5	5
Equinor	Oil & Gas Producers	4	2
Gaztransport et Technigaz	Oil Services	4	5
SBM Offshore	Oil Services	4	5
AstraZeneca	Pharma & Biotech	5	5
Land Securities	Real Estate	5	5
ASML	Semiconductors	5	5
Sage Group	Software & IT Services	5	4
Tele2	Telecom	5	4
DHL Group	Transportation	5	4
Kuehne + Nagel	Transportation	5	4
Getlink	Transportation Infrastructure	4	5
EDP	Utilities & Clean Energy	5	5
Enel	Utilities & Clean Energy	5	5
RWE	Utilities & Clean Energy	5	5

Source: SG Cross Asset Research/Sustainability, Bernstein analysis, Autonomous Research

In the next section 'Sector summary of carbon ratings' each sector analyst discusses in detail the rationale behind their ambition and execution scoring approach and the progress made by companies in setting decarbonization goals in their respective sectors.

Sector analyst summary on carbon ratings

Our sector analysts provide a detailed explanation of their scoring approach, including how they constructed their ratings—such as the criteria and assumptions they used—for both ambition and execution ratings. They also assess the progress companies have made in setting and implementing their decarbonization goals, as well as the execution of these goals.

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Aerospace & Defense

Quality and quantity of carbon data disclosure

The key issue in Aerospace and Defense is emissions generated by air travel. Commercial aerospace companies manufacture or supply aircraft. Carbon emissions from air travel are Scope 3 emissions, specifically product use. Emissions are difficult to measure, as there is varied usage of aircraft by a diverse customer base. On the other hand, defense companies generally do not disclose exact Scope 3 emissions, as those are tied to product usage by their government customers, and such data is usually restricted. There tends to be more clarity on Scope 3 emissions around other aspects of the value chain, such as employee business travel or purchased goods and services. But generally, disclosure of Scope 3 emissions is not consistent across our coverage. Aerospace and Defense companies all disclose metrics on Scope 1 and Scope 2, measuring performance in their annual reports in absolute levels of metric tons of CO2 equivalent. The level of disclosures and targets do differ. Some companies such as Airbus, Safran and Thales clearly and consistently report current-year progress against base-year emissions. Others only report emissions for the current and prior year, so comparisons to the baseline require tracking the results over time and are subject to reporting changes. Some companies have targets with caveats, such as Scope 1 and 2 emissions excluding product testing. The introduction of different disclosures and many different sub targets makes it difficult to understand total emissions.

Carbon targets, rating corporate ambition

European Aerospace and Defense companies all have targets for Scope 1 and 2 emissions reductions that are aligned with the Paris Agreement. They range from 50% to 100% reduction targets vs the baseline, with target years ranging from 2030 to 2050. But in the bigger picture, this is not significant in the evaluation of emissions, as Scope 1 and 2 emissions account for much less than Scope 3. Airbus has estimated that its Scope 1 and 2 emissions are less than 1% of total emissions. The rest are all Scope 3 emissions, with product use (use of commercial aircraft by customers) accounting for over 90% of total emissions. For Safran, Scope 3 accounted for 99.6% of the group's 2025 emissions. For Defense, this proportion tends to be lower given lower utilization but is still significant. According to BAE Systems, about 65% of the industry's emissions come from customer use of products. For Scope 3 emissions, some companies have targets to a certain extent. Thales only includes business travel. Airbus and Safran's reduction targets relate not to absolute Scope 3 emissions but to emissions relative to air traffic. This highlights that carbon emissions from aircraft usage is an issue that needs to be assessed at the industry level. The companies' trajectories are all inherently related and linked, as all large aircraft are manufactured by Airbus and Boeing, and they often share many common suppliers. The industry's targets on carbon emissions help set the direction for major manufacturers and suppliers. At the International Air Transport Association (IATA) Annual General Meeting in 2021, a resolution was passed by member airlines to commit to achieving net-zero carbon emissions from their operations by 2050, a pledge consistent with supporting the Paris Agreement's temperature goal. The net-zero commitment requires a combination of efforts, including 65% from sustainable aviation fuel (SAF), 13% from new technology, 3% from infrastructure and operational efficiencies, and 19% from offsets and carbon capture. For aerospace companies, "new technology" is the addressable component. This is done through the design of newer and more efficient aircraft, and this forms the basis of Scope 3 emission-reduction targets for commercial aerospace companies. The problem with the net-zero ambition is that although only 2-3% of global carbon emissions come from commercial aircraft, aviation is inherently difficult to decarbonize, and given continued growth in the industry, its share of global carbon emissions is likely to increase. In addition, a common misunderstanding is that the impact is often considered greater than that of other sources because aviation emissions are released into the upper atmosphere.

Rating corporate carbon reduction execution plans

Most Aerospace and Defense companies are on track or even ahead of their Scope 1 and 2 emission reduction targets. But because they constitute a very small portion of overall emissions, the real impact is negligible. The true challenge is Scope 3 reductions, which need to be addressed at an industry level. At this point in time, by all measures, the industry is still struggling to make real progress. One near-term opportunity for Scope 3 reductions is replacing legacy airplanes with newer-generation aircraft. The reduction in fuel burn and related carbon emissions for current-generation aircraft amounts to roughly 15-25%. However, Airbus and Boeing are both behind on their delivery targets, held back by supply chain and quality challenges. The result is that airlines have had to fly their older, less efficient aircraft for longer than expected. Developing next-generation aircraft can help reduce emissions, but the typical time between aircraft generations is around 20 years. In the 2010s, we saw new narrowbodies (A320neo and 737 MAX) and widebodies (A350, A330neo, 787). Presently, there are no new clean-sheet aircraft expected from Airbus or Boeing until at least the end of the 2030s. The aircraft and engine manufacturers all continue to spend on research & development to study and develop new aircraft, but such projects are all in the preliminary stages. Beyond delivering more new aircraft and working on next-gen aircraft, decarbonizing aviation largely falls outside the responsibility of aerospace manufacturers. Sustainable aviation fuel (SAF) is seen as the main factor for the aviation industry to lower emissions over the next few decades. SAF is synthesized from sustainable feedstock. For a fuel to qualify as SAF, it must be less carbon-intensive over its lifecycle than a fossil-based fuel and fulfill internationally recognized sustainability criteria. The biggest challenge remains ramping up production to meet growing demand at a time when production costs are prohibitively high. IATA estimates global SAF production at 2.4 billion liters in 2025, which represents only 0.6% of aviation's fuel needs. For reference, in 2019, the aviation industry burned 363 billion liters of jet fuel. It has been estimated that given the expected growth in air travel and the industry's commitment to achieve net-zero CO_2 emissions by 2050, the industry would require an SAF production capacity of 450 billion liters annually in 2050. The major challenge in SAF production is to produce these fuels in a large enough quantity to make them commercially viable while simultaneously ensuring they are produced from feedstock that can be grown or produced without negatively impacting food supplies or water and land use. In an unprecedented and unwelcome development for the industry, in 2024, Air New Zealand withdrew from the Science Based Targets initiative (SBTi) and abandoned its 2030 carbon emission-reduction goal. It cited lack of availability of new, more efficient aircraft as well as of affordable and accessible SAF.

Automobile

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OEM ESG overview:

The auto industry is currently in the midst of an enormous shift in ESG, most notably of course through the electrification of vehicles with the aim of meeting the EU's 2050 net-zero target, in line with the Paris Climate Agreement. Some of our coverage believe that they can achieve this well ahead of schedule. All our coverage has thus far maintained a high level of reliability when it comes to delivering on their goals. Making electricity sources at their factories greener and transitioning from ICE to BEV (battery-electric-vehicle) transmissions will be the key drivers in improving our coverage's ESG standards. This will prove to be tricky because while some companies have achieved margin parity across powertrains, on the whole, BEV products currently offer less attractive margins, hindering the rate at which companies can electrify their product portfolios. EV demand is also not progressing as quickly as once thought; the EU has shown it is willing to be flexible with emissions targets, and the new US administration has removed penalties on automakers for not meeting CAFE standards, while also eliminating consumer incentives to opt for more eco-friendly options. Falling prices and improved infrastructure can act as catalysts to accelerate this process. Elevated oil prices as a result of the US-Iran conflict and the subsequent blockading of the Straits of Hormuz have increased interest in BEVs since the beginning of March, although the impact of high government subsidies in certain European countries cannot be ignored either.

The companies in our coverage that we believe are taking the most meaningful steps to reduce carbon emissions are BMW (BMW.GR) Mercedes-Benz Group (MBG.GR), VW Group (VOW3.GR) and Renault (RNO.FP. All plan to meet net carbon neutrality well ahead of the 2050 deadline put forward by the EU.

BMW has significantly reduced operational emissions, with Scope 1 and 2 at 836,963 tCO2e in 2025, production plants net carbon neutral since 2021, and emissions intensity per vehicle down 97.8% vs 2006. Automotive Scope 3 emissions totaled 127.5 million tCO2e, with a 5.5% yoy reduction largely driven by higher BEV penetration and lower use-phase emissions from electric vehicles. Electrification at BMW is scaling, with 442,072 BEVs (17.9% of sales) and 642,071 electrified vehicles in total (26% of deliveries), reflecting BMW's technology-neutral decarbonization path. Around 99% renewable electricity usage across global production, combined with circular-economy measures such as roughly 30% secondary material content and new battery recycling capabilities, underpins low operational emissions. Strategic initiatives including NEUE KLASSE, over €1bn investment in Steyr for sixth-generation e-drives, and direct sourcing of certified lithium and cobalt reinforce sustainability, digitalization, and ethical supply chains. Science-based climate targets include deep Scope 1 and 2 cuts by 2030, supply chain and use-phase reductions, more than 50% BEV sales mix, at least 60 million tCO2e lifecycle reduction by 2035, and full value-chain climate neutrality by 2050.

Mercedes-Benz targets a net carbon-neutral new vehicle fleet across the full value chain by 2039 ("Ambition 2039"), embedding sustainability into its long-term strategy. The company follows a technology-neutral decarbonization approach, combining battery electric vehicles, plug-in hybrids, and efficient combustion engines to balance flexibility and regulatory compliance. Electrification is scaling steadily: electrified vehicles reached 20.5% of sales in 2025 (368,700 units), supported by a global charging network of about 2.9 million points across 37+ countries. Significant investments underpin the transition, including €8.6bn in R&D for cars, €1.1bn for vans, and €4.1bn in capex, focused on EV platforms, batteries, and software-defined vehicles (MB.OS).

Operations are already decarbonized, with all production plants carbon-neutral since 2022 using 100% renewable electricity, and lifecycle CO2 per car reduced by 28% vs. 2018. Circular economy and supply chain initiatives are expanding, including >96% battery material recovery, increased use of recycled and low-CO2 materials, and 90% of suppliers committed to future CO2-neutral production.

Volkswagen Group targets net carbon neutrality by 2050, aligned with the Paris Climate Agreement. By 2030, it aims to reduce the carbon footprint per customer-km (use phase of vehicles) by 30% versus 2018. The Group plans for global production sites to reach net carbon neutrality by 2040, a decade earlier than previously targeted. This includes a 90% reduction in production-related greenhouse gas emissions by 2040 (vs. 2018), supported by energy efficiency improvements and cleaner energy sourcing.

Volkswagen aims to source 100% of external electricity from carbon-neutral sources by 2030, with

European sites already fully powered by renewable electricity. Under its “Zero Impact Factory” vision, it targets reducing total environmental impacts of production by 37.5% by 2030 and 68.8% by 2040 (vs. 2018), progressing toward full neutrality by 2050.

Renault Group reduced its total carbon footprint by 38% between 2019 and 2025, indicating significant early decarbonization progress. The Group targets Net Zero in Europe by 2040 and globally by 2050, with a 90% reduction across Scopes 1, 2, and 3 versus 2019, complemented by carbon offsetting. Its climate strategy spans the full vehicle lifecycle (design to end-of-life), embedding decarbonization across all business activities since the 2021 Climate Plan. The “FutuREady” roadmap outlined earlier this year includes 16 new EV models, a further 25% cut in energy per vehicle (vs. 2025), 30% circular material usage, and 10–40% battery energy density improvements. 2030 targets include at least a 62% reduction in manufacturing emissions (Scopes 1 and 2) and at least a 27% reduction in value chain emissions (Scope 3), both vs. 2019. Key levers include accelerating electrification, expanding circular economy practices, reducing material and battery footprints, and improving factory energy efficiency.

On the flip side, although they still intend to operate in line with all regulations, over the last year, Stellantis (STLA) and Ferrari (RACE) have, in some respects, made decarbonization *less* of a priority. Stellantis currently plans to reduce absolute GHG emissions (Scopes 1, 2 and 3) by 20-30% by 2030 vs their 2021 baseline and achieve carbon net zero by 2050. These targets are noticeably less ambitious than last year’s when they planned to cut 2030 emissions by 30% and achieve carbon net zero by 2038. Of course, this will have, no doubt, been influenced by emissions penalties in the USA being abolished by the “One, Big, Beautiful Bill Act” in July 2025 which resulted in the elimination of financial penalties for CO2 emissions. Stellantis is actively promoting its higher-emitting HEMI V8 combustion engines in its trucks, SUVs and larger cars, which certainly increases average fleet emissions in North America. Averaged over the last three years, North America provides Stellantis with over 40% of its revenue, with the US comfortably being its most important market.

Ferrari is uniquely positioned when it comes to ESG. This is mainly down to how luxurious its products are, meaning that very few are disposed of (i.e. vehicle life is extremely long). Regardless, from an emissions standpoint, it has made some good progress over the last few years and has given itself reasonably ambitious targets to fulfil on Scopes 1 and 2. It has reduced Scope 1 and 2 emissions by 57% vs 2021, with the aim of reaching a 90% cut by 2030. However, on Scope 3 emissions, which comprise over 90% of its total GHG emissions, it hasn’t given itself a target for 2030, and yoy has only made a 4% reduction in 2025. Furthermore, at its most recent CMD in October 2025, it revised its 2030 vehicle line-up powertrain mix forecast from 20% ICE, 40% hybrid, 40% BEV, to 40% ICE, 40% hybrid, 20% electric. Prioritizing electrification is not on the top of Ferrari’s agenda given it is classified as a Small Vehicle Manufacturer (SVM) in many markets (e.g. EU), meaning it is not subject to the same CO2 regulations as larger automakers. And, like Stellantis, the US is its most important market, and at least for now, it no longer has to concern itself with emissions penalties there either.

Porsche AG has a 2030 target of dropping its Scope 1 and Scope 2 GHG emissions by 76% from 2016, and Scope 3 emissions by at least 42% vs 2023. In 2025, Scope 1 and Scope 2 emissions had already been reduced 70% vs 2016, and Scope 3 emissions 36% below that of 2023.

Porsche is pursuing lower-emission mobility through a mixed powertrain strategy, combining combustion engines, plug-in hybrids, and fully electric vehicles. The company has adjusted its electrification strategy due to market realities, delaying some EV launches and extending combustion and hybrid offerings. Electrified vehicles (BEV + PHEV) represented 34.4% of deliveries in the reporting year, with future growth expected but at a slower pace than previously planned.

Porsche aims to reduce greenhouse gas emissions across the full vehicle lifecycle (supply chain, production, and use phase), aligned with broader decarbonization goals. Scope 3 emissions targets may be revised due to volatile market conditions, evolving regulations, and external dependencies affecting decarbonization pathways. Progress is tracked using the Decarbonization Index (DCI), which measures lifecycle emissions per vehicle across the value chain in CO_2 equivalents.

Valeo's CAP 50 plan, launched in 2021, aims for Net Zero by 2050 with two core 2030 targets focused on operational and value-chain decarbonization. By 2030, Valeo targets a 75% reduction in CO_2 emissions across its own operations (Scopes 1 and 2), versus the 2019 baseline. To achieve this, the company plans around €400m of site energy-efficiency investments from 2020–2030, including solar and wind deployment and LED lighting, while phasing out fossil fuels from operations.

Valeo also targets a 15% reduction in Scope 3 emissions (upstream and downstream) by 2030 by helping suppliers transition to low-carbon solutions and shifting to innovative, lighter and recycled materials.

Its technologies are designed to accelerate EV adoption and lower use-phase emissions, with electrification solutions a key lever for reducing product lifecycle carbon footprint. In parallel, Valeo expects its technologies to avoid 13.6 $MtCO_2e$ of third-party greenhouse gas emissions by 2030, amplifying its impact beyond its own footprint.

Truck OEMs. Firstly, it is important to note that Daimler and Volvo are not exactly comparable, given the different product mix (Daimler only manufactures trucks/buses, while Volvo also makes construction equipment and engines). On ambition, we find Volvo leads Daimler. Volvo includes Scope 3 in its carbon reduction targets, while Daimler does not. Volvo also has a 2040 net zero target verified by the SBTi, while Daimler's aims to just be carbon neutral by 2050 and has not been verified by the SBTi. Daimler does have a slightly more ambitious annualized emissions reduction rate than Volvo when isolating only Scope 1 and 2, and we also note that Daimler incorporates a (lower) market-based estimate for Scope 2 emissions than Bloomberg uses (location-based). On execution: Volvo is ahead of schedule having achieved a 4.2% annualized GHG 1+2 reduction vs its 2.7% target. Daimler has a 2030 GHG Scope 1+2 target which required an annualized 4.7% reduction in emissions between 2021-30: and on its own definition has achieved an 11% annualized reduction since 2021 but we note that, on a consistent definition to Volvo, it is only a 2% annual fall. On emissions by unit (the two companies are not exactly comparable given different product mix), Daimler had a lower absolute Scope 1+2 per unit in 2025 than Volvo. To motivate execution, we note that both companies include number/proportion of electric vehicles sold in management compensation. We also praise Daimler for including CO2 in long-term management incentive components, which Volvo does not. In terms of sustainability of product, diesel trucks clearly have a very harmful environmental impact. Fleets are electrifying, but at a much slower pace than passenger cars as Total Cost of Ownership is higher for a ZEV than a diesel truck today. Comparing Volvo and Daimler, Volvo is also ahead of Daimler: 2.2% of Volvo's deliveries in 2025 were electric, compared to only 1.6% of Daimler's deliveries. While there is some scope for the companies to drive increased ZEV penetration themselves, a lot is determined by external factors outside of their control (battery prices, charging infrastructure, government incentives etc.). Over the coming years, we expect a significant proportion of both companies' capex and R&D to be 'green' as they develop electric powertrains at scale.

Tyremakers. On ambition, Michelin and Pirelli have slightly different carbon reduction targets. Michelin is targeting a 4.3% annual reduction 2019-2030 in Scope 1+2 and 2.6% in Scope 3, while Pirelli is targeting a 6.7% annual reduction 2018-2030 in Scope 1+2 and 2.5% in Scope 3. Both have also been approved by SBTI. Pirelli has set a net zero by 2040 target, while Michelin has only set a net zero target by 2050. On execution, both are ahead of schedule on an annualized basis: Michelin has already achieved its Scope 1+2 2030 target in 2025 and has achieved a 3.8% annualized reduction in Scope 3. Pirelli achieved an 8.6% annualized GHG 1+2 reduction 2019-2025 and 3.9% in GHG 3. To motivate execution, Michelin has included carbon reduction in executive short-term compensation (but not long-term), while Pirelli has included "ESG factors" in both short- and long-term compensation. We expect both to continue dedicating R&D resource to 'green' initiatives, as there is evidence that sustainability criteria are increasingly driving OE fitments (as we discussed in this report). Tyres have a significant impact on the environment, from deforestation as a result of natural rubber use, Tyre Road Wear Particles (TRWPs, fragments created by the friction between the tyre and the road) and end-of-life tyre disposal. Our research shows that optimizing some of these factors can give tyre makers a real competitive advantage, as making tyres more sustainable can improve performance and make the products more attractive to OEMs as they respond to customer demands and look to reduce their own environmental footprint. There are many examples of OEMs choosing a tyre supplier based on their sustainability credentials. For example, Citroën chose Goodyear to supply tyres for its ë-C3, demonstrating a "shared commitment to foster sustainability in the automotive industry". Key sustainability criteria that OEMs look for include sustainable material usage and TRWP emissions, as well as low rolling resistance, which can increase range (EVs) or reduce overall vehicle emissions (ICE).

Beverages

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GHG emissions matter. Beverage alcohol companies continuously seek a 'License to Trade', i.e., the permission of society to continue to manufacture and sell alcohol. However, there is also an opportunity. By reducing energy consumption and its associated GHG emissions, companies can reduce their operating costs, improve business resilience, and boost profitability. Despite these benefits, we have observed a significant reduction in the granularity of our companies' sustainability reporting in the past reporting year.

All brewers follow either the TCFD or the GRI framework of reporting. While Scope 3 accounts for 90% of emissions, all brewers have set a "2040 net zero value chain emissions" target aligned with the Science Based Targets (SBTi). ABI had a mid-term target for 25% reduction in Scope 1, 2 & 3 emissions by 2025, which it exceeded by delivering a 31.9% reduction. ABI provides a breakdown of emissions by region but has notably reduced reporting over the years. Heineken has a net zero Scope 1 & 2 emissions target for 2030, and a mid-term target of reducing Scope 3 emissions by 26% by 2030. Heineken is on track to deliver on the Scope 1 & 2 emissions target, and ahead of timeline to achieve its Scope 3 target. Carlsberg also has a 2030 target for net zero Scope 1 & 2 emissions with a baseline of 2015. As of 2025, Carlsberg was slightly behind on the timeline with a 63% reduction. Against the mid-term value chain emission target of 30% reduction by 2030, they are significantly lagging the timeline.

While performance against targets matters, we think reducing carbon intensity should be the priority for all brewers. In 2025, Carlsberg was the least intensive among brewers in our coverage, with Scope 1 & 2 emission intensity of 2.3 kgCO2e/hl, followed by Heineken at estimated 4.4 kgCO2e/hl, and ABI the last at estimated 5.4 kgCO2e/hl. Overall, looking at the reporting structure, targets, achievement so far, and carbon intensity, we rank Heineken as the best in our coverage on ESG reporting and performance, followed closely by Carlsberg, whereas ABI has slipped to the bottom of the list because of reduced granularity in its disclosures.

Moving to the distillers - The impact of climate change affects spirits through: i) unfavorable crop yields, ii) energy supply and price volatility, and iii) reputational damage to the company for not complying with regulations. Diageo, Pernod Ricard and Campari aim for net zero emissions by 2050 in alignment with SBTi. All three distillers also follow either TCFD or GRI reporting framework.

Diageo also has two mid-term targets – 50% reduction in direct emissions (Scope 1 & 2) by 2030 with net-zero direct emissions by 2040, and a 26% reduction in Scope 3 emissions by 2030. We note that both have been lowered recently. Up to FY25 (FYE June), Diageo performed positively on the Scope 1 & 2 target but is lagging on the timeline. The company saw an increase in Scope 3 emissions during the Covid years but has since reduced emissions, albeit still lagging the timeline. Pernod Ricard has an interim target of reducing Scope 1 & 2 FLAG emissions by 54%, and Scope 1 & 3 emissions by 30.3% for FLAG and 25% for non-FLAG by 2030. The company has been consistently delivering on these targets and has improved its performance to now be on track in FY25 (FYE June). A big plus for Pernod Ricard is sustainability-linked bonds, whose use backs the firm's message of commitment to emissions reduction. However, the complexity of reporting has increased in recent years. Campari has a mid-term target of a 70% reduction in Scope 1 & 2 emissions by 2030, along with a 30% reduction target for total value chain emissions (Scope 1, 2 & 3) both against a baseline of 2019. Until 2025, Campari steadily reduced its Scope 1 & 2 emission intensity ahead of the timeline.

Looking at the reporting structure, targets and their achievement so far, we rank Pernod Ricard as the best distiller in our coverage for the detailed reporting and the sustainability-linked bonds, despite recent complexity, followed by Diageo for consistent reporting and net zero direct emissions target of 2040. We would rank Campari a very close third given the competitive 2030 target, but a lower baseline and relatively less competitive Scope 1 & 2 net zero emissions target.

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Business Services

Quality and quantity of carbon data disclosure; rating the corporate ambition and corporate carbon reduction execution plans

Testing, Inspection, and Certification (Bureau Veritas, SGS, Intertek)

Within the Testing, Inspection, and Certification sector, Bureau Veritas and SGS provide the most granular reporting on Scope 1, 2, and 3 carbon emissions. Their disclosures include detailed breakdowns by emission sources, such as Purchased Goods & Services, Business Travel, and Leased Upstream Assets. Bureau Veritas integrates non-financial data into its annual report, whereas SGS publishes separate sustainability reports. Both companies have set targets to reduce Scope 1 and 2 emissions by approximately 50% over the next decade. Emissions reduction progress has been slow. Bureau Veritas's Scope 1 and 2 emissions have decreased by 3% since 2019, while SGS's emissions have decreased by 8%. Intertek provides less detailed Scope 3 emissions data and has achieved a significant 55% annual reduction in Scope 1 and 2 emissions. None of the three companies currently integrates carbon emissions or ESG factors into their financial reporting frameworks, limiting the financial materiality of these disclosures.

Equipment Rental (Sunbelt)

Sunbelt's sustainability reporting, initiated in 2022, covers all three emission scopes with data segmented by source and geography. The company also reports fleet composition by fuel type, providing insight into its operational footprint. Sunbelt communicates carbon data transparently and has implemented measures such as increased use of electric and hybrid equipment, renewable energy procurement for buildings, and adoption of lower-carbon transport options. Despite these initiatives, Sunbelt's Scope 1 and 2 emissions have increased by 9% since 2023, reflecting operational challenges related to the industrial nature of its equipment. Similar to peers, Sunbelt has not incorporated carbon emissions or ESG factors into its financial reporting.

Distribution (Bunzl)

Bunzl provides comprehensive reporting on Scope 1 and 2 emissions and partial Scope 3 disclosures, reflecting the complexity of its product offerings. The company offers revenue breakdowns by consumables subject to regulation, enabling assessment of exposure to sustainable packaging. Bunzl is undertaking measures to reduce CO2 emissions, including material substitution and investment in hybrid and electric commercial vehicles. Its carbon disclosures are supported by supplementary ESG documentation such as TCFD and SASB indices. Emission reductions have been modest, with an 18% decrease since 2019. The company acknowledges that commercial vehicle electrification is behind schedule, and its 2030 target of a 27.5% reduction will require increased efforts in the coming years.

Credit Agencies (Experian)

Experian's carbon emissions primarily stem from business travel, data centers, and office operations. The company integrates carbon data into its annual report and ESG presentations, reflecting its relatively low environmental impact. Since 2019, Experian has achieved a significant 90% reduction in Scope 1 and 2 emissions. Carbon emissions and ESG factors are not currently integrated into Experian's financial reporting, which limits the financial relevance of these disclosures.

Staffing (Adecco and Randstad)

The staffing sector's carbon emissions mainly arise from business travel and office facilities. Adecco and Randstad provide similar levels of granularity and transparency in reporting Scope 1, 2, and 3 emissions. Both companies have made significant progress in emissions reduction, with Adecco achieving a 28% reduction in Scope 1 and 2 emissions since 2019. Randstad has achieved reductions of 59%. These companies are on track to meet their 2030 emissions targets. As with other sectors, carbon emissions and ESG factors have not been integrated into financial reporting standards.

Pest Control (Rentokil)

Rentokil has reported carbon emissions annually since 2006, with detailed Scope 3 emissions breakdowns covering business travel, transport, distribution, and fumigation. The company tracks environmental efficiency metrics related to vehicle eco-efficiency and property energy use. Rentokil has set a target to reduce Scope 1 and 2 emissions by 100% by 2040. Despite this ambition, Scope 1 and 2 emissions have increased by 61% since 2021, partly due to the acquisition of Terminix. Carbon emissions are not currently financially integrated.

Outsourcing (Teleperformance)

Teleperformance provides granular reporting on Scope 1, 2, and 3 emissions, with disclosures integrated into its annual report and standalone sustainability reports. The company has achieved a 57% reduction in Scope 1 and 2 emissions since 2019. Carbon emissions and ESG factors have not been incorporated into Teleperformance's financial reporting framework.

On a best-in-class basis, SGS and Bureau Veritas clearly lead on ambition, underpinned by best-in-sector Scope 1–3 granularity and credible medium-term reduction targets. By contrast, Rentokil and Sunbelt sit at the lower end, reflecting the difficulty of integrating such ambitions into respective industries. On execution, Experian and Teleperformance stand out, delivering materially better emissions reductions (c.90% and c.57% respectively), highlighting the intrinsic advantage of asset-light operating models. At the other end, Rentokil and Sunbelt screen weakest, with emissions trending in the wrong direction, pointing to the operational constraints inherent in more asset-heavy business models. Overall, the analysis shows as follows: TIC leads on ambition, while asset-light sectors are ahead on execution.

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Capital Goods

Vital enablers — but attractive ESG investments in their own right?

European Capital Goods companies sit at the heart of the energy transition. They provide the electrification equipment, automation, industrial software, and efficiency technologies needed to decarbonize industry, infrastructure, transport, and buildings. The more difficult question for ESG investors is whether these companies are credible decarbonizers themselves.

The answer is not straightforward. While disclosure quality continues to improve, meaningful comparison across peers remains complicated by differing methodologies, baselines, and reporting standards. In addition, while our direct coverage universe has reduced over time, we have broadened the analytical framework by incorporating a selection of European mid-cap industrial companies, expanding the study to a 16-stock sample in order to maintain breadth and strengthen comparative analysis across the sector.

Data quality is improving — but still imperfect

Bloomberg carbon data are becoming materially more reliable for current-year Scope 1 & 2 disclosures, particularly among larger-cap European Capital Goods companies. Across our current 16-stock sample, Bloomberg's 2025 Scope 1 & 2 figures are effectively ~100% accurate, with only minor discrepancies in a limited number of names. This marks a significant improvement versus our prior study, where c.39% of latest-year data points were inaccurate. However, part of this improvement likely reflects both our narrower focus on larger-cap companies with more mature ESG reporting and the inclusion of selected European mid-cap industrials to supplement coverage and provide a broader sector comparison.

The main remaining challenge relates to historical baseline treatment. Bloomberg still struggles to capture revisions and recalculations to prior-year emissions inventories, with several baseline reference points incorrectly reported in our sample due to company restatements or methodology changes. This matters because companies continue to apply differing approaches to baseline years, organizational boundaries, M&A and scope changes, and historical recalculations. As a result, headline emissions reduction claims are not always directly comparable across peers. Similarly, investors increasingly need to distinguish between formally validated SBTi targets, public SBTi commitments, and generic "science-based" or Paris-aligned language.

A practical shortcut to net-zero analysis

Rather than attempting a fully standardized carbon model — which remains resource intensive — we use a simpler comparative framework built around three measures. The first is a Disclosure Score, measuring the credibility and transparency of company targets, with particular focus on SBTi status. The second is a Reduction Score, which measures targeted ambition by calculating the implied annualized reduction rate embedded within company Scope 1 & 2 targets. The third is an Execution Score, measuring achieved emissions reductions versus the company's disclosed baseline.

While imperfect, these measures provide a practical directional framework for comparing operational decarbonization performance across the sector. Expanding the framework to a broader 16-stock universe — including selected European mid-cap industrials — also improves relative comparability and reduces the risk that conclusions are disproportionately influenced by a shrinking large-cap coverage universe.

Key takeaway

European Capital Goods companies continue to offer attractive exposure to the energy transition. However, beneath the broad "green enabler" narrative, operational decarbonization performance varies widely. Some companies combine strong disclosure, ambitious targets, and credible execution. Others still rely on weaker disclosure quality, changing baselines, or ambitious rhetoric unsupported by operational progress.

Accordingly, assessing the sector's carbon investment case increasingly requires investors to look beyond headline emissions reductions and focus instead on disclosure quality, target credibility, consistency of execution over time, and the comparability of reporting practices across both large-cap and mid-cap industrial peers.

Chemicals

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Quality and quantity of carbon data disclosure

Within our European Chemicals coverage, the sector performs reasonably well in both the quality and quantity of carbon data disclosure. All companies report Scope 1-3 emissions and have established emissions reduction targets for Scope 1-2, which are verified by the Science Based Targets initiative (SBTi) except for BASF.

Methodology

We rank companies using a scoring system based on two key metrics:

1. Carbon reduction strategy: This metric is based on the percentage of targeted annualized emissions reduction. Scores range on a sliding scale where a target of -1% annual reduction scores 1, and targets of -5% or more score 5. Companies without SBTi validation have half a point deducted.

2. Execution: This metric measures the gap between targeted and achieved annualized reductions. Companies performing worse than their target score 1; those meeting their target score 2; exceeding targets by 1-2 percentage points score 3; exceeding by 3-4 points score 4; and exceeding by 5 points or more score 5.

Carbon targets: rating the corporate ambition

Most companies have SBTi-verified, near-term emission reduction targets for 2030, averaging a -37% cut from baseline. Ambition varies, with Akzo Nobel targeting the highest reduction (-52% vs 2018) and Air Liquide the lowest (-33% vs 2020). All aim for net-zero by 2050, but none have verified targets yet. The average BSG ambition score is 2.9 out of 5, led by Arkema and Akzo Nobel scoring 4 due to strong annual reduction rates of -4.41% and -4.30% respectively.

Rating the corporate carbon reduction execution plans

The chemicals sector has generally performed well against its emissions goals, achieving an average reduction of 25% by 2025 relative to base levels. Companies are actively investing in sustainability initiatives to drive these reductions. For example, Air Liquide plans to invest €8bn by 2035 in low-carbon and renewable hydrogen, with its ADVANCE strategy allocating €16bn (2022+2025) towards the energy transition, including 3GW of electrolysis capacity by 2030. Akzo Nobel aims for 100% renewable electricity by 2030 and had already reached 69% by 2025, with full renewable power in Europe and North America. Across the sector, companies are reducing greenhouse gas emissions, phasing out coal, adopting renewable feedstocks, and implementing sustainability roadmaps. These efforts, backed by transparent reporting and ambitious targets, are driving long-term sustainability across the sector. The average BSG execution rating is 2.8, led by Arkema and Syensqo scoring highest at 4 for strong credibility. Despite its lower score of 1, Akzo Nobel has reduced Scope 1+2 emissions by -47%, quicker than Scope 3 emissions (-20%), and set highly ambitious targets.

Food & HPC

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Scope 3 emissions continue to dominate across the European Food and Household & Personal Care (HPC) sector, driven by structurally embedded sources such as agricultural sourcing, dairy production, packaging, and logistics. By contrast, Scope 1 and 2 emissions are smaller and more controllable, with most companies already making strong progress through renewable energy procurement, energy efficiency, and operational improvements. As a result, many firms are on track to eliminate Scope 1 and 2 emissions by 2030. The core decarbonzation challenge therefore lies in Scope 3, where progress is slower, more complex, and dependent on long-term transformation across supply chains, particularly in agriculture, requiring sustained investment and collaboration with suppliers.

While ambition is broadly consistent across the sector, with all companies disclosing emissions, aligning with SBTi frameworks, and targeting net zero by 2050, execution against internal reduction targets varies meaningfully. Nestlé, Unilever, and Danone remain the largest absolute emitters due to their scale and exposure to agriculture. Nestlé is targeting a 50% reduction by 2030 (vs 2018) and had already achieved a 24.5% reduction by 2025, surpassing its near-term target and indicating delivery slightly ahead of required levels. Unilever has set comprehensive targets, including a 100% reduction in Scope 1 and 2 emissions by 2030 (vs 2015), alongside Scope 3 targets of 42% reduction in energy and industrial emissions and 30.3% reduction in FLAG emissions (vs 2021). Progress on Scope 1 and 2 is advanced, with emissions reduced by approximately 77% vs baseline, but Scope 3 remains more challenging, with current plans covering around 78% of required reductions and leaving a 22% scaling and innovation gap, suggesting execution risk despite strong ambition. In contrast, Danone has demonstrated steady progress, with emissions declining from peak levels reached in 2019 and continuing to fall through 2025. Scope 3 emissions represent approximately 96% of total emissions, with dairy-related methane a key contributor. The company has achieved a 29.8% reduction in methane emissions vs 2020 and exceeded its regenerative agriculture sourcing target, reaching 42% of key ingredients, supporting delivery broadly in line with its required reduction trajectory.

Beyond the largest emitters, some variation in execution is evident across the sector when assessed against company-specific targets. A number of companies, including L'Oréal, Reckitt, Haleon, and Orkla, are progressing well, with emissions reductions broadly in line with or slightly above their required levels in some cases. Others, such as Beiersdorf, are delivering reductions broadly in line with their targets. Meanwhile, companies such as Lindt remain somewhat below target trajectories, although supported by credible decarbonization strategies. Overall, while ambition is largely consistent, execution outcomes continue to vary across the sector.

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1. Quality and Quantity of Carbon Data Disclosure

Whitbread has the best disclosure in the peer group, with clear Scope 1, 2 and 3 reporting, both market-based and location-based Scope 2, and good linkage between methodology, boundaries and target progress.

TUI provides broad disclosure, but its data is harder to interpret given its mix of airlines, cruises, hotels, retail and destination services. Scope 3 quality is also weaker, with only 21% based on primary data.

IHG provides adequate disclosure, including franchised hotels, but the franchise-heavy model limits granularity because data relies partly on owner reporting and estimates.

Overall, Whitbread offers the cleanest disclosure, TUI the broadest but most complex dataset, and IHG sufficient visibility but weaker data control.

2. Carbon Targets – Rating Corporate Ambition

Whitbread scores an ambition rating of 5/5. Its SBTi-validated targets include a 99.6% absolute Scope 1 and 2 reduction by 2040, a 90% Scope 3 reduction by 2050, and 2030 interim targets across Scope 1, 2, 3 and FLAG emissions. This is the strongest framework given its breadth, validation and clear milestones.

TUI scores an ambition rating of 4/5. Its SBTi-validated targets cover airlines, cruises and hotels, including a 24% airline CO_2e/RPK reduction, 27.5% cruise emissions reduction and 46.2% Hotels & Resorts reduction by 2030 versus 2019. The targets are credible, but delivery depends on hard-to-abate transport decarbonization.

IHG also scores an ambition rating of 4/5. Its target is a 46% absolute reduction in Scope 1, 2 and 3 emissions by 2030 from 2019. The scope is broad, but less operationally secure because franchised hotels account for a large share of emissions.

Overall, Whitbread has the strongest target framework, while TUI and IHG have credible but structurally harder pathways.

3. Execution of Carbon-Reduction Plans

Whitbread scores an execution rating of 5/5. By 2025/26, it had achieved a 48.3% absolute Scope 1 and 2 reduction, a 63.0% Scope 1 and 2 intensity reduction, a 21.0% Scope 3 reduction and a 40.2% FLAG reduction, already ahead of its 2030 FLAG target. Delivery is supported by low-carbon rooms, heat pumps and solar PV across 221 hotels.

TUI scores an execution rating of 4/5. Total location-based emissions declined 2.2% year-on-year in 2025, with Scope 1 down 3.8%, market-based Scope 2 down 14.0% and Scope 3 broadly flat. Fleet renewal, SAF use and hotel solar PV support are progressing, but the score is not higher because the absolute reductions remain modest and aviation / cruise emissions are inherently difficult to reduce.

IHG scores an execution rating of 2/5. Market-based emissions rose from 6.25m tCO_2e in 2019 to 6.72m tCO_2e in 2025, despite a 46% reduction target by 2030. To reach the target, emissions would need to fall to c.3.37m tCO_2e by 2030, almost 50% below the 2025 level. Efficiency initiatives such as Green Engage and Low Carbon Pioneers have not yet translated into absolute reductions.

Overall, Whitbread is the execution leader, TUI is progressing but still constrained by its emissions mix, and IHG is materially behind the required pathway

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Infrastructure & Construction

Quality and quantity of carbon data disclosure

Across our European infrastructure coverage, the source of emissions is overwhelmingly Scope 3.

Airports. For airports, including Aena, Groupe Aéroports de Paris (AdP) and Flughafen Zurich (FHZ), the provision of the infrastructure to enable air transport emits relatively little (less than 1% of overall emissions are Scope 1 + 2). However, the exposure to airlines' emissions, due to the requirement to burn fuel in order to generate thrust from their engines, puts indirect emissions at a massively higher level. In addition, airports also generate Scope 3 emissions through their large-scale investment programs: the running and production of the resources and materials used in expanding and maintaining aviation infrastructure require emission-prone processes. Disclosure among the covered airports is good, with detailed breakdown across Scope 1-3 emissions and emission reduction strategies.

Construction. For construction companies, including ACS, Eiffage, Ferrovial, Hochtief and Vinci, Scope 3 emissions also matter a lot more than Scope 1 and 2 emissions (less than 10% of total emissions on average are Scope 1 + 2). Like airports, it is the running and production of the resources and materials used in expanding and maintaining infrastructure that emit the most. Producing building materials, such as cement, glass and steel, emits a significant share of global emissions – and our construction companies are among the world's largest consumers of these products. In addition, the operation of toll road networks and airports further contributes to indirect emissions, as both cars and planes use mostly non-renewable fuel to power their engines. Disclosure among the covered construction companies is good, with detailed breakdown across Scope 1-3 emissions and emission reduction strategies.

Surface transportation. Beyond airports and construction companies, our coverage includes surface transportation group GetLink, which owns and operates the EuroTunnel connecting France and the UK through a below-sea-level, electrified railway system. As a result, this company has generally low direct and indirect emissions and is our coverage's lowest emitter.

Assessing the decarbonization ambition

All companies in our coverage have expressed a willingness to reduce their direct emissions and have developed detailed emission reduction strategies.

Airports. The airport operators have strong commitments to reducing absolute emissions and becoming net carbon-neutral by 2045 at the latest, with some aiming for 2035 or even 2026. Spanish airport operator Aena has committed to achieving carbon neutrality for its direct emissions (i.e., only Scope 1 + 2) at its domestic airport operations and at its international assets, such as Luton in the UK, by 2026. The company targets net-zero direct emissions at the group level by 2040. French airport operator AdP targets net-zero direct emissions (again, only Scopes 1 + 2), at its domestic airports Paris-Orly and Paris-Le Bourget as well as in Delhi and Hyderabad (operated through GMR) from 2030 onwards. The company's Paris-Charles de Gaulle airport will be carbon-neutral from 2035, while six further international airports will achieve this by 2050. Swiss airport operator FHZ has the goal of reaching net-zero emissions by 2040. Lastly, German airport operator Fraport aims to operate on a carbon-free, net-zero emissions basis at Frankfurt Airport and all fully consolidated group airports by 2045 without offsetting (i.e., "zero carbon means we will achieve this target without offsetting our emissions"). The group also has an interim goal to cut group emissions to 95k tons CO_2 and Frankfurt Airport emissions to 50k tons CO_2 by 2030 vs 1990.

The key strategies identified by the airport operators to reduce direct emissions are: 1) the instalment of green/renewable energy sources; 2) the implementation solutions to reduce the heating and cooling needs of the airport infrastructure; and 3) fleet electrification.

Construction. Comparing the decarbonization strategies of individual construction companies is challenging due to the different reporting criteria and base years. However, all companies have committed to carbon neutrality for their direct emissions by 2050 latest, with specific interim targets to be reached by 2025-2030. The cumulative reduction of emissions across our construction coverage is c.40% by 2030 on average, which breaks down to c.3.5% reduction p.a. over 12 years.

Surface transportation. Getlink has defined the most ambitious emissions target across our coverage, with a cumulative reduction of 54% by 2030 over 11 years.

Still, carbon emissions across our coverage are predominantly Scope 3, and targets to reduce indirect emissions appear less developed. While both airports and construction companies have communicated their ambition to reduce indirect emissions on average by 30%, or 2.5% p.a., by 2030 over 12 years, not all companies have defined clear objectives other than the broader carbon neutrality, including direct emissions, by 2050. This is mainly due to the inherent emissions of the aviation and building material sectors. New technology is not expected to meaningfully decarbonize these sectors.

Rating the corporate carbon reduction execution plans

All companies in our coverage have clearly defined strategies to reduce their direct carbon emissions and achieve carbon neutrality in the years ahead. Aena, FHZ and GetLink are particularly ambitious, with Getlink targeting reduced emissions by more than -50% by 2030 vs 2019, and both Aena and FHZ targeting complete carbon neutrality.

However, these strategies often do not include Scope 3 emissions, which are much more difficult to reduce. Airports do not control the capital spending decisions of airlines, nor the fuel used to power the planes. The biggest challenge in decarbonizing the aviation industry is the available quantity of SAF: if SAF is not produced (and it is currently at levels way below actual demand), airlines cannot use it, and airports are unable to reduce their indirect emissions. Airports can merely help this process by providing the required infrastructure to store and use SAF. Airports' Scope 3 emissions are thus extremely hard to abate, given that even the most cutting-edge aviation technology (i.e., more fuel-efficient engines) does not present a viable solution for the vast majority of the aviation sector's carbon emissions. Acknowledging this fact, FHZ has not even set itself a target to reduce Scope 3 emissions.

Construction companies face the same issue, given their operations of toll roads and airports. They are dependent on the decarbonization of their customers, which is increasingly happening but not something the companies can actively impact. In addition, building materials used in the construction of infrastructure are produced through emission-heavy processes. Emissions-friendlier alternatives are scarce and currently not produced at a level where they meet demand.

When looking at the existing progress on reducing carbon emissions, Aena and Ferrovial have performed particularly well, with the reduced emissions exceeding their ambitious targets. AdP and FHZ, meanwhile, are lagging their strategic targets. All other companies are on track to achieve their targets.

Luxury Goods

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Luxury companies continue to embrace sustainability, environmental and societal responsibility, inclusivity and diversity. These continue to be key planks in luxury's current strategic framework, particularly in relation to younger consumer cohorts, ensuring that the sector remains proactively engaged. For more, please see our recent Black Book on ESG (Global Luxury Goods: In Search of Ethical Luxury). Carbon targets, rating the corporate ambition

Disclosure and ambition transparency across our coverage remains high overall. We use an expanded series of metrics to measure transparency. The methodology is explained in ESG "MAQ": Measuring the Unmeasured – Global Luxury Goods companies seem to be all on board the ESG bandwagon. However, disclosure norms do seem to be shifting, with fewer companies providing CDP disclosures in 2025, for example.

Carbon strategy ambitions continue to favor those with significant scale. EssilorLuxottica and LVMH continue to lead our reduction coefficient ratings. Both companies start from significantly larger Scope 1 and 2 emission footprints, largely reflecting their greater scale compared to peers. This means that more can and has been done in terms of GHG emissions reduction.

Greater vertical integration also affords a better grip on emissions throughout the supply chain. EssilorLuxottica, Richemont, Swatch Group, and Hermès enjoy a high degree of vertical integration relative to peers, which has likely enabled more ambitious annualized emissions reduction targets.

Rating the corporate carbon reduction execution plans

Most luxury brands continue to score relatively poorly in terms of execution. On Scope 1+2 GHG emission reduction performance, only Burberry and LVMH have delivered in line with their emissions reduction ambitions, so far.

Hermès and Kering lead on their share of green capital expenditure, which align with the EU's Taxonomy regulations. LVMH has made slightly less headway. On the other hand, Moncler, EssilorLuxottica, Brunello Cucinelli, Burberry, Richemont, and Swatch Group have deployed significantly lower to no green capex. Poor performance from the latter three may reflect the fact that they are domiciled outside of the EU.

Across the sector, management incentives continue to incorporate ESG and CSR targets. We highlight that Kering has increased the weight of its 'Climate' criteria within new CEO Luca de Meo's variable remuneration framework from 5% to 10%. Burberry, on the other hand, has pushed back its net-zero ambitions by 10 years to 2050.

Media & Internet

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Our analysis centers on the disclosure of Scope 1, 2 and 3 emissions, and our findings highlight consistent patterns across our Media & Internet coverage. The entire coverage has low Scope 1 & 2 intensities (we also used market-based data when location-based data was not available). Consumer-facing digital models tend to have significantly higher Scope 3 GHG emissions than classic traditional Media models. Although Scope 3 emissions tend to dominate a company's overall GHG footprint, we focus only on Scope 1 (direct emissions, e.g. heating) and location-based Scope 2 (e.g. indirect emissions from purchased energy) as companies have greater control over these emissions. Some companies, such as Delivery Hero, have adopted "carbon intensity" policies for their Scope 3 emissions (usually defined as relative to revenue) to account for the fast-growing nature of their businesses; however, for the purposes of this analysis, we have excluded these targets.

Publicis and AutoTrader have the highest ambitions; AutoTrader and ITV are executing best on their goals.

Publicis and AutoTrader are the two companies that stand out due to their very ambitious targets of reducing their GHG emissions (Scope 1+2+3) by 97% and 90% respectively until 2040; based on an annualized reduction rate this equates to c.5%. While AutoTrader has also been stellar in executing its ambitious goals, 17pp ahead of its annualized target, Publicis has been underperforming its target by c. 3pp and we see ITV as a best-in-class executer of its emission targets as they are 9% ahead of their targets, on an annualized basis.

ITV's Scope 1+2 reductions (c. 63% below their baseline-year 2019), have been driven by building consolidation, the decommissioning of boilers at one of their hubs, and wider energy efficiency measures.

As the best executer, AutoTrader has been reducing its Scope 1+2 emissions significantly by moving all of its data centers to cloud providers, reducing overall electricity/gas usage by 50%, and switching 100% of its fleet to EV/low-emission vehicles. In order to influence its suppliers' emissions, AutoTrader is developing climate-related supplier selection criteria.

Those performing least well

Starting in 2025, CTS Eventim introduced a carbon-reduction policy that only covers Scope 1+2 emissions. The policy is aimed at reducing carbon emissions by 42% until 2030 vs 2025 (with the commitment to evaluate Scope 3 emission goals in 2026). Given that Eventim has not implemented any incentives in its management compensation packages to reduce carbon emissions, or committed to Net Zero goals, we view CTS Eventim as the weakest in execution and ambition (although no results have been released yet).

Delivery Hero's execution of its emission targets has been very weak as the emissions have increased by 35% vs the baseline-year 2022 and on an annualized basis, Delivery Hero is 17pp behind its target which is the lowest in our coverage. However, to decrease its emissions, Delivery Hero has committed to achieving 100% renewable electricity across owned and leased infrastructure by 2030 through green tariffs.

Medtech

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Among the Stoxx600 healthcare companies analyzed, there is a marked contrast between best and worst practices in carbon emission reduction strategies and their execution. Across our Consumer Medtech coverage, Straumann stands out as a best-practice example, having established comprehensive Scope 1, 2, and 3 targets aligned with long-term net-zero ambitions. With a base year of 2021 and a target year of 2040, Straumann aims for a 90% reduction in Scope 1, 2, and 3 emissions, corresponding to an annualized reduction rate of approximately 5%. The company's strong execution rating of 5 reflects effective alignment between its ambitious strategy and actual emission reductions, supported by credible targets validated by the Science Based Targets initiative (SBTi). This demonstrates a well-integrated approach combining clear objectives, credible capital allocation, and transparent progress monitoring.

In contrast, Demant exemplifies poor execution despite an SBTi-validated carbon reduction strategy. The company targets a 90% reduction of Scope 1-3 emissions by 2050 and a substantial reduction of 46% by 2030. However, Demant's Scope 1+2 emissions have increased by 21% since the 2019 base year, revealing a significant gap between stated goals and actual performance. This disconnect is underscored by a low execution rating of 1 and a lack of credible low-carbon capital expenditure, highlighting challenges translating strategy into effective action, which results in weaknesses in implementation and oversight.

Focusing specifically on the traditional European medtech sector, Philips stands out as a best-practice example through its strong execution of solid reduction targets. Philips has set a carbon reduction goal of approximately 49% by 2030, which is more ambitious compared to peers as shown by the highest reduction coefficient amongst Medtech companies. The company is lowering carbon emissions in line with its planned annualized reduction rate, earning a top execution rating of 5 out of 5. Until 2025, Philips had already achieved a 20% reduction in emissions, very much in line with its planned target. This strong performance is supported by transparent progress monitoring and credible investments in low-carbon initiatives, demonstrating effective alignment between strategic goals and operational outcomes. Philips' carbon emission reduction performance highlights the critical importance of execution quality in driving meaningful emission reductions. This positions the company as a benchmark for sustainability leadership within the medtech sector, showing that steady, credible progress can be more impactful than ambition alone.

This comparison between best and worst practices underscores a fundamental lesson for the EU medtech industry: ambition alone is insufficient to achieve climate goals. Robust execution, credible investment in low-carbon technologies, and transparent monitoring and accountability mechanisms are essential to drive meaningful and sustained carbon emission reductions. Companies that integrate these elements effectively are better positioned to lead the transition to a low-carbon future and deliver on their sustainability commitments.

Notably, none of the companies in European Medtech analyzed currently links executive compensation to ESG or carbon reduction targets, which may undermine accountability and hinder the realization of emission reduction ambitions.

In summary, best practices among Medtech companies are characterized by clear, science-based targets, credible investment in low-carbon initiatives, and transparent execution, while worst practices reveal ambition without effective delivery or accountability mechanisms.

Metals & Mining

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Quality and quantity of carbon data disclosure; carbon targets, rating the corporate ambition

▪ Boliden still leads the sector in corporate ambition, with the only fully validated SBTi 1.5°C-aligned target among our covered European names. It aims to cut 42% absolute GHG emissions for Scope 1-2, and 30% for Scope 3 by 2030. Sustainability targets, such as GHG emissions reduction, make up 20% of LTIP.

▪ Antofagasta ranks second, with reduction targets for all three scopes (1–3). The company aims for a 10% cut for Scope 3 by 2030, and 50% cut for Scope 1 and 2 emissions by 2035. While described as science-based, these targets have not been validated by the SBTi. Climate and broader ESG metrics are embedded in executive pay, with a defined weighting in long-term incentives.

▪ Glencore sits third, reflecting in part its decision to retain its coal business. It has set group-wide industrial emissions reduction targets (Scopes 1–3) of 15% by 2026, 25% by 2030, and 50% by 2035, along with a 2050 net zero ambition (subject to policy support). Overall, the disclosure is satisfactory, covering all three scopes, but lacks detail on NOx and SO_2, and might not have SBTi-approved targets in foreseeable future. ESG performance is factored into CEO compensation.

▪ Anglo American is fourth, with targets for Scopes 1, 2, and 3, though its Scope 3 target is set for 2040. Its subsidiary, De Beers, has SBTi-approved near-term and net-zero targets, but there is no group-level SBTi commitment. ESG metrics account for roughly 20% of both long-term incentives and annual bonus structures.

- Rio Tinto is ranked fifth due to the absence of Scope 3 targets, despite providing detailed plans and disclosures for Scopes 1, 2 and 3. The company aims to reduce absolute Scope 1 and 2 GHG emissions by 15% by 2025 and by 50% by 2030 (relative to 2018 levels) and reach net zero emissions by 2050. In 2025, 10% of the short-term incentive plan (STIP) and 20% of the long-term incentive plan (LTIP) were weighted towards decarbonization, including the progress of carbon abatement projects.

Execution against targets

- **Boliden** reported 2025 Scope 1–2 emissions of 946kt CO_2e, flat year-on-year. It retains multiple levers to close the gap, including increased electrification, greater use of renewables, and ongoing efficiency improvements. Planned initiatives - such as transitioning mine vehicles to electric from 2027 - should support progress over time.

- **Antofagasta** has reduced 37% of its Scope 1 and 2 GHG emissions since 2020 (vs. 50% reduction target by 2035). Hence, they are on track to achieve target. However, CO_2e emissions intensity has increased (again) by 9% yoy in 2025 to 1.91 Mt CO_2e driven by higher fuel consumption due to an increase in open pit mining activities.

- **Glencore** is on track to achieve both short- and long-term targets, with 2025 combined emissions (Scopes 1, 2, and 3 excl. EVR) at 399.9 Mt CO_2e (-4% yoy). The decline was primarily due to the managed decline of coal production, with some additional contribution from lower energy use and efficiency gains at industrial assets. It had reduced c.28% total Scope 1, 2 and 3 industrial emissions by 2025, and hence is ahead of 2026's 15% reduction target.

- **Rio Tinto** reported 2025 Scope 1 and 2 emissions (adjusted equity basis) of 31.5Mt CO_2e, and after applying 1.2 Mt CO2e carbon credit, net emissions was 30.4 Mt which was -17% vs. 2018 levels (against -15% target by 2025). The next target is 50% reduction by 2030, which leans heavily on off-the-shelf solutions like renewable energy deals, but success hinges on advancing solutions for the Pacific Aluminum smelters. Discussions are progressing, but any delays might risk the 2030 target. Post-2030, reduction might come from carbon-free aluminum production through the ELYSIS plant, hydrogen calcination, and battery electric haul trucks.

- **Anglo American** reported 2025 Scope 1 and 2 emissions of 6.3 MtCO_2e (excl. divested business), a 14% reduction yoy. Hence, it is on track to meet its interim target of 30% reduction by 2030 against its 2020 baseline. It has sold its met coal business to Dhilmar Limited (closing date expected Q1 2027), which should contribute to lower absolute GHG emissions next year. It is "improving energy efficiency – using less energy, consuming less diesel, and therefore generating fewer emissions, to produce each tonne of product."

Non-alcohol Beverages

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GHG emissions matter. Beverage companies are facing increasing scrutiny from regulators, investors and consumers to reduce their environmental footprint. However, there is also an opportunity. By reducing energy consumption and its associated GHG emissions, companies can reduce their operating costs, improve business resilience, and boost profitability.

Looking at bottlers and brewers, CCH positions ESG at the core of its strategy. CCH has set two key midterm targets for 2030 (against a baseline of 2019): a 46.2% reduction in direct emissions (Scope 1 & 2), alongside a 28% reduction in Scope 3 emissions, both on the pathway to net zero by 2040. However, progress against these targets is slightly behind schedule. For Scope 1 & 2, CCH reached a 21% reduction by 2025, just below the 25% required to remain aligned with the target, while for Scope 3 it achieved only 12% versus a 15% interim milestone.

Royal Unibrew has reinforced its ESG ambitions, with renewed SBTi validated targets covering

Scope 1, 2 and 3 and a stronger commitment to reach net zero by 2040, including raising its 2030 total emissions reduction midterm target from 50% to 60% (against a 2019 baseline). However, despite this stronger ambition, execution was impacted by a setback in 2024, when absolute emissions increased due to the acquisitions of Vrumona in The Netherlands and San Giorgio in Italy, and higher business activity. As a result, progress is now lagging the expected trajectory, with only a 20% reduction achieved by 2025 versus the 32% required to stay on track toward its 2030 target. While Royal Unibrew points to additional ESG initiatives across raw materials, transportation, traded goods and packaging, these remain undefined, limiting visibility on their effectiveness in closing the gap to its targets.

Looking at the reporting structure, targets and their achievement so far, we view CCH and Royal Unibrew as comparable in terms of ambition, with both companies setting SBTi-aligned targets and net zero commitments. However, we rank CCH better on execution, supported by more tangible initiatives such as supplier collaboration, increased use of recycled materials, as well as investments in energy efficient equipment.

Oil Services

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Quality and quantity of carbon data disclosure

The carbon assessment of the Oil Services sector needs to account for the structural differences between asset-heavy and asset-light companies. On the one hand, asset-heavy companies (e.g. Subsea7, Saipem) operate fleets and yards with high direct emissions, which are generally well reported but inherently difficult to decarbonize. On the other hand, asset-light companies (e.g. Technip Energies) are primarily engineering and technology providers, with much smaller operational footprints. Their Scope 1 & 2 GHG emissions are also well disclosed, and the reported reductions are typically more consistent over time.

Carbon targets, rating the corporate ambition

All companies have high ambitions for Scope 1 & 2 emissions. SBM Offshore (asset heavy) aims to be net zero as early as 2030. Technip Energies (asset light) targets 90% Scope 1 & 2 reduction by 2030. Saipem and Subsea 7 target a 50% reduction in their emissions by 2035, a somewhat ambitious goal, in our view. We keep our ambition scores unchanged versus last year.

Rating the corporate carbon-reduction execution plans

GTT is best in class. It targets a 55% reduction in Scope 1 & 2 GHG emissions (c.6% p.a.) and a c.33% reduction in Scope 3 emissions (c.3% p.a.) by 2033. GTT's execution is ahead of plan for its Scope 1 & 2 GHG emissions, with reductions of c.10% and c.17%, respectively. We assign GTT an execution score of 5, reflecting the combination of achievable targets and consistent outperformance.

SBM Offshore is ahead of schedule. It aims for net zero by 2030 on Scope 1 & 2 emissions (and by 2050 on Scopes 1, 2 & 3), hence a needed 3% annual reduction in GHG by 2030. It has achieved an impressive -12% p.a. reduction over the past nine years and has already reached its target ahead of schedule, hence our execution score of 5.

Vallourec is on track. It targets a 25% reduction in its total carbon footprint (Scopes 1, 2 and 3) by 2030, implying a c.3% annual reduction from a 2021 baseline. In addition, Vallourec has set carbon intensity targets, including a 30% reduction in CO_2 emissions per ton of rolled steel and finished tubes by 2030. Since 2021, Vallourec has delivered a reduction of c.5% per year in its total carbon footprint, above the trajectory implied by its 2030 target. We assign Vallourec an execution score of 4.

Saipem is on track. This asset-heavy company aims for a 50% reduction in its Scope 1 & 2 GHG emissions by 2035 (c.4% p.a.). Its current performance is broadly in line with this trajectory, with Saipem's emissions decreasing at c.3% p.a. Given Saipem's retrofit and renewal initiatives to achieve its decarbonization targets, we assign Saipem an execution score of 3.

Tenaris is on track. Tenaris targets a 30% reduction in carbon intensity for Scope 1, 2 and 3 emissions by 2030 compared to a 2018 baseline, measured as CO_2 emissions/ton of processed steel. The company does not disclose a formal absolute emissions reduction target. It has delivered reductions broadly in line with the required trajectory (c.3% p.a.). We assign Tenaris an execution score of 3.

Technip Energies might fall short of its target. Technip Energies targets a 90% reduction in its Scope 1 & 2 GHG emissions by 2030, implying a c.23% annual reduction from 2021. Progress has been strong in absolute terms (c.46% reduction achieved in 2025 vs 2021); however, the current reduction pace of c.14% remains below the required trajectory. We reiterate our execution score of 2.

Rubis. Rubis aims to reduce by 20% its Scope 1 & 2 GHG emissions by 2030. This trajectory implies a reduction of c.2% p.a. Since 2019, Rubis has fallen short of the required trajectory, effectively increasing its Scope 1 & 2 GHG emissions by c.1% p.a. We assign Rubis an execution score of 1.

Subsea 7, worst in class. The company is aiming for a 50% reduction in its Scope 1 & 2 GHG emissions by 2035 and net zero by 2050. However, Subsea7's Scope 1 & 2 emissions have increased c.12% compared to the base year (2018). Given that there have not been any additional intermediate targets and there is still no clear path towards the required trajectory, our execution score of 1 remains unchanged.

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Real Estate

The Real Estate sector contributes c.42% to C02 emissions globally, of which 15% relates to embodied carbon and the remaining 27% to buildings' construction and operational carbon during use, according to 2024 data from GRESB. Therefore, the sector plays an important role in managing and reducing the global carbon footprint. Across Real Estate, carbon intensity varies by sub-sector and is largely dependent on the utilization of a property by its intended users. Residential and Offices are the most carbon-intense sub-sectors at 116kg C02/sqm/yr and 35kg C02/sqm/yr respectively, and Self-storage is one of the least carbon-intensive sub-sectors at 3kg C02/sqm/yr, according to a 2024 study by KMPG and EPRA.

Quality and quantity of carbon data disclosure

ESG disclosure by the listed Real Estate sector is, in our view, of a high standard, especially among the larger landlords, which account for most of the 26 companies under our coverage, and this is reflected in high scores for many companies. These high standards are underpinned by the disclosure requirements of the industry body, EPRA (European Public Real Estate Association), which gives accreditation based on its criteria.

Carbon targets and corporate ambition

In addition to disclosure, we have also scored our coverage based on the execution of their carbon reduction strategies. Our approach incorporates various factors, including awarding a higher score to companies that include Scope 3 (emissions produced by tenants) in the scope of their targets. We also take into account the target percentage reduction; and the time required to achieve it, rewarding those companies with more ambitious targets over shorter time frames.

Where companies have several targets, we focus on the near-term target – German residential companies tend to have such targets. Several companies are well on track with their reduction ambitions, including LEG and Gecina. Other notable examples include self-storage company Shurgard, which has relatively high ambitions and had already achieved a 67% reduction in carbon emissions by year-end 2025 (vs a 2017 baseline). This reflects the relatively low property utilization and the high degree of control the landlord has over power usage.

Across our coverage, companies tend to have a similar approach to carbon reduction: property refurbishment (modernization) and redevelopment, including improving thermal insulation and enhancing the efficiency of heating and lighting systems, and considering on-site renewable energy production. Across the Residential, Retail, Office and Healthcare sub-sectors, it is also important that landlords provide tenants with information and recommendations/incentives to lower their own energy consumption, or switch to more renewable energy sources as a way of reducing total emissions.

What does this mean for Real Estate?

Strong ESG credentials have increasingly become critical for business success in real estate. They can affect real estate developers' ability to secure planning permission for new schemes, a landlord's ability to attract tenants and command higher rents, and the ability to optimize prices achieved on disposal.

In the UK, there is growing evidence of a rental premium being achieved on properties with higher environmental credentials as assessed by the local market standard, Energy Performance Certificates (EPC). *The UK government has implemented future EPC rating requirements as part of its Minimum Energy Efficiency Standards (MEES) – all rented non-domestic buildings >1k sqm are now required to achieve a minimum EPC rating of B from 2031*. Green-certified buildings typically sell for a premium relative to comparable non-certified ('brown') assets, with ESG performance increasingly reflected in valuation premiums and discounts. Building accreditations are increasingly being reported by listed real estate companies to show the percentage of portfolios that have been certified and the grade stratification, including BREEAM and LEED certification systems. In Central London, 70% of April 2026 office take-up was for assets with an excellent/outstanding BREEAM rating, highlighting the continuing shift in demand towards top-rated assets.

Retail

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Most players in the Food and Apparel retail sector have net-zero commitments in place, but the quality of disclosure, ambitions and current progress towards the targets vary significantly. We outline some of the key practices (both good and bad) that we see within our coverage.

Aligning with recognized frameworks such as the SBTi and backed by quantified interim milestones across all relevant scopes, is a useful first screen for ambition. For example, Inditex is committed to reducing emissions by 53% by 2030, alongside a 20% cut by 2027 across Scopes 1–3, which is SBTi-approved. Of the Food names, Carrefour distinguishes itself with a 32% Scope 3 reduction target by 2030, through levers such as regenerative agriculture and low-carbon product development. Similarly, M&S has made a commitment that all UK farms supplying it will adopt regenerative agricultural practices by 2030, e.g. reduced ploughing and tillage, and soil-health management to increase carbon sequestration.

The stronger performers underpin targets with granular decarbonization roadmaps. H&M integrates material switch programs and large-scale supplier-focused renewable energy projects into its plan, and has scaled contracted renewable volumes (e.g. via PPAs and certificates) to support both its own operations and key tiers of its supply chain. ABF has strengthened divisional transition plans for Primark, Twinings Ovaltine and ABF Sugar, with SBTi-validated targets and defined levers such as plant retrofits, fuel switching and supplier-level energy-efficiency programs. Puma has formalized a climate transition plan linked to SBTi-approved near-term targets, detailing measures such as onsite solar at Tier 1 and 2 factories and the phasing out of coal in high-risk sourcing countries. Among food retailers, Axfood stands out for coupling a relatively conservative emissions baseline with a very operationally focused plan (energy-efficient stores, logistics optimization, and supplier collaboration on agricultural emissions) and reasonably transparent reporting.

We continue to see embedded climate considerations in capital allocation as a positive indicator of commitment. ABF is one of the few companies to quantify this at group level, earmarking a defined share of capex for climate strategy (c.6%) and disclosing sizeable spend on low-carbon manufacturing retrofits. Ahold Delhaize and H&M use internal carbon prices in investment appraisal, effectively turning emissions into a financial constraint in capex decisions. Several others, including M&S and Axfood, increasingly reference climate as a formal criterion in store refurbishment, logistics, and energy projects. In contrast, peers such as Sainsbury's, Zalando, Ocado and B&M still do not disclose how much of their capex is directed towards decarbonization, making it difficult to judge whether investment levels match stated ambitions.

Circularity and sustainable materials are becoming important differentiators in general retail. Zalando and Inditex both target lower-impact textiles and stronger supplier engagement on sustainable fabrics, while H&M, Puma and Primark are leaning on material mix changes and product design (durability, recyclability) as core Scope 3 levers. M&S uses 100% "responsibly sourced" cotton (e.g., Better Cotton, organic and Fairtrade) for its clothing; in its denim range, it produces clothing with reduced water use (>80% less) and higher recycled material content. B&M, by contrast, provides limited detail on how product level or materials choices will contribute to its decarbonization pathway.

Despite progress, several weaknesses persist. Many firms still provide no breakdown of "green capex" or the share of total investment devoted to climate projects, undermining confidence in delivery of long-dated targets. Scope 3 strategies are often vague. JD Sports and Zalando, whose models depend heavily on third-party brands, give limited detail on how they will influence upstream product footprints through brand selection, minimum standards or data requirements. Next continues to outline long-term supply chain ambitions without disaggregating them by Scope 3 category, making progress hard to track. In Food Retail, Ocado focuses on logistics efficiency and packaging, but offers fewer quantitative details than leaders such as Tesco or Axfood. B&M's climate narrative remains especially high level, with narrow coverage of value-chain emissions.

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Semiconductors

Quality and quantity of carbon data disclosure

We find that our European semiconductor coverage continues to provide a high quality and quantity of carbon disclosure, with ASML, BESI and Infineon all disclosing Scope 1, Scope 2 and Scope 3 emissions in their latest sustainability reporting. ASML's 2025 Annual Report discloses net and gross Scope 1 and 2 emissions, net Scope 3 emissions, Scope 3 intensity per €m of gross profit, supplier commitment metrics and product energy-use KPIs. ASML also reports under the European Sustainability Reporting Standards. BESI's 2025 Sustainability Statement and SASB/GRI reference tables disclose Scope 1 & 2, Scope 3, renewable electricity, fuel consumption and supply-chain sustainability. Infineon's 2025 Sustainability Report includes climate-change reporting, sustainability targets and disclosure of Scope 1, 2 and 3 emissions, with SBTi-recognized targets covering own operations and the value chain. Overall, disclosure is regular, increasingly standardized and sufficient for yoy comparison.

Carbon targets, rating the corporate ambition

ASML remains the most ambitious of the three companies. It targets GHG neutrality across its value chain by 2040, achieved 0 kt net Scope 1 and 2 emissions in 2025, reports 11.5 Mt net Scope 3 emissions, and has a Scope 3 intensity KPI of 0.67 kt CO_2e per €m of gross profit versus a 2025 SBTi target of 0.93 kt. ASML's roadmap covers operations, supply chain, customer use-phase impacts and product energy efficiency. BESI's ambition is also strong: the company states that it has developed short- and long-term targets through 2050, has reduced Scope 1 & 2 emissions intensity by 98% since 2019, and increased renewable electricity usage to 99% from 18%. However, BESI's targets appear less externally benchmarked than ASML's and Infineon's. Infineon made the clearest step-up in ambition: in 2025, SBTi recognized its Scope 1 and 2 target as aligned with the 1.5°C pathway and validated its Scope 3 supplier-engagement target. Under this target, 72.5% of suppliers by emissions covering purchased goods and services, capital goods and upstream transportation and distribution should have science-based targets by 2029. Infineon also retains its own goal of Scope 1 and 2 carbon neutrality by the end of fiscal year 2030.

Carbon targets, rating the corporate ambition

Execution remains strong across all three names. ASML reached 0 kt net Scope 1 and 2 emissions in 2025, reduced gross Scope 1 and 2 emissions to 26 kt, lowered net Scope 3 emissions to 11.5 Mt from 12.0 Mt in 2024, and improved Scope 3 intensity to 0.67 kt CO_2e per €m gross profit. BESI continues to execute well: it reports a 98% reduction in Scope 1 & 2 emissions intensity since 2019, a 67% reduction in fuel-consumption intensity. Infineon reports an 84% reduction in Scope 1 and 2 emissions versus 2019, 100% renewable electricity and continued use of levers such as energy efficiency and green electricity. Scope 3 remains the main challenge, but engagement is more formalized: ASML reports supplier commitment progress, BESI reports supply-chain sustainability initiatives, and Infineon has an SBTi-validated supplier target. Considering ambition and execution, ASML still deserves the top score given its broad Scope 1–3 roadmap, 2040 value-chain neutrality target and strong execution. BESI remains very strong operationally, especially on Scope 1 and 2, but we would keep it one notch below ASML due to less visible external validation of its targets. Infineon has improved meaningfully and now has a stronger SBTi-backed ambition, but its Scope 3 target remains supplier-engagement based rather than a full absolute value-chain reduction target.

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Software & IT Services

Quality and quantity of carbon data disclosure

In general, companies provide comprehensive and detailed data series for the various areas affected by GHG emissions (this was not the case four years ago). Even though calculation methods have sometimes changed, companies have taken care to update the data series.

Carbon targets, rating the corporate ambition

The most discriminating feature in our assessment is the emission scopes covered by a company's targets, i.e. Scopes 1 & 2 or Scopes 1, 2 & 3, and whether Scope 3 is taken into account fully or only partially.

Software and IT services companies have, in most cases, had their targets validated by SBTi and aim to reduce their GHG emissions by around 5% (Capgemini, Sopra Steria, Arcadis, Sage) per year on Scopes 1, 2 and 3.

These companies could, in general, achieve their Scope 1 & 2 objectives relatively smoothly within the targeted time frames as their carbon footprints are relatively limited in size. The largest emission source generally comes from travel due to the companies' consulting activities. In our view, the deployment of generative AI and AI agents at client sites does not, in itself, alter the scale of their greenhouse gas emissions, given that most software and service companies are not IT infrastructure providers for their end customers and have not themselves chosen to build their own IT infrastructure for training and running AI models.

In this context, the most ambitious companies are those that have committed to a significant reduction in their GHG emissions – including the entire Scope 3 (indirect emissions) – within a short period of time, while aiming for strong growth in business.

Rating the corporate carbon-reduction execution plans

Beyond the objectives set by the companies (the key differentiating criterion being the targeted annual reduction), we also assess the credibility of each company's pathway to achieving the objectives.

In general, Software and IT Services companies can rapidly reduce their Scope 1 & 2 targets as they have a limited data-center footprint (mostly leveraging their partnerships with cloud infrastructure providers, such as AWS, Microsoft Azure and GCP) and have largely adopted renewable energy for their offices. Most companies have also tied top management's compensation to ESG targets.

Many of the companies we cover perform relatively well on this metric. We have therefore highlighted the laggards, either because their disclosures are insufficient (Reply) or because their targets are not ambitious enough (Dassault Systèmes). SAP stands out somewhat, with a target of reducing its GHG emissions by more than 10% per year.

Telecom

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Carbon targets, rating the corporate ambition

Despite its vital importance in connecting consumers and businesses, through both wireless and wireline networks, the telecommunications industry is only responsible for an estimated 2% of global emissions. If we break down these emissions further, by source, we find that Scope 1 emissions are very limited and typically come from a company's own vehicle fleet. Scope 2 emissions are more significant (around one-quarter of total) and come from electricity usage to power a telco's network. Finally, Scope 3 emissions (indirect emissions from the value chain) are the most significant, usually making up at least three-quarters of total emissions.

Our analysis suggests all telcos save Vodafone, Telenor, and Cellnex beat their targets this year.

Many telcos are resorting to carbon-offsetting strategies (such as purchasing renewable energy certificates, RECs) or signing long-term power purchase agreements (PPAs) with alternative energy providers. Gradual efficiency upgrades can be made to data centers to reduce their emissions, but two ongoing industry transitions are central to the story of carbon abatement by telcos: fiber-to-the home and 5G. The former is a crucial innovation as, according to French regulator Arcep, new fiber uses around 90% less electricity than its predecessor, copper. However, the transition from 4G to 5G has its caveats. Despite 5G being more efficient (i.e. transmitting more data per kWh of electricity), the relentless growth in data demand from consumers means there is a need to increase the density of antennas, leading to overall higher energy consumption. The switch-off of legacy 2G and 3G networks will mitigate this increase, but on a net basis, emissions are projected to rise.

Majority of European telcos have set multiple, ambitious emission-reduction targets

In alignment with globally defined standards, all the European telecom companies have long-term goals of becoming net-zero emitters of greenhouse gases. The companies tend to focus on reducing market-based emissions as management has more comprehensive levers to control these than location-based emissions. Some companies are more ambitious than others. For example, Tele2 aims to be net zero by 2035 (receiving a 5/5 rating for Ambition), while the rest push their targets out to 2040-2050. We do not think that any of the companies warrant an Ambition rating of less than 3/5 as the vast majority aim for between a -2% and -7% annual emissions reduction, a range commended by the SBTI. We credit companies such as Orange and Vodafone for having specific targets for their emerging market operations, as this separation allows for greater accountability.

Rating the corporate carbon-reduction execution plans: Most telecoms are making meaningful progress in reducing GHG emissions

Our analysis examined the abatement of location-based Scope 1 & 2 emissions rather than market-based emissions, as the latter is affected by the purchase of RECs and PPAs. Ten of our coverage companies appear to be in line with or exceeding their targeted annual emissions reductions, with Deutsche Telekom, Tele2 and Nokia leading the way. We were impressed by Nokia's progress in reducing emissions by 14% annually compared to its -2% target, with some of this reduction originating from reaching a 99% share of renewable energy in its own factories. By contrast, three telcos have recorded an increase in their location-based Scope 1 & 2 emissions. Specifically, Telenor and Vodafone have recorded an increase in the average emissions intensity of the electricity used from the grid. This could be explained as the trend was impacted by high exposure to emerging markets where there is a greater reliance on diesel generators caused by the absence, in some geographies, of a reliable grid.

Transport

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Airlines

Airlines' emissions are overwhelmingly Scope 1, as planes must burn fuel to generate thrust from their engines. In this sector, emissions are extremely hard to abate. While in ground transportation, physics permits the replacement of combustion engines with batteries as a source of energy, at present batteries cannot deliver the requisite energy density to power commercial flight. Even the most cutting-edge technology only envisages planes with less than 100 seats, and a range of just 500 miles, by the early 2030s: less than one-half the passenger capacity and less than one-sixth the range of an A320, and therefore not a viable solution for the vast majority of aviation's carbon emissions. Sustainable Aviation Fuel (SAF) offers the only feasible route to decarbonization, but we have significant concerns over the availability of both fuel and feedstock.

The global airline industry group IATA has committed to a net-zero target by 2050, implying that a cumulative total of 21.2 gigatons of carbon will be abated between now and 2050. The path is dependent principally on SAF, providing 65% of total decarbonization. A further 13% is to come from new aircraft and 3% from operational measures. That leaves 19%, which will come from carbon capture & storage (11%) and offsets (8%). Here, airlines must be cognizant of the quality of offsets: sometimes doing nothing has proved to be a better solution than using offsets. easyJet, for example, admirably began offsetting all its carbon emissions from 2019. However, following accusations of "greenwashing", it terminated this practice during the pandemic, transitioning from using Verified Emission Reduction (VER) certificates for all flights to a new roadmap that prioritizes absolute carbon reduction overcompensation.

Several European airlines have announced their own targets for decarbonization and/or SAF usage that exceed the mandates from the European Commission. On SAF specifically, Ryanair leads with a targeted 12.5% drop-in by 2030, followed by Air France-KLM, IAG and Wizz Air at

10%. We view IAG as the leader in actively working towards more sustainable aviation, and the company currently has the highest SAF blend ratio at 3.3%.

Logistics

The logistics industry breaks down neatly into the asset-light companies (DSV and Kuehne+Nagel) and the asset-heavy ones (DHL and Maersk).

The asset-light companies are in the business of managing supply chains for their customers. They do not control the capital spending decisions for ships and planes, nor the fuel used: all of this is subcontracted to airlines and shipping lines, and over 95% of their emissions are Scope 3, with only a small amount Scope 1 (own trucks) and Scope 2 (energy in warehouses). Forwarders cannot force customers into more expensive, lower-carbon transportation. Consequently, there is limited ability for these companies to shape their own decarbonization journey.

Asset-heavy groups, by contrast, have a much greater ability to decarbonize. For DHL, The decarbonization strategy is heavily centered on its Express division, which is the primary driver of the group's Scope 1 emissions due to its extensive aircraft fleet. The division accounted for 5m tons of Scope 1 CO_2e out of the groups 6m tons in 2025. DHL has the highest SAF blending target of any aviation business we know, at 30% drop-in by 2030, and bears the financial risks of decarbonization. We also see more alignment between decarbonization goals and executive compensation at DHL than anywhere else in our coverage. Much of the rest of DHL's total emissions are Scope 3 emissions in the forwarding business, similar to the asset-light companies as described above.

Maersk already provides the lowest-carbon form of international goods transportation: container shipping. By way of comparison, airfreight typically emits some 1kg of carbon per ton-km, while container shipping emits just 6 grams. Yet the group continues to seek improvement. Notably, Maersk has evolved its objectives into a more aggressive "Net Zero by 2040" ambition, moving its target a full decade ahead of the industry's general 2050 timeline. In addition, for 2030, Maersk has committed to a 35.00% absolute reduction in Scope 1 emissions and a 22.00% absolute reduction in total Scope 3 emissions. It has aggressively invested in "future-proofing" its fleet through dual-fuel technologies, with 21 dual-fuel vessels already in operation as of the first quarter of 2026. Maersk has an emission-reduction target of 92% from 2022 to 2040 across Scopes 1-3, and executive compensation is linked to ESG, which carries a 20.00% weighting in the LTI (Long-Term Incentive) program.

Air Tech

Air tech companies are responsible for much lower carbon emissions than airlines. Scope 2 emissions are significant, given the need to power computation. Since 2022, Amadeus has also enhanced Scope 3 emissions measurement. In 2025, its targets were reevaluated by the SBTI. Amadeus targets a 42% reduction by 2030 from a 2022 base year across Scopes 1 and 2, and net zero by 2050. Amadeus also has absolute-reduction targets of 90% from 2022 to 2050 across Scopes 1-3 and executive compensation is linked to ESG. Additionally, these ESG-related targets account for 12.0% of the total performance plan, with five specific sustainability targets included in the annual incentive scheme for all employees, including executive directors.

Utilities & Clean Energy

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Many of the companies within our Utilities & Clean Energy coverage play a crucial role in advancing global climate goals by developing, implementing or facilitating sustainable energy solutions. Our coverage spans a broad spectrum of companies, including renewable energy generators, electricity transmission and distribution (T&D) providers, gas T&D companies, water utilities, and integrated utilities that operate across the entire value chain - from thermal and renewable generation to T&D and energy retail. This diversity complicates direct comparisons, as companies differ significantly in their emissions profiles and starting points, largely depending on their position within the power and gas value chain. For instance, pure-play renewable generators have minimal emissions, electricity T&D companies primarily report Scope 2 emissions from line losses and have relatively low Scope 1 emissions, energy retailers typically have a large share of Scope 3 emissions, while firms with conventional power generation exhibit high Scope 1 emissions and can see dramatic improvements through plant closures or asset divestitures.

Rating the ambition of reduction targets

We assessed the ambition of our companies' targets using two equally weighted variables. The first is the reduction coefficient, which measures a company's ambition in reducing Scope 1 and 2 GHG emissions, considering the annualized reduction rate in their strategy weighted by their baseline emissions; this metric is then scored by quintile. However, this method may disadvantage companies with already low baseline emissions. The second variable is an ordinal target disclosure score, reflecting whether a company's interim reduction strategy covers relevant emission scopes and aligns with SBTi. Of the 26 companies assessed, 19 have set targets approved by SBTi, 3 have made commitments to SBTi and the remaining 4 have neither aligned their targets nor made commitments to SBTi.

Eight companies in our coverage achieved the highest possible score of 5, demonstrating best-in-class ambition. This group includes integrated companies such as Enel, EDP/R, Engie, Iberdrola and SSE, as well as E.ON (a regulated utility), RWE (a renewable generator), and Veolia. In contrast, Enagas and Italgas, two gas distribution companies, received the lowest scores of 2. Although these two companies set annualized reduction targets broadly in line with the peer average of 5%, their ambition scores were penalized due to their low baseline emissions, reflecting their smaller size, as well as their lack of alignment with SBTi standards.

Rating the execution of reduction targets

We also assessed the execution of our companies' targets using two equally weighted variables. The first is the performance rating, which measures the underperformance or overperformance of reported 2025 Scope 1 and 2 emissions against each company's announced target, based on a linear pathway; this metric is then scored by quintile. The second is a green capex score derived from the share of 2025 reported capex that is EU taxonomy aligned, or the latest available figure if 2025 data is unavailable. EU taxonomy-aligned capex refers to investments that make a substantial contribution to at least one of the climate and environmental objectives, while also ensuring no significant harm to the remaining objectives and meeting minimum standards on human rights and labor standards. This metric indicates the extent to which our companies are currently channeling investments into sustainable activities, offering an insight into their transition trajectory, whereas focusing solely on carbon emission metrics reflects only past performance and the existing asset mix. In our assessment, the range of taxonomy-aligned capex observed was from 6.3% (Enagas) to 100% (Elia), with an average of 78%.

Four companies in our coverage, Enel and EDP/R (integrated utilities), and RWE and Orsted (renewable generators), achieved the highest possible score of 5. These companies are all overperforming against their announced emission-reduction targets and have high levels of EU taxonomy-aligned capex, averaging at 96%. In contrast, Enagas, Snam and Veolia received the lowest score of 2. Enagas has increased its Scope 1+2 emissions by 3% compared to its baseline while having the lowest share of EU taxonomy-aligned capex. While Snam and Veolia are on track to achieve their announced targets, these two companies were mainly impacted by low levels of EU taxonomy-aligned capex at 34% and 47% respectively.

SG carbon score for the Stoxx 600 equity index

We calculate the carbon score for the STOXX 600 as of 22 May 2026, which includes 601 European companies. However, not all companies provide carbon data (GHG emissions for Scopes 1 and 2), and some companies report highly abnormal values that distort the final score calculation. These companies are therefore excluded from the scoring process.

In total, 22 companies are excluded, including 3 with particularly outlying values: EDP Renewables, Hannover Rueck SE, and Investor AB-B SHS. Although EDP Renewables is excluded as a standalone entity, its score is still reflected through its inclusion in the scoring of its parent company, EDP. This leaves 579 companies included in the scoring. It is worth noting that, of these 22 companies, only Syensqo and Zabka group are covered by Bernstein/Autonomous analysts.

List of STOXX 600 companies excluded from Carbon Scoring

TICKER	COMPANY NAME	EXCLUSION REASON
ABVX FP Equity	ABIVAX SA	Missing carbon data
AMV0 GY Equity	AUMOVIO SE	Missing carbon data
CSG NA Equity	CSG NV	Missing carbon data
CVC NA Equity	CVC CAPITAL PARTNERS PLC	Missing carbon data
EXO NA Equity	EXOR NV	Missing carbon data
GALD SE Equity	GALDERMA GROUP AG	Missing carbon data
KMAR NO Equity	KONGSBERG MARITIME AS	Missing carbon data
LTMC IM Equity	LOTTOMATICA GROUP SPA	Missing carbon data
MANTA FH Equity	MANDATUM OYJ	Missing carbon data
MICC NA Equity	MAGNUM ICE CREAM CO NV/THE	Missing carbon data
OCTVSDB SS Equity	OCTAVE INTELLIGENCE PLC-SDR	Missing carbon data
PAH3 GR Equity	PORSCHE AUTOMOBIL HLDG-PRF	Missing carbon data
R3NK GY Equity	RENK GROUP AG	Missing carbon data
SDZ SE Equity	SANDOZ GROUP AG	Missing carbon data
SUNB LN Equity	SUNBELT RENTALS HOLDINGS INC	Missing carbon data
SYENS BB Equity	SYENSQO SA	Missing carbon data
VSURE SS Equity	VERISURE PLC	Missing carbon data
ZAB PW Equity	ZABKA GROUP SA	Missing carbon data
ZEG LN Equity	ZEGONA COMMUNICATIONS PLC	Missing carbon data
EDPR PL Equity	EDP RENEWABLES SA	Outlier
HNR1 GY Equity	HANNOVER RUECK SE	Outlier
INVEB SS Equity	INVESTOR AB-B SHS	Outlier

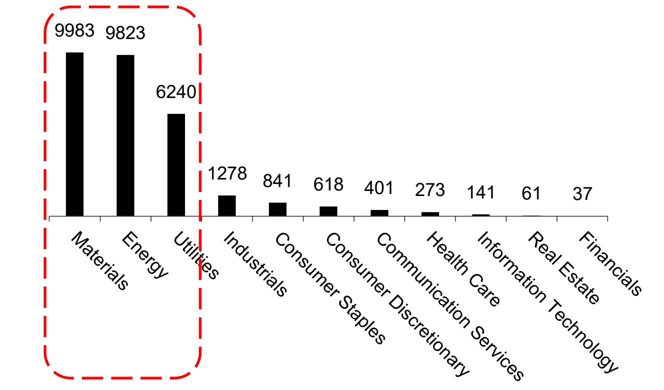
Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

2025 carbon intensity

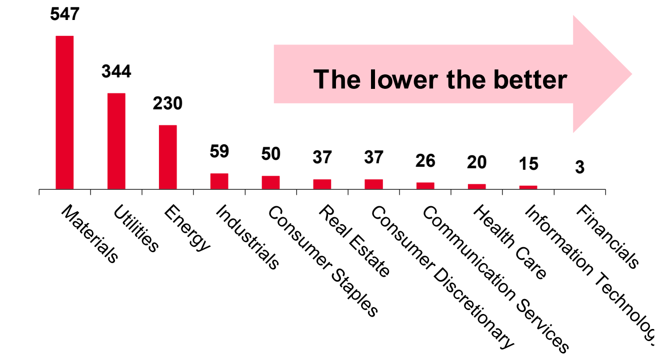
Carbon Intensity is calculated as the ratio of a company's GHG emissions in Scopes 1 + 2 in thousands of metric tons of carbon dioxide equivalent (CO2e) to its consolidated revenue in billions of euros, both in 2025.

The sectors with the highest carbon intensity are also those generating the largest volumes of GHG emissions. Materials, Energy, and Utilities dominate both in terms of absolute emissions—approximately 9,983, 9,823, and 6,240 thousand metric tons of CO_2e respectively—and in carbon intensity, with Materials and Energy posting the highest levels at 547 and 344, followed by Utilities at 230. By contrast, sectors such as Financials, Information Technology, and Health Care remain at much lower levels across both metrics. In the case of Materials, Energy, and Utilities, this link between carbon intensity and emissions reflects the intrinsic nature of the industries, where carbon-intensive activities directly translate into higher overall emissions.

Stoxx 600: 2025 GHG emissions in Scopes 1+2 on average by sector in thousands of metric tons of carbon dioxide equivalent (CO2e)



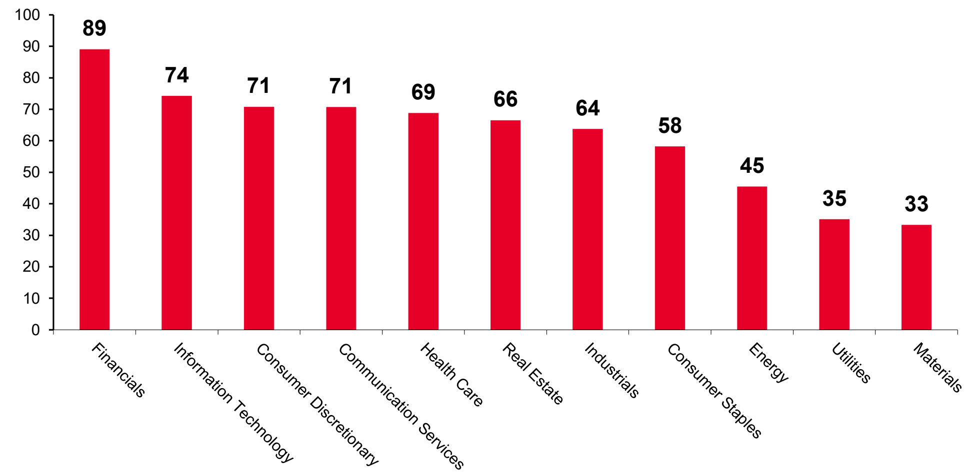
Stoxx 600: 2025 carbon intensity on average by sector



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

Financials stand out with the highest score at 89, followed by Information Technology at 74, indicating relatively lower carbon intensity profiles due mostly to their low GHG emissions. In contrast, more carbon-intensive industries have lower scores, with Energy at 45, Utilities at 35, and Materials at 33. **This distribution highlights a clear sector divide, where service-oriented and technology-driven sectors tend to perform better, while traditionally resource-intensive industries lag on carbon intensity.**

Carbon Stoxx 600: carbon intensity score on average by sector

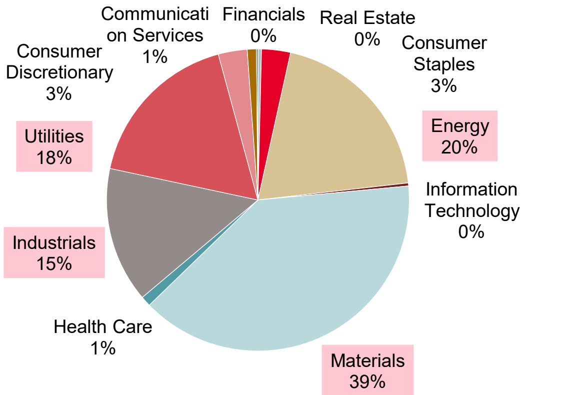


Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

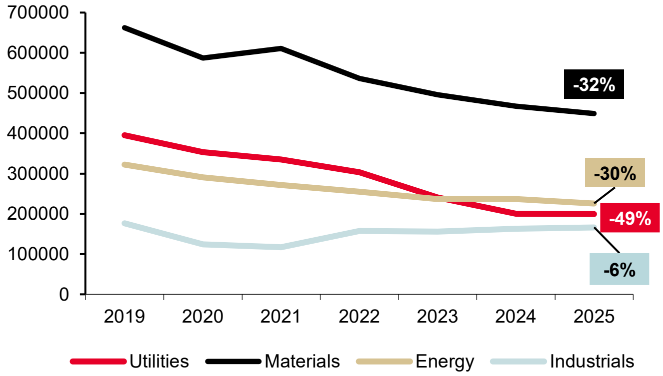
2019-25 carbon intensity trend

The data highlights a high concentration of emissions within a limited number of sectors in 2025, with Materials (39%), followed by Energy (20%), Utilities (18%), and Industrials (15%) combined accounting for the vast majority of the STOXX 600 carbon footprint, while all other sectors remain marginal contributors. Over the 2019–2025 period, an overall downward trend emerges among these highest-emitting sectors, reflecting ongoing decarbonization efforts. The reduction is particularly significant for Utilities (-49%), while Materials (-32%) and Energy (-30%) also delivered substantial declines. The decrease for Industrials, however, was more modest (-6%), suggesting a slower transition.

Sector breakdown of STOXX 600 Scope 1 + 2 emissions (2025)



Change in Scope 1 + 2 GHG emissions (in thousands of metric tons of carbon dioxide equivalent (CO2e)) in top-polluting STOXX 600 sectors (2019–2025)

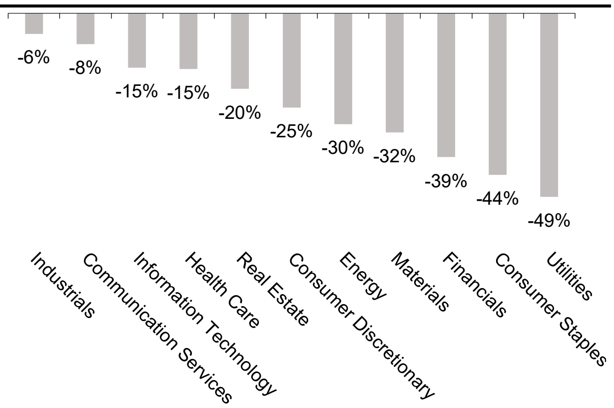


Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

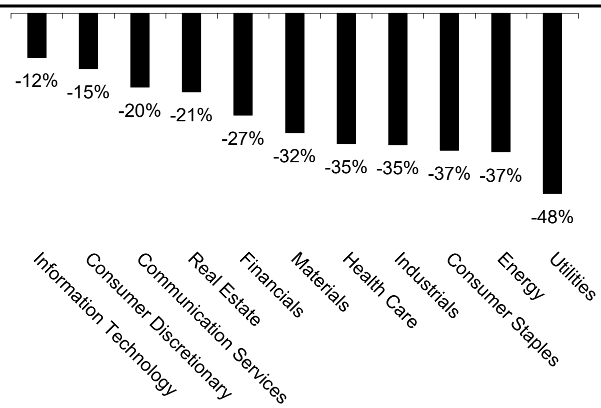
Overall, both absolute emissions and carbon intensity decline across all STOXX 600 sectors between 2019 and 2025, reflecting broad-based progress in decarbonization. **The biggest drops were in traditionally high-emitting sectors such as Utilities, Consumer Staples, and Energy, while lower-emitting sectors also saw steady, albeit more moderate, improvements**. However, Industrials stand out: although their absolute GHG emissions decline relatively less compared to other sectors, their carbon intensity improves more significantly, leading to a relative rise in their positioning when considering intensity trends.

Bernstein analysts highlight the strategic role of the utilities sector in the energy transition, noting that ***"Many of the companies within our Utilities & Clean Energy coverage play a crucial role in advancing global climate goals by developing, implementing or facilitating sustainable energy solutions."***

Stoxx 600: GHG emissions in Scopes 1 & 2, trend 2019-2025 (%)



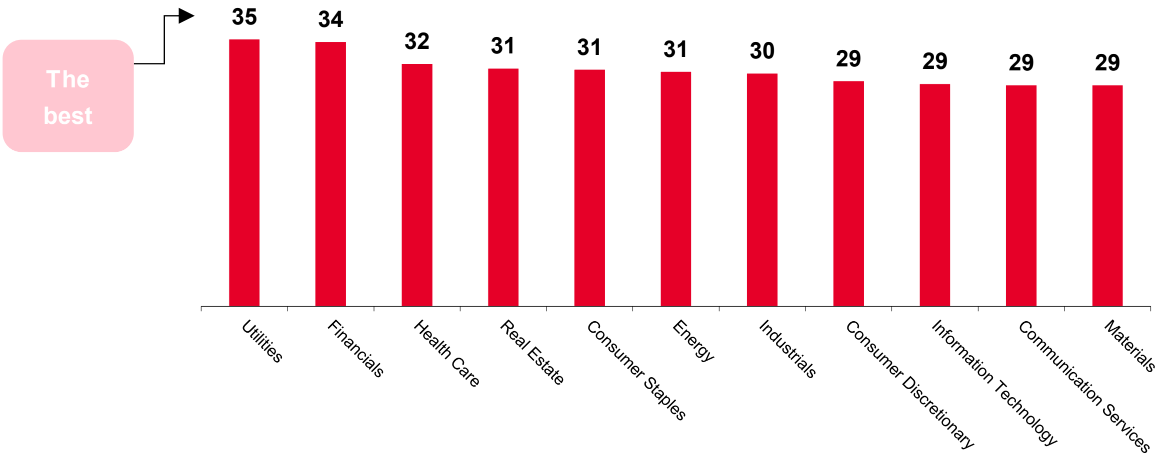
Stoxx 600: Carbon intensity in Scopes 1 & 2, trend 2019-2025 (%)



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

The results show that Utilities scores best, which is consistent with its significant drop in carbon intensity and leading trend performance. In contrast, Financials ranks second in the final score, despite a weaker average trend. This is because the scoring methodology captures the distribution of individual values rather than the sector average. As a result, Financials enjoys the advantage of many moderately acceptable values after transformation, which improves its relative position in the ranking.

Stoxx 600: Carbon intensity trend 2019-2025, score by sector



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg

2025 ambition score

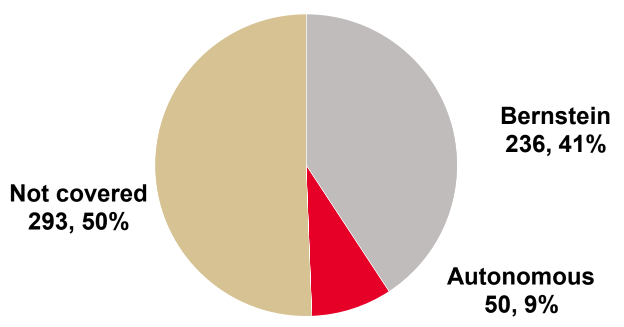
Coverage spans roughly half of the STOXX 600 universe, with 579 companies included in the carbon score calculation. In 2025, 238 ratings were received from Bernstein, while Autonomous ambition ratings were carried over from 2024**. Across sectors, ambition levels remain broadly consistent at the higher end, with Real Estate standing out as the top-scoring sector**, slightly ahead of the rest. This leading positioning is also supported by Bernstein analysts, who note:

"We think that ESG disclosure by the listed real estate sector is of a high standard, especially among the larger landlords, which form the basis of our 26 companies under coverage and is reflected in our high scores for many companies."

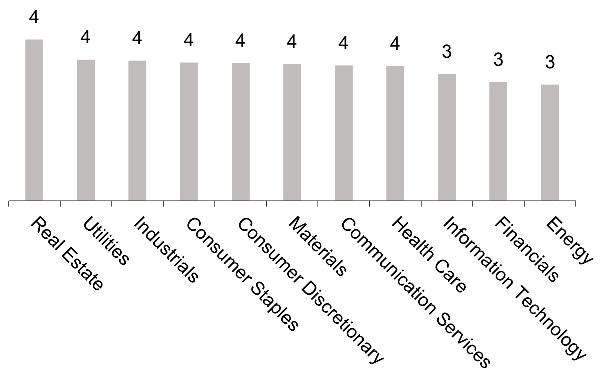
The high ambition level observed in the Real Estate sector can be explained by several intrinsic factors, including significant regulatory and environmental pressure, a well-established ESG disclosure framework, and the adoption of demanding carbon-reduction targets. As highlighted by Bernstein analysts: "***The Real Estate sector contributes approximately 42% to CO2 emissions globally…***", underlining the scale of the challenge and the resulting pressure to act.

In addition, the sector boasts a robust ESG framework, as ***"ESG disclosure by the listed real estate sector is of a high standard…"***, enabling better tracking and management of emissions. Finally, companies demonstrate ambitious target-setting practices, with analysts noting that their approach includes ***"awarding a higher score to companies that include Scope 3… and the target percentage reduction and time to achieve the target,"*** reflecting a strong commitment to comprehensive and accelerated decarbonization.

Bernstein / Autonomous coverage of 579 STOXX 600 companies used for the carbon score calculation - 2025



Stoxx 600: Bernstein / Autonomous ambition ratings by sector (1: the worst, 5: the best)

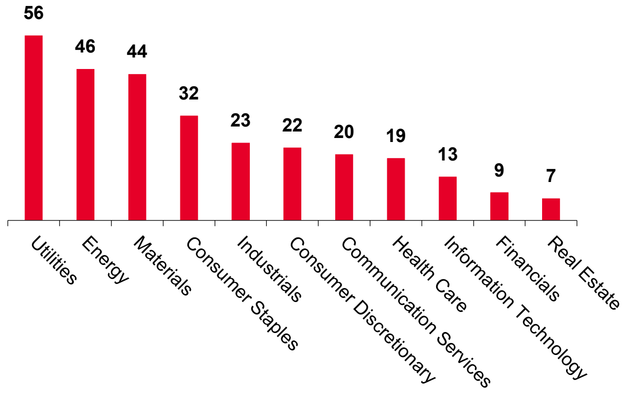


Source: SG Cross Asset Research/Invest in the Carbon Transition, Bernstein/Autonomous Research

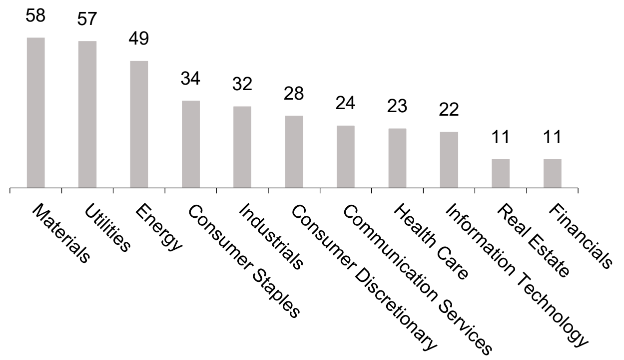
Utilities, Energy, and Materials stand out as the leading sectors in both reduction efforts and ambition. Despite being among the most emission-intensive industries, these sectors are demonstrating the strongest commitment to change by delivering the highest reduction scores, with Utilities at 56, Energy at 46, and Materials at 44.

At the same time, they also top the ambition ranking, with Materials (58), Utilities (57), and Energy (49) setting the most forward-looking targets. This dual leadership highlights the fact that the sectors with the greatest environmental impact are also those deploying the most effective strategies and showing the highest level of commitment to decarbonization, positioning them at the forefront of the transition.

Stoxx 600: Reduction score by sector



Stoxx 600: Ambition score by sector



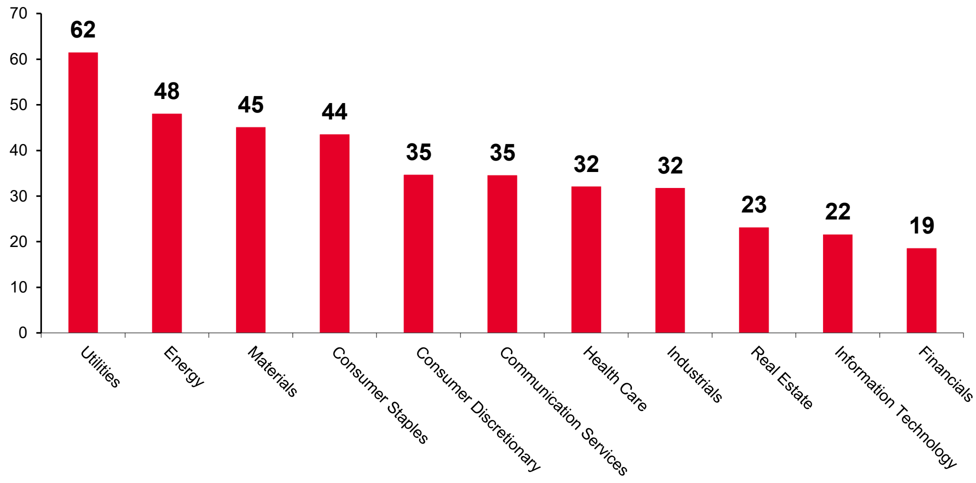
Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Utilities, Energy, and Materials clearly lead the ambition ranking with scores of 62, 48, and 45, showing that despite being the most carbon-intensive sectors, they are making the strongest efforts by combining high reduction performance with strong future commitments. Consumer Staples follows closely at 44, while sectors like Consumer Discretionary and Communication Services sit in the middle at 35, reflecting more moderate progress.

At the lower end, Real Estate (23), Information Technology (22), and Financials (19) lag behind, indicating both weaker current reductions and lower ambition. Overall, the pattern suggests that the most polluting sectors are also those taking the most decisive action and deploying the most effective strategies.

In the case of Real Estate, this lower score can be partly explained by more long-term reduction strategies, with companies typically targeting gradual improvements over extended time horizons rather than aggressive short-term reductions, resulting in lower reduction coefficients despite overall sound decarbonization plans.

Stoxx 600: Ambition score by sector (the higher the better)



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Utilities ranks second on average rating, first on reduction, and second overall on ambition, partly driven by a number of companies within the sector demonstrating strong decarbonization commitments, as highlighted by analysts:

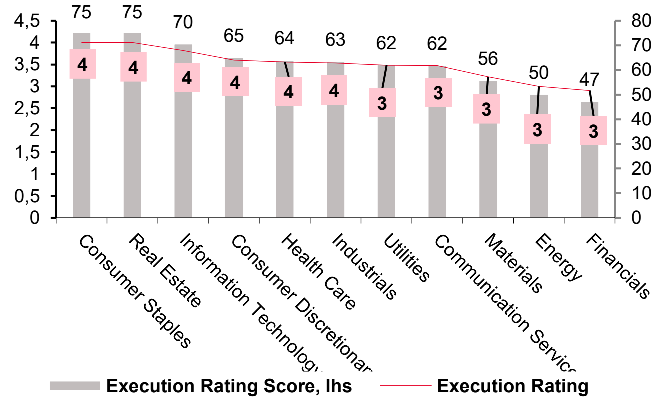
– ***"Of the 26 companies assessed, 19 have set targets approved by SBTi, 3 have made commitments to SBTi and the remaining 4 have neither aligned their targets nor made commitments to SBTi."***

– ***"Eight companies in our coverage achieved the highest possible score of 5, demonstrating best-in-class ambition."***

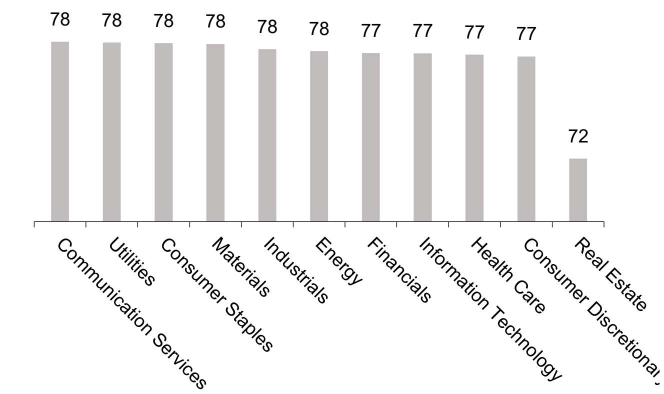
2025 execution score

The distribution of ratings broadly follows the same pattern as the execution rating scores across sectors, with a consistent alignment on average. Consumer staples and real estate stand out as the best-performing sectors, with very good execution scores and favorable ratings. In contrast, financials has the weakest execution performance, reflecting lower scores, which can be explained by the complexity and specific nature of its activities.

Stoxx 600: Execution ratings (rhs) & execution rating score (lhs) by sector (1: the worst, 5: the best)



Stoxx 600: Performance score by sector (the higher the better)

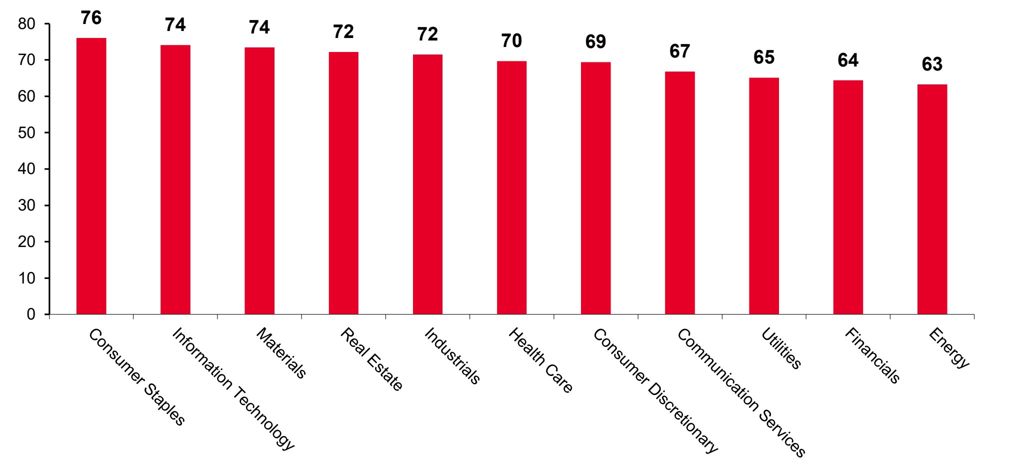


Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Bernstein/Autonomous Research

The execution scores are relatively close across sectors, indicating a fairly limited dispersion overall. Consumer staples stands out as the top-performing sector, leading with the highest score. Most other sectors cluster within a narrow range, reflecting broadly similar execution levels.

It is also worth noting that the score for financials is somewhat flattered by the use of performance scores for companies without ratings. These performance scores are relatively high, so they contribute to lifting the overall execution score of the sector despite weaker underlying execution dynamics.

Stoxx 600 Execution score by sector (the higher the better)



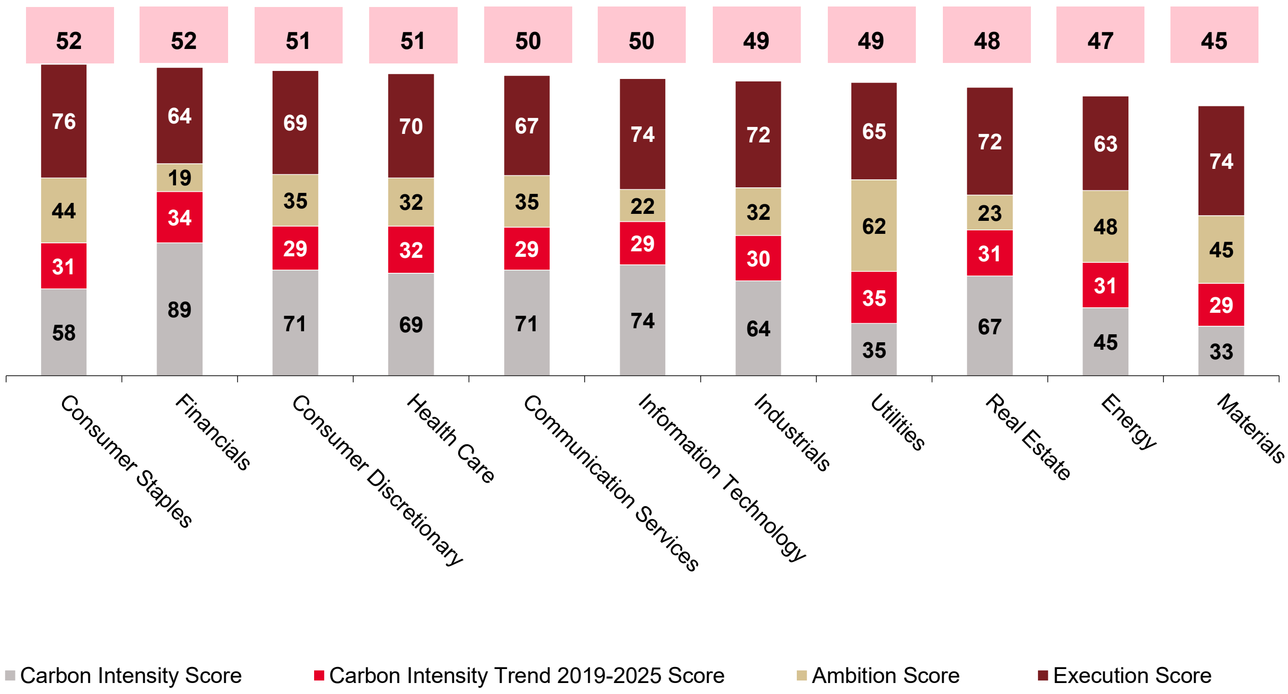
Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Bernstein/Autonomous Research

2025 carbon score

Finally, the STOXX 600 carbon score is calculated as the average of four sub-scores: carbon intensity, carbon intensity trend, ambition, and execution. **What is particularly insightful is the combined score, as a weaker component can be offset by better scores in other categories**.

For instance, Utilities has a relatively low carbon intensity scores but scores more highly on ambition and execution, which helps compensate overall. In contrast, Consumer Staples ranks at the top of the index, largely driven by a significantly higher execution score compared to other sectors.

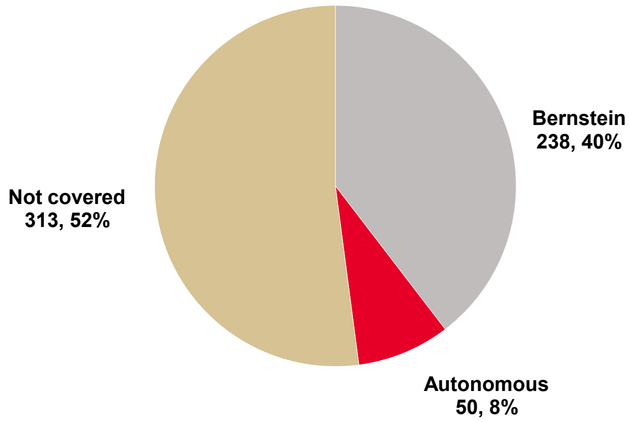
Stoxx 600: SG carbon score average by sector (the higher the better)



Source: SG Cross Asset Research/*Invest in the Carbon Transition*, *Bloomberg*, Bernstein/Autonomous Research

List of companies in the Stoxx600 index covered by Bernstein/Autonomous

Stoxx 600: Bernstein coverage (22 May 2026) & Autonomous coverage (19 June 2025)



Source: SG Cross Asset Research/Invest in the Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Coverage list (1/6)

Ticker	Company name	Sector
BG AV Equity	BAWAG GROUP AG	Financials
DNP PW Equity	DINO POLSKA SA	Consumer Staples
BP/ LN Equity	BP PLC	Energy
DSY FP Equity	DASSAULT SYSTEMES SE	Information Technology
REY IM Equity	REPLY SPA	Information Technology
PEO PW Equity	BANK PEKAO SA	Financials
RBI AV Equity	RAIFFEISEN BANK INTERNATIONA	Financials
BAS GY Equity	BASF SE	Materials
ALC SE Equity	ALCON INC	Health Care
AM FP Equity	DASSAULT AVIATION SA	Industrials
ARGX BB Equity	ARGENX SE	Health Care
RTO LN Equity	RENTOKIL INITIAL PLC	Industrials
RUI FP Equity	RUBIS	Utilities
AI FP Equity	AIR LIQUIDE SA	Materials
BEI GY Equity	BEIERSDORF AG	Consumer Staples
ENG SQ Equity	ENAGAS SA	Utilities
IG IM Equity	ITALGAS SPA	Utilities
ORK NO Equity	ORKLA ASA	Consumer Staples
RKT LN Equity	RECKITT BENCKISER GROUP PLC	Consumer Staples
TTE FP Equity	TOTALENERGIES SE	Energy
HSBA LN Equity	HSBC HOLDINGS PLC	Financials
SHBA SS Equity	SVENSKA HANDELSBANKEN-A SHS	Financials
BIRG ID Equity	BANK OF IRELAND GROUP PLC	Financials
PKO PW Equity	PKO BANK POLSKI SA	Financials
GMAB DC Equity	GENMAB A/S	Health Care
SUBC NO Equity	SUBSEA 7 SA	Energy
BPE IM Equity	BPER BANCA SPA	Financials
AIBG ID Equity	AIB GROUP PLC	Financials
ATCOA SS Equity	ATLAS COPCO AB-A SHS	Industrials
KNEBV FH Equity	KONE OYJ-B	Industrials
ENR GY Equity	SIEMENS ENERGY AG	Industrials
ABI BB Equity	ANHEUSER-BUSCH INBEV SA/NV	Consumer Staples
PRU LN Equity	PRUDENTIAL PLC	Financials
TEN IM Equity	TENARIS SA	Energy
STAN LN Equity	STANDARD CHARTERED PLC	Financials
BNP FP Equity	BNP PARIBAS	Financials
LISP SE Equity	CHOCOLADEFABRIKEN LINDT-PC	Consumer Staples
ORSTED DC Equity	ORSTED A/S	Utilities
SCR FP Equity	SCOR SE	Financials
UCG IM Equity	UNICREDIT SPA	Financials
EMG LN Equity	MAN GROUP PLC/JERSEY	Financials
BMW GY Equity	BAYERISCHE MOTOREN WERKE AG	Consumer Discretionary
ACA FP Equity	CREDIT AGRICOLE SA	Financials
ADM LN Equity	ADMIRAL GROUP PLC	Financials
AIR FP Equity	AIRBUS SE	Industrials
BA/ LN Equity	BAE SYSTEMS PLC	Industrials
BC IM Equity	BRUNELLO CUCINELLI SPA	Consumer Discretionary
BME LN Equity	B&M EUROPEAN VALUE RETAIL PL	Consumer Discretionary
BNZL LN Equity	BUNZL PLC	Industrials
BT/A LN Equity	BT GROUP PLC	Communication Services
BVI FP Equity	BUREAU VERITAS SA	Industrials
DEMANT DC Equity	DEMANT A/S	Health Care

Source: SG Cross Asset Research/Invest in The Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Coverage list (2/6)

Ticker	Company name	Sector
DTE GY Equity	DEUTSCHE TELEKOM AG-REG	Communication Services
ELISA FH Equity	ELISA OYJ	Communication Services
ERICB SS Equity	ERICSSON LM-B SHS	Information Technology
FR FP Equity	VALEO	Consumer Discretionary
GRF SQ Equity	GRIFOLS SA	Health Care
HLN LN Equity	HALEON PLC	Health Care
INF LN Equity	INFORMA PLC	Communication Services
ITRK LN Equity	INTERTEK GROUP PLC	Industrials
LDO IM Equity	LEONARDO SPA	Industrials
LR FP Equity	LEGRAND SA	Industrials
MONC IM Equity	MONCLER SPA	Consumer Discretionary
NDX1 GY Equity	NORDEX SE	Industrials
NOKIA FH Equity	NOKIA OYJ	Information Technology
NOVOB DC Equity	NOVO NORDISK A/S-B	Health Care
OR FP Equity	L'OREAL	Consumer Staples
ORA FP Equity	ORANGE	Communication Services
P911 GY Equity	DR ING HC F PORSCHE AG	Consumer Discretionary
PNN LN Equity	PENNON GROUP PLC	Utilities
RAA GY Equity	RATIONAL AG	Industrials
RAND NA Equity	RANDSTAD NV	Industrials
SBRY LN Equity	SAINSBURY (J) PLC	Consumer Staples
SFZN SE Equity	SIEGFRIED HOLDING AG-REG	Health Care
SN/ LN Equity	SMITH & NEPHEW PLC	Health Care
SOON SE Equity	SONOVA HOLDING AG-REG	Health Care
STLAM IM Equity	STELLANTIS NV	Consumer Discretionary
SW FP Equity	SODEXO SA	Consumer Discretionary
TEF SQ Equity	TELEFONICA SA	Communication Services
TEL NO Equity	TELENOR ASA	Communication Services
TELIA SS Equity	TELIA CO AB	Communication Services
TEP FP Equity	TELEPERFORMANCE	Industrials
ULVR LN Equity	UNILEVER PLC	Consumer Staples
UU/ LN Equity	UNITED UTILITIES GROUP PLC	Utilities
VER AV Equity	VERBUND AG	Utilities
VK FP Equity	VALLOUREC SA	Energy
VOD LN Equity	VODAFONE GROUP PLC	Communication Services
ZAL GY Equity	ZALANDO SE	Consumer Discretionary
AGS BB Equity	AGEAS	Financials
LLOY LN Equity	LLOYDS BANKING GROUP PLC	Financials
SWEDA SS Equity	SWEDBANK AB - A SHARES	Financials
ISP IM Equity	INTESA SANPAOLO	Financials
ABN NA Equity	ABN AMRO BANK NV-CVA	Financials
HSX LN Equity	HISCOX LTD	Financials
AALB NA Equity	AALBERTS NV	Industrials
BARC LN Equity	BARCLAYS PLC	Financials
CFR SE Equity	CIE FINANCIERE RICHEMO-A REG	Consumer Discretionary
CNA LN Equity	CENTRICA PLC	Utilities
COLOB DC Equity	COLOPLAST-B	Health Care
DIE BB Equity	D'IETEREN GROUP	Consumer Discretionary
ELI BB Equity	ELIA GROUP SA/NV	Utilities
ERF FP Equity	EUROFINS SCIENTIFIC	Health Care
EVD GY Equity	CTS EVENTIM AG & CO KGAA	Communication Services
JYSK DC Equity	JYSKE BANK-REG	Financials
NXT LN Equity	NEXT PLC	Consumer Discretionary
RACE IM Equity	FERRARI NV	Consumer Discretionary
RHM GY Equity	RHEINMETALL AG	Industrials

Source: SG Cross Asset Research/Invest in The Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Coverage list (3/6)

Ticker	Company name	Sector
RIO LN Equity	RIO TINTO PLC	Materials
SHELL NA Equity	SHELL PLC	Energy
SPM IM Equity	SAIPEM SPA	Energy
UMG NA Equity	UNIVERSAL MUSIC GROUP NV	Communication Services
VWS DC Equity	VESTAS WIND SYSTEMS A/S	Industrials
EBS AV Equity	ERSTE GROUP BANK AG	Financials
AV/ LN Equity	AVIVA PLC	Financials
BEZ LN Equity	BEAZLEY PLC	Financials
MNG LN Equity	M&G PLC	Financials
BAMI IM Equity	BANCO BPM SPA	Financials
DANSKE DC Equity	DANSKE BANK A/S	Financials
BMPS IM Equity	BANCA MONTE DEI PASCHI SIENA	Financials
NEX FP Equity	NEXANS SA	Industrials
ASSAB SS Equity	ASSA ABLOY AB-B	Industrials
SCHP SE Equity	SCHINDLER HOLDING-PART CERT	Industrials
CPR IM Equity	DAVIDE CAMPARI-MILANO NV	Consumer Staples
PRY IM Equity	PRYSMIAN SPA	Industrials
RXL FP Equity	REXEL SA	Industrials
AGN NA Equity	AEGON LTD	Financials
DNB NO Equity	DNB BANK ASA	Financials
WPP LN Equity	WPP PLC	Communication Services
ENX FP Equity	EURONEXT NV	Financials
KBC BB Equity	KBC GROUP NV	Financials
LSEG LN Equity	LONDON STOCK EXCHANGE GROUP	Financials
NEXI IM Equity	NEXI SPA	Financials
AAL LN Equity	ANGLO AMERICAN PLC	Materials
ABF LN Equity	ASSOCIATED BRITISH FOODS PLC	Consumer Staples
AC FP Equity	ACCOR SA	Consumer Discretionary
ADP FP Equity	ADP	Industrials
AKE FP Equity	ARKEMA	Materials
AKZA NA Equity	AKZO NOBEL N.V.	Materials
ARCAD NA Equity	ARCADIS NV	Industrials
BESI NA Equity	BE SEMICONDUCTOR INDUSTRIES	Information Technology
BN FP Equity	DANONE	Consumer Staples
BRBY LN Equity	BURBERRY GROUP PLC	Consumer Discretionary
CA FP Equity	CARREFOUR SA	Consumer Staples
CAP FP Equity	CAPGEMINI SE	Information Technology
CARLB DC Equity	CARLSBERG AS-B	Consumer Staples
CLNX SQ Equity	CELLNEX TELECOM SA	Communication Services
COV FP Equity	COVIVIO	Real Estate
CPG LN Equity	COMPASS GROUP PLC	Consumer Discretionary
DGE LN Equity	DIAGEO PLC	Consumer Staples
DIM FP Equity	SARTORIUS STEDIM BIOTECH	Health Care
EL FP Equity	ESSILORLUXOTTICA	Health Care
ELE SQ Equity	ENDESA SA	Utilities
ELIS FP Equity	ELIS SA	Industrials
FDJU FP Equity	FDJ UNITED	Consumer Discretionary
FORTUM FH Equity	FORTUM OYJ	Utilities
GJF NO Equity	GJENSIDIGE FORSIKRING ASA	Financials
GTT FP Equity	GAZTRANSPORT ET TECHNIGA SA	Energy
HEIO NA Equity	HEINEKEN HOLDING NV	Consumer Staples
HO FP Equity	THALES SA	Industrials
IAG LN Equity	INTL CONSOLIDATED AIRLINE-DI	Industrials
IHG LN Equity	INTERCONTINENTAL HOTELS GROU	Consumer Discretionary

Source: SG Cross Asset Research/Invest in The Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Coverage list (4/6)

Ticker	Company name	Sector
JD/ LN Equity	JD SPORTS FASHION PLC	Consumer Discretionary
JMT PL Equity	JERONIMO MARTINS	Consumer Staples
KER FP Equity	KERING	Consumer Discretionary
KPN NA Equity	KONINKLIJKE KPN NV	Communication Services
KSP ID Equity	KINGSPAN GROUP PLC	Industrials
LOTB BB Equity	LOTUS BAKERIES	Consumer Staples
MC FP Equity	LVMH MOET HENNESSY LOUIS VUI	Consumer Discretionary
ML FP Equity	MICHELIN (CGDE)	Consumer Discretionary
MRL SQ Equity	MERLIN PROPERTIES SOCIMI SA	Real Estate
NG/ LN Equity	NATIONAL GRID PLC	Utilities
NN NA Equity	NN GROUP NV	Financials
NOVN SE Equity	NOVARTIS AG-REG	Health Care
NTGY SQ Equity	NATURGY ENERGY GROUP SA	Utilities
NWG LN Equity	NATWEST GROUP PLC	Financials
PIRC IM Equity	PIRELLI & C SPA	Consumer Discretionary
PNDORA DC Equity	PANDORA A/S	Consumer Discretionary
PUM GY Equity	PUMA SE	Consumer Discretionary
RED SQ Equity	REDEIA CORP SA	Utilities
REL LN Equity	RELX PLC	Industrials
RNO FP Equity	RENAULT SA	Consumer Discretionary
ROP SE Equity	ROCHE HOLDING AG	Health Care
SAF FP Equity	SAFRAN SA	Industrials
SAN FP Equity	SANOFI	Health Care
SAP GY Equity	SAP SE	Information Technology
SBMO NA Equity	SBM OFFSHORE NV	Energy
SOP FP Equity	SOPRA STERIA GROUP	Information Technology
SRG IM Equity	SNAM SPA	Utilities
STMN SE Equity	STRAUMANN HOLDING AG-REG	Health Care
SVT LN Equity	SEVERN TRENT PLC	Utilities
TRN IM Equity	TERNA-RETE ELETTRICA NAZIONA	Utilities
VOLVB SS Equity	VOLVO AB-B SHS	Industrials
VOW3 GY Equity	VOLKSWAGEN AG-PREF	Consumer Discretionary
G IM Equity	GENERALI	Financials
EZJ LN Equity	EASYJET PLC	Industrials
III LN Equity	3I GROUP PLC	Financials
SAMPO FH Equity	SAMPO OYJ-A SHS	Financials
AZA SS Equity	AVANZA BANK HOLDING AB	Financials
STJ LN Equity	ST JAMES'S PLACE PLC	Financials
AMBUB DC Equity	AMBU A/S-B	Health Care
BLND LN Equity	BRITISH LAND CO PLC	Real Estate
CTEC LN Equity	CONVATEC GROUP PLC	Health Care
DHER GY Equity	DELIVERY HERO SE	Consumer Discretionary
DTG GY Equity	DAIMLER TRUCK HOLDING AG	Industrials
EDEN FP Equity	EDENRED	Financials
EQNR NO Equity	EQUINOR ASA	Energy
GET FP Equity	GETLINK SE	Industrials
GFC FP Equity	GECINA SA	Real Estate
IFX GY Equity	INFINEON TECHNOLOGIES AG	Information Technology
LGEN LN Equity	LEGAL & GENERAL GROUP PLC	Financials
LONN SE Equity	LONZA GROUP AG-REG	Health Care
MBG GY Equity	MERCEDES-BENZ GROUP AG	Consumer Discretionary
RMS FP Equity	HERMES INTERNATIONAL	Consumer Discretionary
RR/ LN Equity	ROLLS-ROYCE HOLDINGS PLC	Industrials
TE FP Equity	TECHNIP ENERGIES NV	Energy

Source: SG Cross Asset Research/Invest in The Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Coverage list (5/6)

Ticker	Company name	Sector
TUI1 GY Equity	TUI AG	Consumer Discretionary
UHR SE Equity	SWATCH GROUP AG/THE-BR	Consumer Discretionary
URW FP Equity	UNIBAIL-RODAMCO-WESTFIELD	Real Estate
WDP BB Equity	WAREHOUSES DE PAUW SCA	Real Estate
AED BB Equity	AEDIFICA	Real Estate
TEG GY Equity	TAG IMMOBILIEN AG	Real Estate
SU FP Equity	SCHNEIDER ELECTRIC SE	Industrials
CCH LN Equity	COCA-COLA HBC AG-DI	Consumer Staples
DG FP Equity	VINCI SA	Industrials
FGR FP Equity	EIFFAGE	Industrials
RBREW DC Equity	ROYAL UNIBREW	Consumer Staples
ROCKB DC Equity	ROCKWOOL A/S-B SHS	Industrials
SGO FP Equity	COMPAGNIE DE SAINT GOBAIN	Industrials
ABBN SE Equity	ABB LTD-REG	Industrials
MKS LN Equity	MARKS & SPENCER GROUP PLC	Consumer Staples
SGRO LN Equity	SEGRO PLC	Real Estate
CS FP Equity	AXA SA	Financials
TRYG DC Equity	TRYG A/S	Financials
AD NA Equity	KONINKLIJKE AHOLD DELHAIZE N	Consumer Staples
AMS SQ Equity	AMADEUS IT GROUP SA	Consumer Discretionary
ANTO LN Equity	ANTOFAGASTA PLC	Materials
AUTO LN Equity	AUTOTRADER GROUP PLC	Communication Services
AZN LN Equity	ASTRAZENECA PLC	Health Care
BOL SS Equity	BOLIDEN AB	Materials
DHL GY Equity	DHL GROUP	Industrials
DSV DC Equity	DSV A/S	Industrials
EDP PL Equity	EDP SA	Utilities
ENEL IM Equity	ENEL SPA	Utilities
ENGI FP Equity	ENGIE	Utilities
ENI IM Equity	ENI SPA	Energy
EOAN GY Equity	E.ON SE	Utilities
GETIB SS Equity	GETINGE AB-B SHS	Health Care
GSK LN Equity	GSK PLC	Health Care
HEIA NA Equity	HEINEKEN NV	Consumer Staples
HMB SS Equity	HENNES & MAURITZ AB-B SHS	Consumer Discretionary
IBE SQ Equity	IBERDROLA SA	Utilities
ITV LN Equity	ITV PLC	Communication Services
ITX SQ Equity	INDUSTRIA DE DISENO TEXTIL	Consumer Discretionary
KGX GY Equity	KION GROUP AG	Industrials
KNIN SE Equity	KUEHNE + NAGEL INTL AG-REG	Industrials
LAND LN Equity	LAND SECURITIES GROUP PLC	Real Estate
LEG GY Equity	LEG IMMOBILIEN SE	Real Estate
LI FP Equity	KLEPIERRE	Real Estate
LIGHT NA Equity	SIGNIFY NV	Industrials
MAERSKB DC Equity	AP MOLLER-MAERSK A/S-B	Industrials
MRK GY Equity	MERCK KGAA	Health Care
NESN SE Equity	NESTLE SA-REG	Consumer Staples
PHIA NA Equity	KONINKLIJKE PHILIPS NV	Health Care
PUB FP Equity	PUBLICIS GROUPE	Communication Services
RI FP Equity	PERNOD RICARD SA	Consumer Staples
RMV LN Equity	RIGHTMOVE PLC	Communication Services

Source: SG Cross Asset Research/Invest in The Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Coverage list (6/6)

Ticker	Company name	Sector
RWE GY Equity	RWE AG	Utilities
RYA ID Equity	RYANAIR HOLDINGS PLC	Industrials
SGE LN Equity	SAGE GROUP PLC/THE	Information Technology
SHL GY Equity	SIEMENS HEALTHINEERS AG	Health Care
SIE GY Equity	SIEMENS AG-REG	Industrials
SPIE FP Equity	SPIE SA	Industrials
SSE LN Equity	SSE PLC	Utilities
TEL2B SS Equity	TELE2 AB-B SHS	Communication Services
TSCO LN Equity	TESCO PLC	Consumer Staples
VIE FP Equity	VEOLIA ENVIRONNEMENT	Utilities
VNA GY Equity	VONOVIA SE	Real Estate
WKL NA Equity	WOLTERS KLUWER	Industrials
WTB LN Equity	WHITBREAD PLC	Consumer Discretionary
FBK IM Equity	FINECOBANK SPA	Financials
GLEN LN Equity	GLENCORE PLC	Materials
LHA GY Equity	DEUTSCHE LUFTHANSA-REG	Industrials
ASML NA Equity	ASML HOLDING NV	Information Technology
AXFO SS Equity	AXFOOD AB	Consumer Staples
EXPN LN Equity	EXPERIAN PLC	Industrials
ASRNL NA Equity	ASR NEDERLAND NV	Financials
SYENS BB Equity	SYENSQO SA	Materials
ZAB PW Equity	ZABKA GROUP SA	Consumer Staples

Source: SG Cross Asset Research/Invest in The Carbon Transition, Bloomberg, Bernstein/Autonomous Research

Report completed on 10 July 12:36 CET

List of covered companies of the Bernstein equity research analyst(s) for which they certify their views under the US Securities and Exchange Commission's Regulation Analyst Certification: see page 73 onwards.

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RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

Bernstein brand

Bernstein rates stocks based on forecasts of relative performance for the next 6-12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index (EDM) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid-Cap Price Return Index USD (JP) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid-Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

· Outperform: Stock will outpace the market index by more than 15 pp

· Market-Perform: Stock will perform in line with the market index to within +/-15 pp

· Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments

Autonomous brand

The Autonomous brand rates stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of ratings:

· Outperform (OP): Stock will outpace the relevant index by more than 10 pp

· Neutral (N): Stock will perform in line with the market index to within +/-10 pp

· Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas. Not Rated (NR) is applied to companies that are not under formal coverage.

Horizon and classification

For both brands, recommendations are based on a 12-month time horizon.

DISTRIBUTION OF RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS/ RATINGS AND PRICE TARGET HISTORY

This research publication covers six or more companies. For price chart and other company disclosures, please visit https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action or you can write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

CONFLICTS OF INTEREST

SG and/or an affiliate(s) acted as active Joint Bookrunner in Aeroports de Paris S.A bond issuance (EUR 500 M / 9 yr)

SG and/or an affiliate(s) acted as Joint Lead Manager in Banco BPM S.p.A.'s bond issue (EUR 500m (WNG) 11.5NC6.5 Green)

SG and/or its affiliates acted as Joint Lead Manager in Banca Monte Dei Paschi di Siena's bond issue (€500mn WNG 3NC2 Senior Preferred, RegS)

SG and/or its affiliates acted as Global Coordinator in the Coca-Cola HBC Finance BV's new issue (EUR, RegS, 8y) guaranteed by Coca-Cola HBC AG

SG and/or its affiliates acted as Global Coordinator in Dassault Systemes's bond issuance (EUR benchmark 5Y)

SG and/or an affiliate(s) acted as Join Bookrunner in Erste Group Bank's bond issue, and acted as Dealer Manager on the concurrent tender offer on their outstanding EUR 500m 5.125% PerpNC-Oct-25 Additional Tier 1 (ISIN XS1961057780).

SG and/or an affiliate(s) acted as Joint Bookrunner in Erste Group Bank AG's Green Bond issue (EUR Benchmark RegS Perpetual 7.3Y Additional Tier 1) and as Dealer Manager for Erste Group Bank AG's bond tender offer (Targeted notes: XS2108494837 & AT0000A2L583).

SG and/or an affiliate acted as Joint Bookrunner in Elis' bond issue (EUR 400m, 6y).

SG and/or an affiliate(s) acted as Joint Active Bookrunner in ENGIE SA's Dual Tranche Green bond issue (EUR Benchmark, 6yr, 12yr).

SG and/or its affiliates acted as Joint Active Bookrunner in Mercedes-Benz International Finance BV's dual tranche bonds issue guaranteed by Mercedez-Benz Group AG (EUR, long 3y & 6y, RegS)

SG and/or an affiliate acted as Global Coordinator in Compagnie Générale des Établissements Michelin SCA's bond issue (EUR 1bn, 7y and 12y).

SG and/or an affiliate(s) is acting as passive bookrunner in relation to Novartis' USD6bn multi-part bond offering.

SG and/or its affiliates are advising Bouygues on its acquisition with Free-Groupe iliad and Orange part of Altice's activities in France

SG and/or its affiliates acted as Joint Bookrunner for Renault's bond issue (EUR, 5y).

SG and/or an affiliate(s) is acting as advisor to Sopra Steria in relation to the proposed disposal of its banking software activities to Axway Software.

SG and/or affiliate(s) acted as Joint bookrunner in Technip Energies N.V.'s bond issue (EUR500M WNG 7Y).

SG acted as Active Joint Bookrunner in UNIVERSAL MUSIC GROUP's bond issue (dual-tranche EUR 500m WNG - 4Y & 10Y).

SG is acting as M&A advisor to ArcelorMittal in its acquisition of a strategic stake in Vallourec

SG and/or an affiliate acted as Joint Bookrunner in Vonovia SE's bond issue (EUR 2bn, triple-tranche, 5y, 8.5y and 12y).

SG and/or its affiliates beneficially own 0.5% or more of [the total issued share capital/any class of common equity securities] with a net [long/short] position of: Rolls-Royce Holdings PLC, Anglo American PLC, AB InBev, ABN AMRO, Koninklijke Ahold Delhaize NV, AEGON, Ageas, AIB, Airbus SE, Akzo Nobel NV, Arcadis, ASR, Auto Trader Group, BAE Systems PLC, Banco BPM, Barclays, BE Semiconductor Industries NV, Beazley, Bank of Ireland, B&M European Value Retail SA, Banca Monte Paschi, Bunzl, BPER Banca, Burberry Group PLC, Carrefour SA, Coca-Cola Hellenic, Cellnex Telecom SA, Centrica PLC, ConvaTec Group PLC, Danske Bank, Vinci, Delivery Hero SE, Sartorius Stedim Biotech, DNB, EDP - Energias de Portugal SA, EssilorLuxottica SA, Elisa, Man Group, Enel SpA, Siemens Energy AG, Euronext, CTS Eventim, Fineco, FDJ United, Eiffage, Generali, Gecina, Glencore PLC, Grifols, Gaztransport & Technigaz, Haleon plc, Hiscox, International Consolidated Airlines Group SA, Infineon Technologies AG, InterContinental Hotels Group PLC, 3i Group PLC, Informa, Intesa Sanpaolo, Intertek Group PLC, KION Group, Kone, Koninklijke KPN NV, Kingspan, Leonardo SpA, LEG Immobilien AG, Deutsche Lufthansa AG, Signify, Lloyds Banking Group, Michelin, Moncler SpA, Nexans, NN Group, Nokia, Naturgy Energy Group SA, NatWest Group, Next PLC, Orange SA, Pandora, Pennon Group, Prudential, Prysmian SpA, Puma, Ferrari NV, Randstad, RELX PLC, Rheinmetall AG, Pernod Ricard SA, Rio Tinto PLC, Reckitt, Rightmove, Renault SA, Rockwool B, Rubis, Rexel, Safran SA, Sampo, Sanofi, SBM Offshore NV, SCOR, Sage, Sopra Steria Group, Spie, Saipem SpA, Snam SpA, St James's Place, Stellantis NV, Subsea 7 SA, Syensqo, Telenor ASA, Tele2, Teleperformance, Terna, Tryg, Tesco PLC, TUI AG, Unicredit, Unilever, Unibail-Rodamco-Westfield, Veolia, Vallourec SA, Vonovia SE, Vodafone Group PLC, Wolters Kluwer NV and Whitbread PLC.

SG and/or its affiliates beneficially own 1% or more of a class of common equity securities of the following companies: Rolls-Royce Holdings PLC, Anglo American PLC, AEGON, Ageas, AIB, ASR, Auto Trader Group, Barclays, BE Semiconductor Industries NV, Beazley, BAWAG, Bank of Ireland, B&M European Value Retail SA, Banca Monte Paschi, BP PLC, Burberry Group PLC, Cellnex Telecom SA, Centrica PLC, ConvaTec Group PLC, Delivery Hero SE, Enel SpA, Siemens Energy AG, Experian PLC, FDJ United, Grifols, Gaztransport & Technigaz, Haleon plc, Hiscox, International Consolidated Airlines Group SA, Infineon Technologies AG, Italgas SpA, 3i Group PLC, Informa, Intertek Group PLC, Koninklijke KPN NV, Leonardo SpA, LEG Immobilien AG, Signify, Lloyds Banking Group, LSE Group, Merlin Properties, Nexans, Nexi, NN Group, NatWest Group, Next PLC, Orange SA, Prudential, Pernod Ricard SA, Rightmove, Renault SA, Rockwool B, Rubis, Rexel, Sampo, SBM Offshore NV, SCOR, Sage, St James's Place, Subsea 7 SA, Technip Energies, Teleperformance, Unicredit, Unibail-Rodamco-Westfield, Vallourec SA, Vonovia SE, Vodafone Group PLC, Wolters Kluwer NV and Whitbread PLC.

Bernstein Autonomous LLP or BSG France SA, beneficially, has either a net long or short position of 0.5% or more of the total issued share capital of a class of common equity securities of the following MiFID eligible securities: Anglo American PLC, Aalberts, Aedifica, AIB, Arkema, Ambu A/S, Argenx SE, Auto Trader Group, BAE Systems PLC, Beazley, BAWAG, Burberry Group PLC, Compass Group PLC, Danske Bank, Erste Group, Elis, Euronext, Experian PLC, Genmab A/S, Gaztransport & Technigaz, Hiscox, InterContinental Hotels Group PLC, 3i Group PLC, Informa, Nexans, NN Group, NatWest Group, Prudential, Prysmian SpA, Ferrari NV, Royal Unibrew, RELX PLC, Reckitt, Roche Holding AG, Ryanair Holdings PLC, Saipem SpA, Straumann Holding AG, TAG Immobilien AG, Tele2, Tryg and Tesco PLC.

AB and/or its affiliates beneficially own 1% or more of a class of common equity securities of the following companies: AIB, Auto Trader Group, BAE Systems PLC, Beazley, Burberry Group PLC, Euronext, Genmab A/S, Gaztransport & Technigaz, Informa, NN Group, Prysmian SpA, Ferrari NV, Royal Unibrew, Saipem SpA, TAG Immobilien AG, Tryg and Tesco PLC.

Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from Rolls-Royce Holdings PLC, AB InBev, ABN AMRO, Accor SA, Credit Agricole, Koninklijke Ahold Delhaize NV, Aéroports de Paris, AEGON, Airbus SE, Arkema, Akzo Nobel NV, Amadeus IT Group, SA, Assa Abloy AB, Aviva, Banco BPM, Barclays, Bank of Ireland, Banca Monte Paschi, BPER Banca, Burberry Group PLC, BT Group PLC, Bureau Veritas SA, Carrefour SA, Capgemini, Carlsberg A/S, Coca-Cola Hellenic, Cellnex Telecom SA, Centrica PLC, AXA, Danske Bank, Vinci, DHL Group, DNB, Dassault Systèmes, Deutsche Telekom AG, Erste Group, Edenred, EDP - Energias de Portugal SA, Elis, Enel SpA, Enagas, Engie SA, ENI SpA, Euronext, E.ON SE, Equinor ASA, Eiffage, Fortum OYJ, Valeo SA, Generali, Getlink, Gecina, Heineken NV, HSBC, International Consolidated Airlines Group SA, Italgas SpA, 3i Group PLC, Intesa Sanpaolo, KBC, Kering SA, KION Group, Kuehne + Nagel International AG, Deutsche Lufthansa AG, Klepierre, Lloyds Banking Group, Legrand SA, LVMH Moet Hennessy Louis Vuitton SE, Merck KGaA, Merlin Properties, Nestle SA, Nokia, Novartis AG, Naturgy Energy Group SA, NatWest Group, L'Oréal, Orange SA, Bank Pekao, Koninklijke Philips NV, PKO BP, Rheinmetall AG, Pernod Ricard SA, Rio Tinto PLC, Reckitt, Renault SA, Rubis, Rexel, Sanofi, Siemens Healthineers AG, Siemens AG, Smith & Nephew PLC, Spie, Snam SpA, Stellantis NV, Schneider Electric SA, Sodexo SA, Technip Energies, Telefonica SA, TAG Immobilien AG, Teleperformance, Terna, TUI AG, Unicredit, Universal Music Group, Unibail-Rodamco-Westfield, Verbund, Veolia, Vonovia SE and Volkswagen AG.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Anglo American PLC, ABB Ltd, Associated British Foods PLC, AB InBev, ABN AMRO, Accor SA, Credit Agricole, Koninklijke Ahold Delhaize NV, Aéroports de Paris, AEGON, Ageas, Air Liquide SA, AIB, Airbus SE, Arkema, Akzo Nobel NV, Dassault Aviation SA, Amadeus IT Group, SA, Antofagasta PLC, ASR, Assa Abloy AB, Aviva, Avanza, AstraZeneca PLC, BAE Systems PLC, Banco BPM, Barclays, BASF SE, Beiersdorf, BAWAG, Bank of Ireland, Banca Monte Paschi, Bayerische Motoren Werke AG, Danone, BNP Paribas, Boliden AB, BP PLC, BPER Banca, Burberry Group PLC, BT Group PLC, Bureau Veritas SA, Carrefour SA, Capgemini, Carlsberg A/S, Coca-Cola Hellenic, Cie Financiere Richemont SA, Cellnex Telecom SA, Centrica PLC, Covivio SA, Compass Group PLC, Davide Campari-Milano NV, AXA, Danske Bank, Vinci, DHL Group, D'Ieteren, Sartorius Stedim Biotech, DNB, DSV A/S, Dassault Systèmes, Deutsche Telekom AG, Daimler Truck AG, Erste Group, Edenred, EDP - Energias de Portugal SA, EssilorLuxottica SA, Endesa SA, Elis, Man Group, Enel SpA, Enagas, ENI SpA, Siemens Energy AG, Euronext, Equinor ASA, Eurofins Scientific SE, Ericsson, easyJet PLC, Fineco, Fortum OYJ, Valeo SA, Generali, Getlink, Gecina, Glencore PLC, Gaztransport & Technigaz, Heineken NV, Hennes & Mauritz AB, Thales SA, HSBC, Hiscox, International Consolidated Airlines Group SA, Italgas SpA, 3i Group PLC, Intesa Sanpaolo, Jyske Bank, KBC, Kering SA, KION Group, Kuehne + Nagel International AG, Leonardo SpA, LEG Immobilien AG, Legal & General, Klepierre, Lloyds Banking Group, Legrand SA, LSE Group, AP Moller - Maersk A/S, Mercedes-Benz Group AG, LVMH Moet Hennessy Louis Vuitton SE, Michelin, M&G Plc, Merlin Properties, Nexans, Nexi, National Grid PLC, NN Group, Nokia, Novartis AG, Novo Nordisk A/S, Naturgy Energy Group SA, NatWest Group, L'Oréal, Orsted A/S, Dr Ing hc F Porsche AG, Koninklijke Philips NV, Pirelli & C SpA, Pennon Group, Prudential, Publicis Groupe SA, Ferrari NV, Randstad, Raiffeisen, Redeia, RELX PLC, Pernod Ricard SA, Rio Tinto PLC, Reckitt, Renault SA, Roche Holding AG, Rubis, Rexel, Ryanair Holdings PLC, Safran SA, Sampo, Sanofi, SAP SE, SBM Offshore NV, SCOR, Handelsbanken, Siemens Healthineers AG, Siemens AG, Smith & Nephew PLC, Sopra Steria Group, Spie, Saipem SpA, Snam SpA, Stellantis NV, Schneider Electric SA, Sodexo SA, Swedbank, Technip Energies, Telefonica SA, TAG Immobilien AG, Teleperformance, Terna, TUI AG, Unicredit, Swatch Group AG/The, Universal Music Group, Unibail-Rodamco-Westfield, Verbund, Veolia, Vallourec SA, Vonovia SE, Volvo AB, Wolters Kluwer NV and Zabka Group SA.

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Associated British Foods PLC, Credit Agricole, Admiral, AEGON, Air Liquide SA, AIB, Arkema, ASR, Aviva, Barclays, Beiersdorf, BAWAG, Bank of Ireland, Bayerische Motoren Werke AG, BNP Paribas, Carlsberg A/S, AXA, Danske Bank, Sartorius Stedim Biotech, DNB, Dassault Systèmes, Erste Group, Edenred, Endesa SA, Euronext, E.ON SE, Eiffage, Generali, Glencore PLC, Hennes & Mauritz AB, HSBC, Hiscox, Informa, Intesa Sanpaolo, Industria de Diseno Textil SA, Jyske Bank, KBC, LEG Immobilien AG, Legal & General, Lindt & Sprüngli, Lloyds Banking Group, AP Moller - Maersk A/S, Mercedes-Benz Group AG, M&G Plc, Merck KGaA, Merlin Properties, National Grid PLC, NatWest Group, Next PLC, Orange SA, Prysmian SpA, Redeia, Roche Holding AG, Sampo, SCOR, Handelsbanken, Siemens AG, Swedbank, Teleperformance, Terna, Verbund, Vonovia SE and Vodafone Group PLC.

Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: ABN AMRO, Credit Agricole, Admiral, AEGON, Aviva, BAE Systems PLC, Banco BPM, Barclays, BAWAG, Bank of Ireland, Banca Monte Paschi, BNP Paribas, AXA, Danske Bank, DNB, Erste Group, Man Group, Fineco, Generali, HSBC, Hiscox, Intesa Sanpaolo, Jyske Bank, KBC, Legal & General, Lloyds Banking Group, M&G Plc, Merlin Properties, NN Group, Novartis AG, Novo Nordisk A/S, PKO BP, Prudential, Raiffeisen, Hermes International, Sampo, SCOR, Handelsbanken, Swedbank and Unicredit.

Bernstein and/or affiliates expect to receive or intend to seek compensation for investment banking services in the next three months from Rolls-Royce Holdings PLC, Anglo American PLC, Accor SA, Credit Agricole, Aéroports de Paris, Ageas, Air Liquide SA, Airbus SE, Akzo Nobel NV, Amadeus IT Group, SA, Assa Abloy AB, AstraZeneca PLC, Banco BPM, Barclays, Bank of Ireland, Banca Monte Paschi, Bayerische Motoren Werke AG, BPER Banca, Carrefour SA, Coca-Cola Hellenic, Cellnex Telecom SA, Centrica PLC, Danske Bank, Vinci, Dassault Systèmes, Deutsche Telekom AG, Erste Group, EDP - Energias de Portugal SA, EssilorLuxottica SA, Elia Group, Elis, Enagas, Engie SA, ENI SpA, Euronext, easyJet PLC, Eiffage, Fortum OYJ, Valeo SA, Generali, Getlink, Gecina, Thales SA, HSBC, Italgas SpA, Intesa Sanpaolo, KBC, Kering SA, KION Group, Kuehne + Nagel International AG, Klepierre, Lloyds Banking Group, Legrand SA, Michelin, Nexans, Novartis AG, Naturgy Energy Group SA, L'Oréal, Orange SA, Bank Pekao, Pirelli & C SpA, Rheinmetall AG, Pernod Ricard SA, Renault SA, Rubis, Rexel, Sanofi, Cie de St-Gobain, Siemens Healthineers AG, Smith & Nephew PLC, Spie, Saipem SpA, Snam SpA, Stellantis NV, Schneider Electric SA, Technip Energies, Telefonica SA, Teleperformance, TUI AG, Unicredit, Universal Music Group, Unibail-Rodamco-Westfield, Verbund, Veolia, Vonovia SE, Vodafone Group PLC and Volkswagen AG.

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